

LCOS-SLM
Liquid Crystal Based Spatial Light Modulator

SLM-200

Operation Manual



Notes to Users

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1. Introduction

1.1 Description

The santec spatial light modulator (SLM) is based on reflective liquid crystal on silicon (LCOS) microdisplay technology. The SLM has an active matrix display with 1920 x 1200 resolution (WUXGA) and a 0.63 diagonal which enables the SLM to modulate optical phases freely and generate arbitral 2D phase patterns (gratings, phase mask, holograms, etc.) on a LCOS pixel-by-pixel basis. The phase response of SLM is adjusted even for different wavelengths (450 to 1600 nm) automatically using proprietary GUI software with built-in adjustable gamma control. The signal is addressed via a standard DVI (Digital Visual Interface) signal e.g. by a PC's graphics.

The SLMs are characterized by ultra-low phase fluctuation from high drive frequency over 1kHz and 10bit high resolution addressing. These abilities are suitable for various applications including laser material processing, optical switching devices, wavefront correction and pulse shaping.

1.2 Features

- Stable operation with no mechanical motion
- All-in-one compact design
- Pure, linear and precise phase control: Addressing bit depth 10 Bit
- Telecom qualified high reliability: Telcordia GR-1221 Core
- Ultra-low phase fluctuation: Typ. $< 0.001\pi$ rad.
- Wide wavelength range (450 to 1600 nm) by adjustable gamma control



(a) All-in-one model



(b) Separate model

Figure 1: SLM body.

1.3 Care

The unit should not be operated in any of the following circumstances:

- If water or other liquid is spilled on the unit.
- When the unit is exposed to moisture.
- When the unit is cleaned or moved.
- If damage is found to the main cord.
- When the unit condenses.

Pay attention to the following guidelines for safe, trouble-free operation:

- Do not place in direct sunlight.
- Do not place in an environment with high temperature and high humidity.
- Do not place in an environment with dust, dirt, salt, or corrosive gas.
- Do not place in an area subject to large vibrations.
- Do not place in an area subject to noise by electric field, magnetic field, etc.
- Do not place in an area where the unit is subject to falling objects.
- Do not place in an area with possible exposure to water.

General usage guidelines:

- All electrical interfaces have to be connected without applied voltage.
- It is required to boot up the instrument AFTER connecting all devices, even if the DVI output instrument is available.
- Do not touch the surface of the display because the LCD has the potential of damage by static electricity.
- If you plan to illuminate the SLM with high power light or shorter wavelength light like ultraviolet, please consult the santec service site.
- If you handle the LCOS unit, please wear a wristband for ESD on an earth grounded table as shown in figure 2.

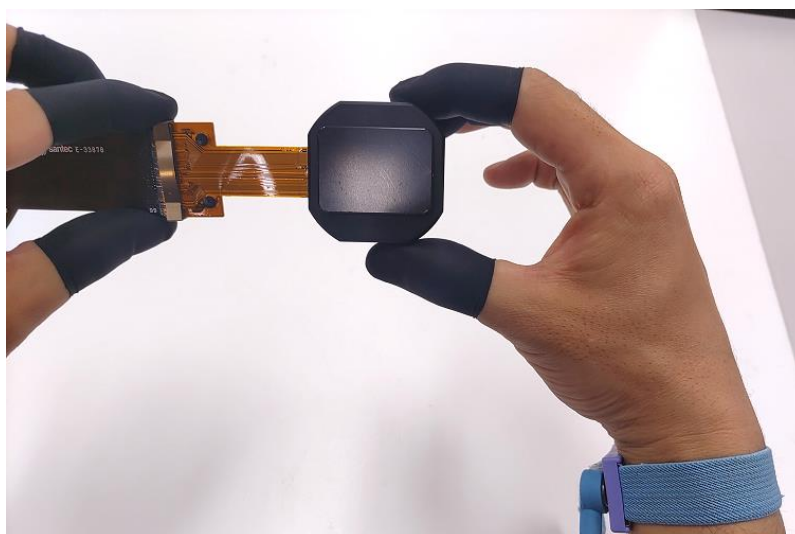


Figure 2: LCOS unit handling.

1.4 Warranty

The warranty of the product is based on santec document (Doc: SA-15-0308).

2. Package

2.1 Package contents

This product is composed of the SLM body and accessories.

The following parts are included:

● All-in-one model

- (1) LCOS-SLM body
- (5) AC adapter
- (6) DVI/HDMI cable
- (7) USB3.0 Type A/B cable
- (8) SMB/BNC cable
- (9) Polarizer (Wavelength: 450~700 nm)
- (10) USB flash drive (Control software GUI / User operational manual)

● Separate model

- (1) LCOS-SLM body
- (2) LCOS unit
- (3) Bracket for LCOS unit
- (4) Attachment for LCOS unit
- (5) AC adapter
- (6) DVI/HDMI cable
- (7) USB3.0 Type A/B cable
- (8) SMB/BNC cable
- (9) Polarizer (Wavelength: 450~700 nm)
- (10) USB flash drive (Control software GUI / User operational manual)



Figure 3: Contents

2.2 Shipping package

● Shipping box

1	Vender name
2	Model number
3	Serial number



Figure 4: Shipping box

● Lower case

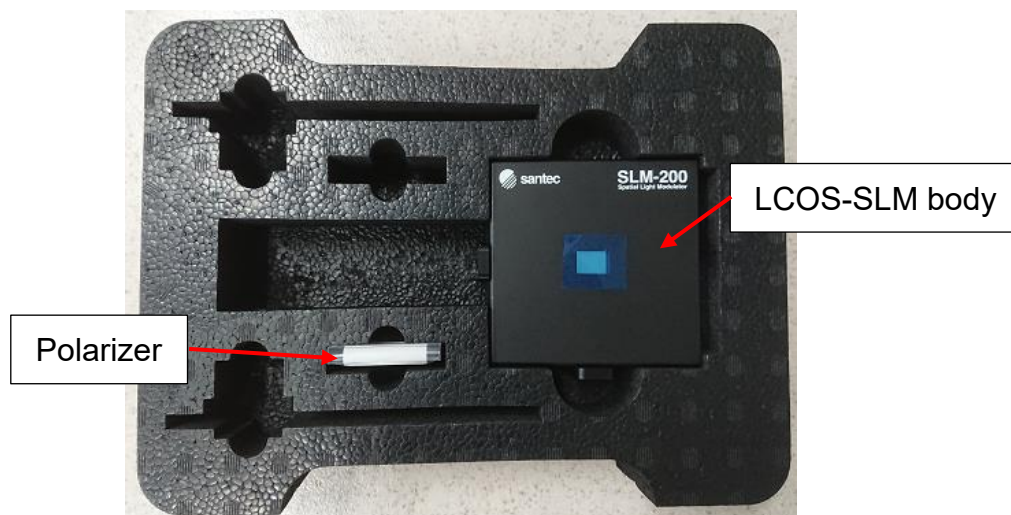


Figure 5: Lower case storage (All-in-one model)

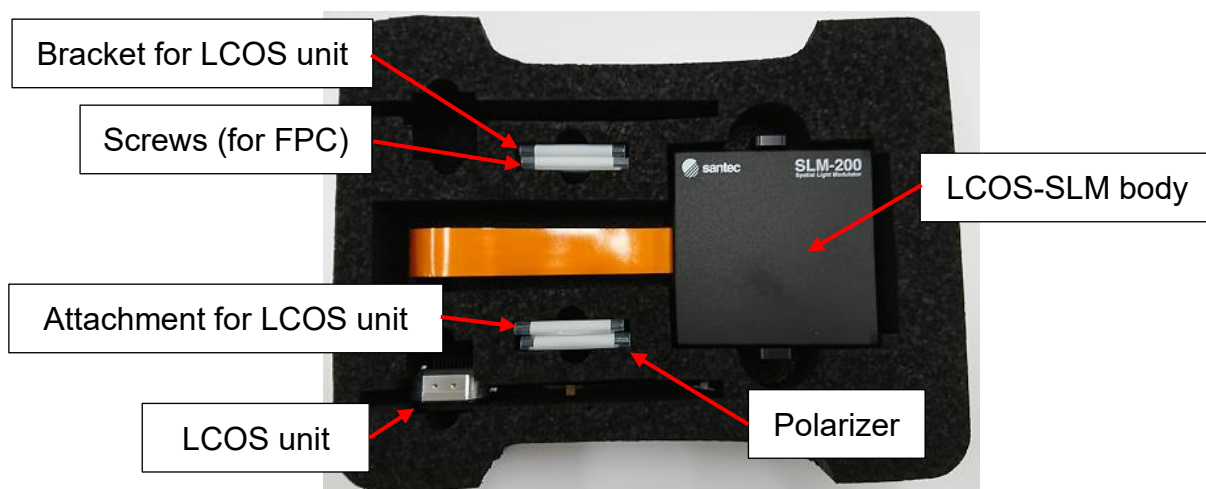


Figure 6: Lower case storage (Separate model)

● Upper case



Figure 7: Upper case storage

3. Display specifications

3.1 Parameters

Table 1: Absolute Maximum Ratings

<Optical Characteristics>

Item	Min	Max	Units	Notes
Optical power handling ¹⁾	Typ. 10		W/cm ²	@1550nm CW, 2.0mm beam diameter

1) The value is not guaranteed.

<Electrical characteristics>

[5VDC power supply] *LCOS-SLM body_DC power jack

Item	Min	Max	Units	Notes
Input voltage	-	5.5	V	-

[Trigger input] *LCOS-SLM body_SMB connector

Item	Min	Max	Units	Notes
Input voltage	-	5	V	-

[LCOS unit]

Item	Min	Max	Units	Notes
ESD	±150		V	200pF, 0Ω Apply three times at room temperature (non-operation)

<Operating characteristics>

Item	Min	Max	Units	Notes
Operating temperature	15	35	degC	No condensation
Operating humidity	-	80	%	No condensation

<Storage characteristics>

Item	Min	Max	Units	Notes
Storage temperature	0	40	degC	No condensation
Storage humidity	-	80	%	No condensation

Table 2: Optical Characteristics

Item	Min	Max	Units	Notes
Wavelength range	450	1600	nm	(Refer to appended table about AR coating option)
Panel size	(H)15.36 x (V)9.60		mm	Active area
Panel resolution ¹⁾	(H)1920 x (V)1200		pixel	-
Pixel size / pitch	7.8 / 8.0		μm	-
Panel reflectivity	Typ. >90		%	Depending on specified wavelength range
Aperture ratio	95		%	-
Gray level	10 (1024)		bit	-
Response time ²⁾	Typ. 300		ms	Depending on LCOS phase pattern
LCOS drive frequency	1200		Hz	-
Phase depth	2π	-	rad.	-
Phase stability	Typ. <0.002 π		rad.	-

1) Specification on the defect pixels are no object.

2) Response time is not affected by Frame rate.

Table 3: Electrical and Mechanical characteristics

Item	Min	Max	Units	Notes
Interface	Digital Video Interface (DVI-D)* USB3.0 (typeB) Trigger IN/OUT (SMB)		-	*10-bit using RGB 8-bit, 3 colors
Frame rate	60 or 120		Hz	Factory default setting
Input voltage to AC adapter	100	240	VAC	-
Mating durability (LCOS unit connector)	-	50	cycle	Insertion / Extraction cycles
Dimensions	LCOS unit	45 x 45 x 25.7	mm	Separate model only
	SLM body	117.6 x 117.6 x 33.7		-
Weight	LCOS unit	0.13	kg	Separate model only
	SLM body	0.42		-
Control software	GUI software and SDK for Windows		-	C#, Python, Matlab, Labview

Table 4: AR coating options

Item	Parameter							Units
Ordering code number	-01	-02	-03	-04	-12	-14	-21	-
AR coat range	450-550	750-850	1000-1100	1500-1600	400-700	450-550 / 1500-1600	450-1600	nm
reflectance ¹⁾	<0.5				<1.5	<0.6	<2.5	%

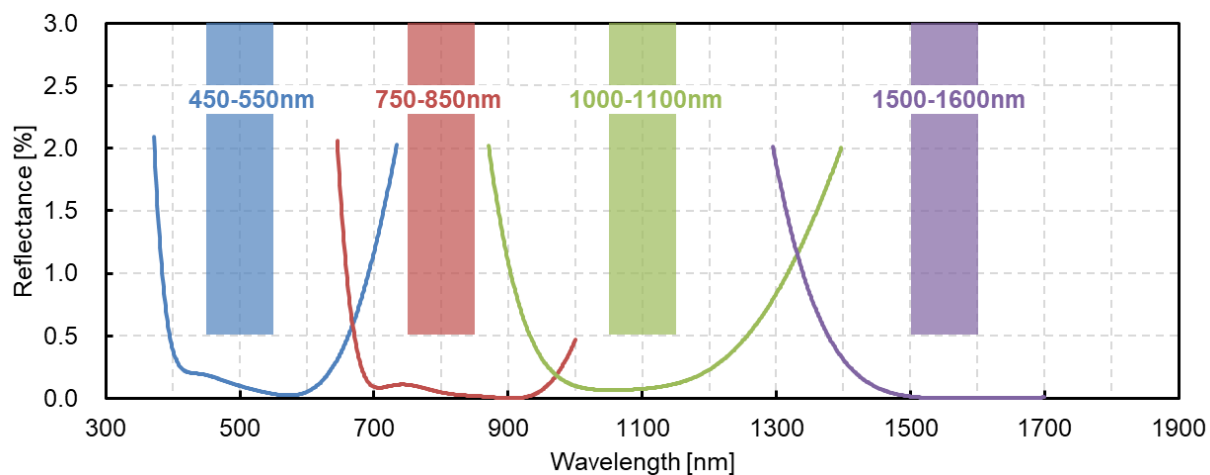
1) Angle of incidence = 0 degree

Seven types of AR coating are available as shown in Figure 8, 9.

- (1) 450-550 nm
- (2) 750-850 nm
- (3) 1000-1100 nm
- (4) 1500-1600 nm
- (5) 400-700 nm
- (6) 450-550 / 1500-1600 nm
- (7) 450-1600 nm

■ Incident angle = 0 degree

○ Single-Band



○ Wide-Band & Dual-Band

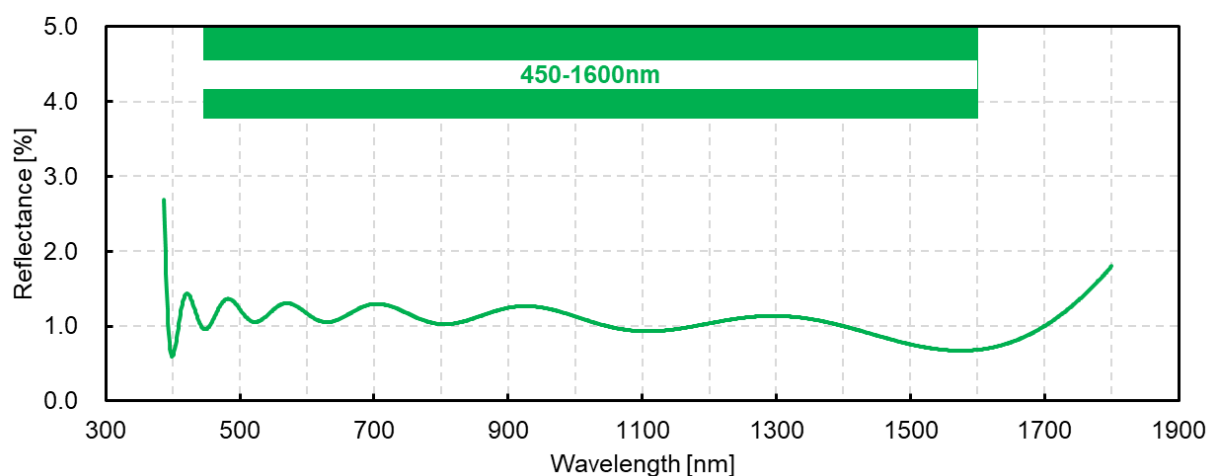
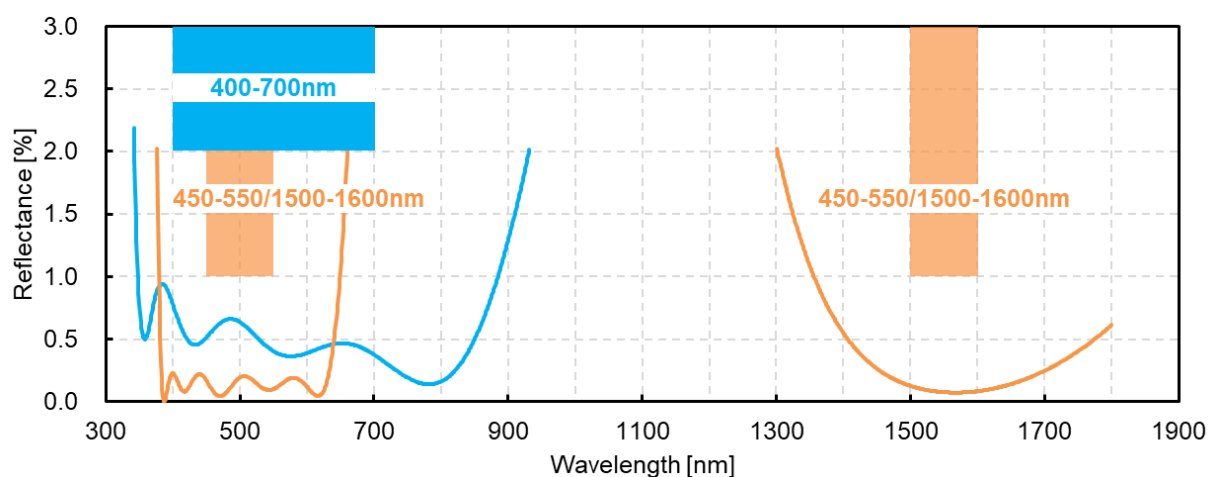
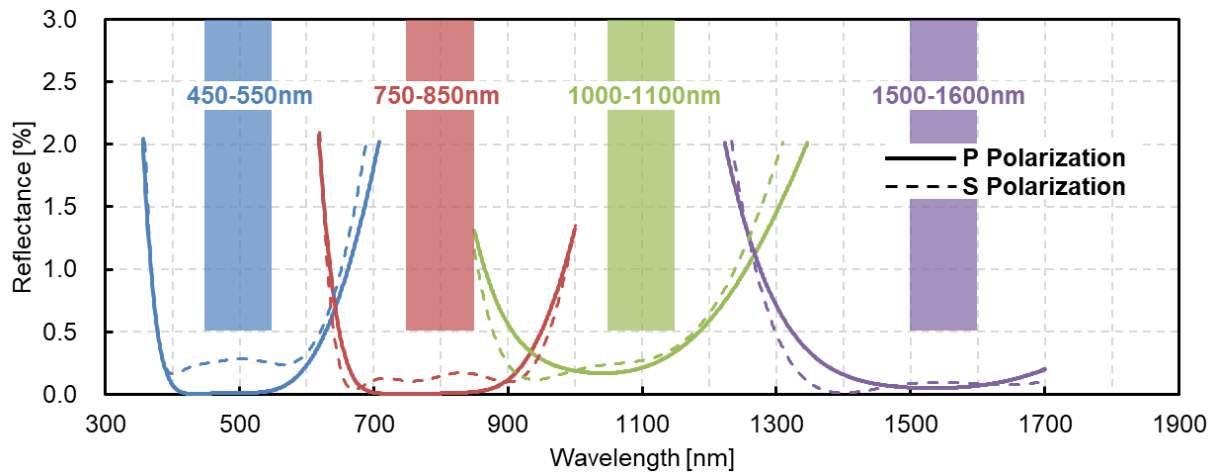


Figure 8: AR reflectivity (angle: 0 degree).

■ Incident angle = 30 degree S & P Polarization

○ Single-Band

Reference data



○ Wide-Band & Dual-Band

Reference data

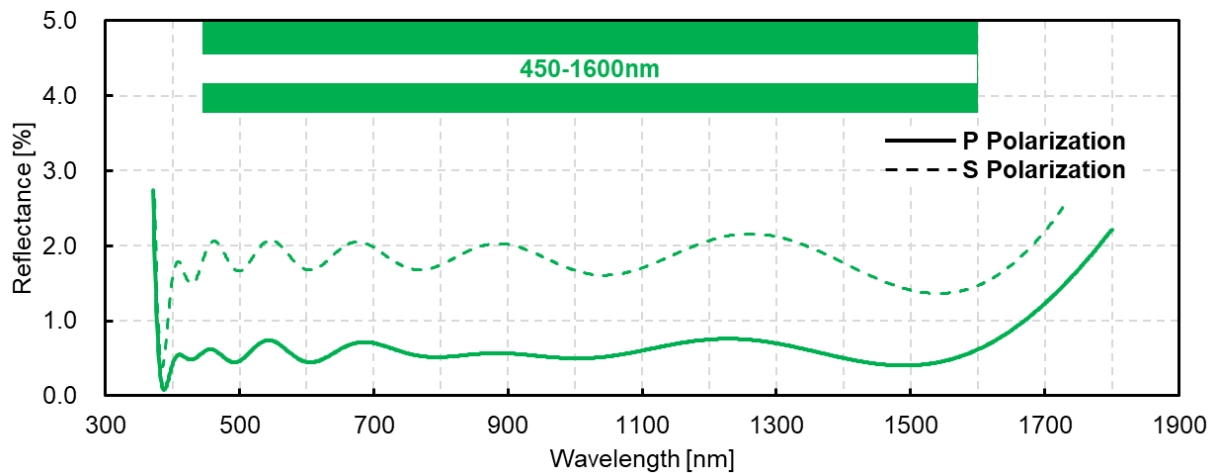
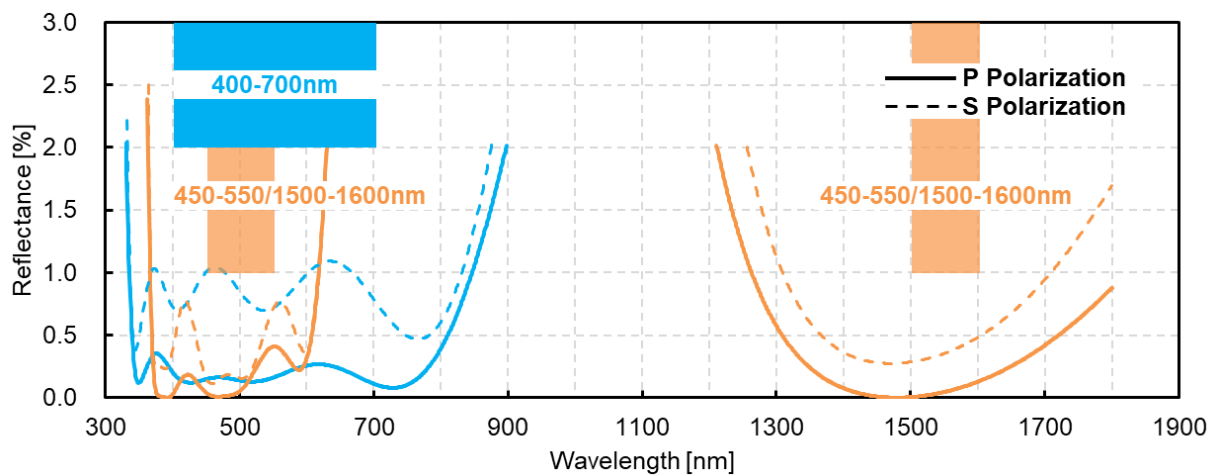


Figure 9: AR reflectivity (angle: 30 degree).

3.2 Package schematic

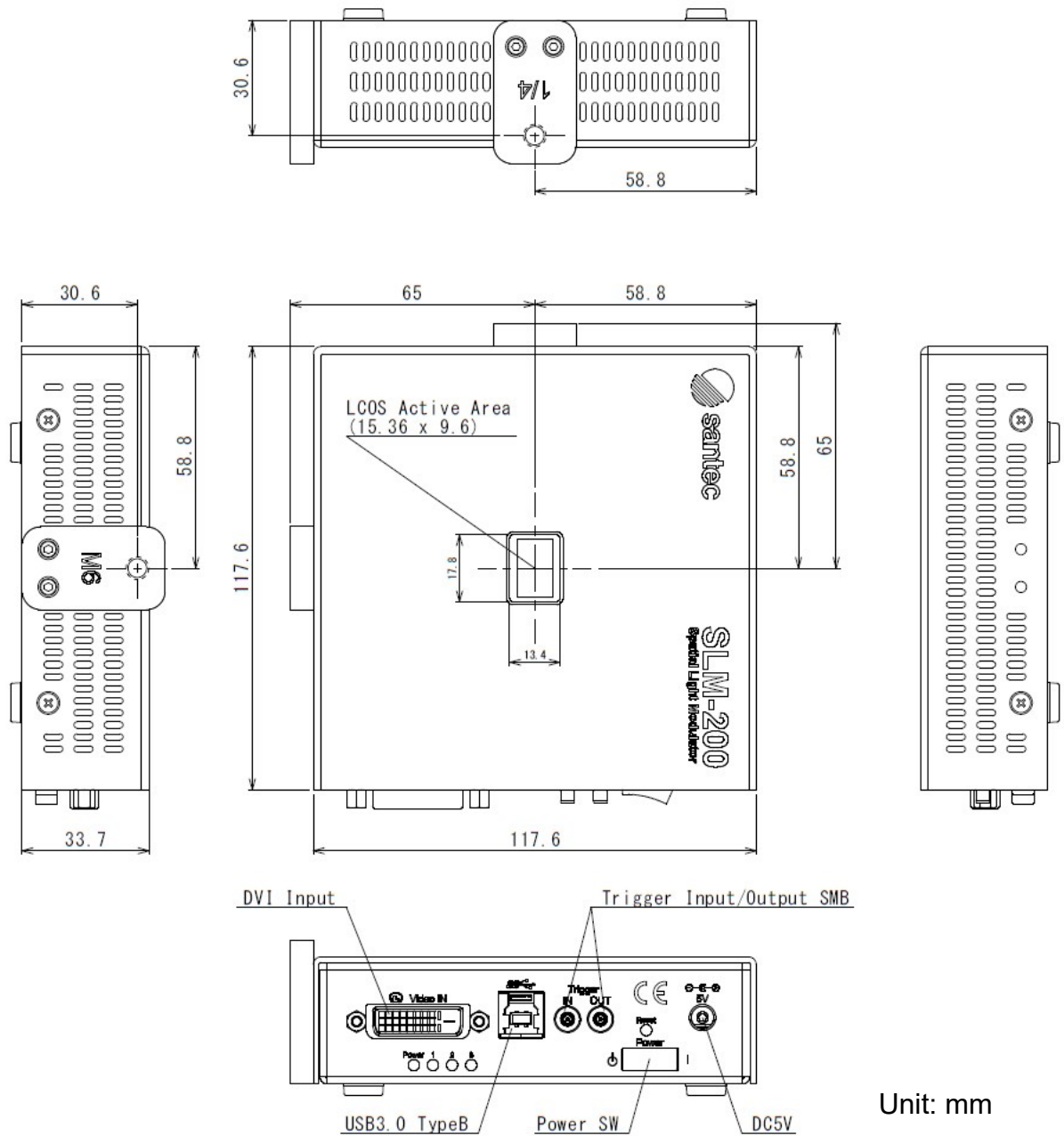
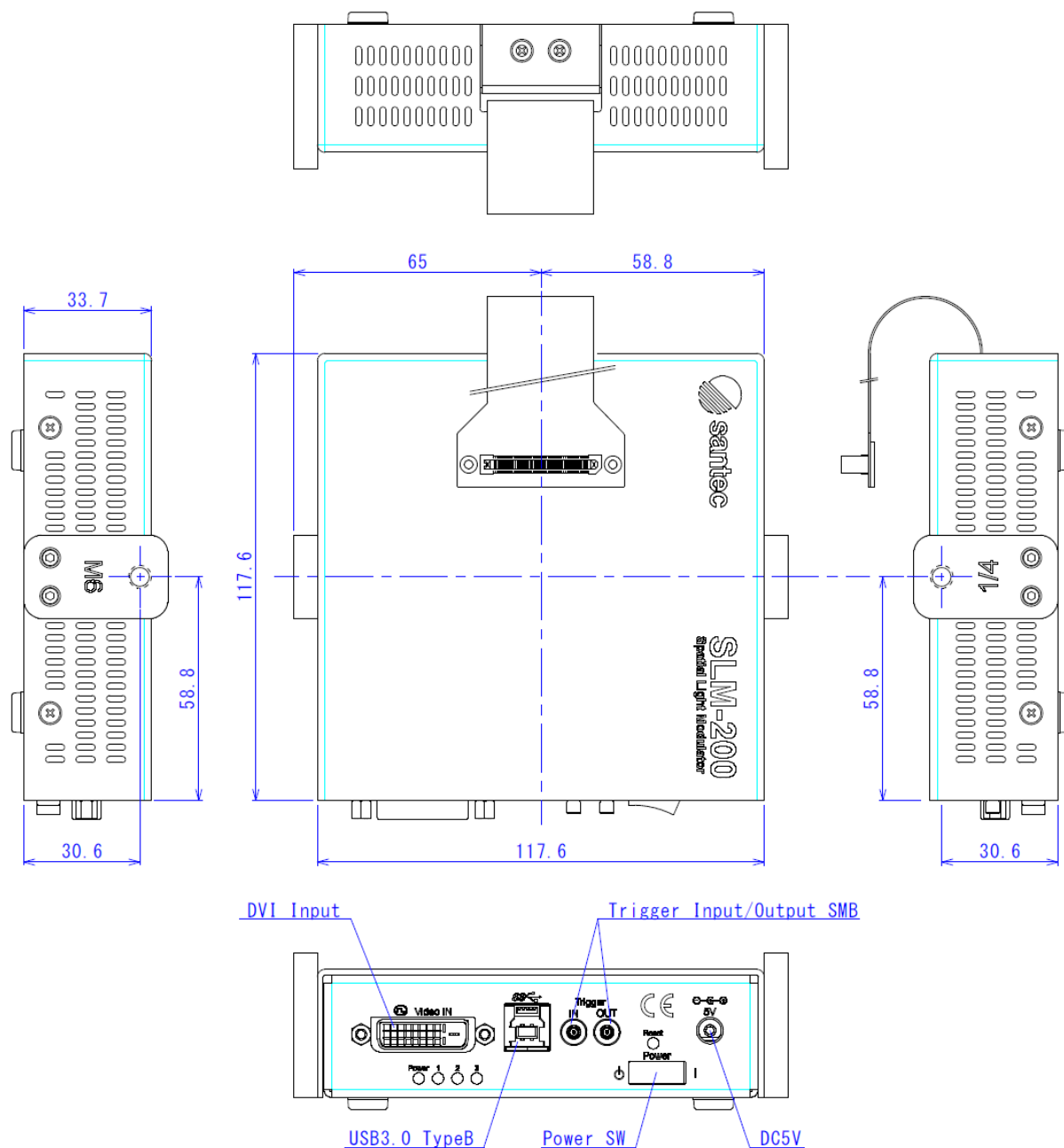


Figure 10: Dimensions (All-in-one model).



Unit: mm

Figure 11: Dimensions (Separate model).

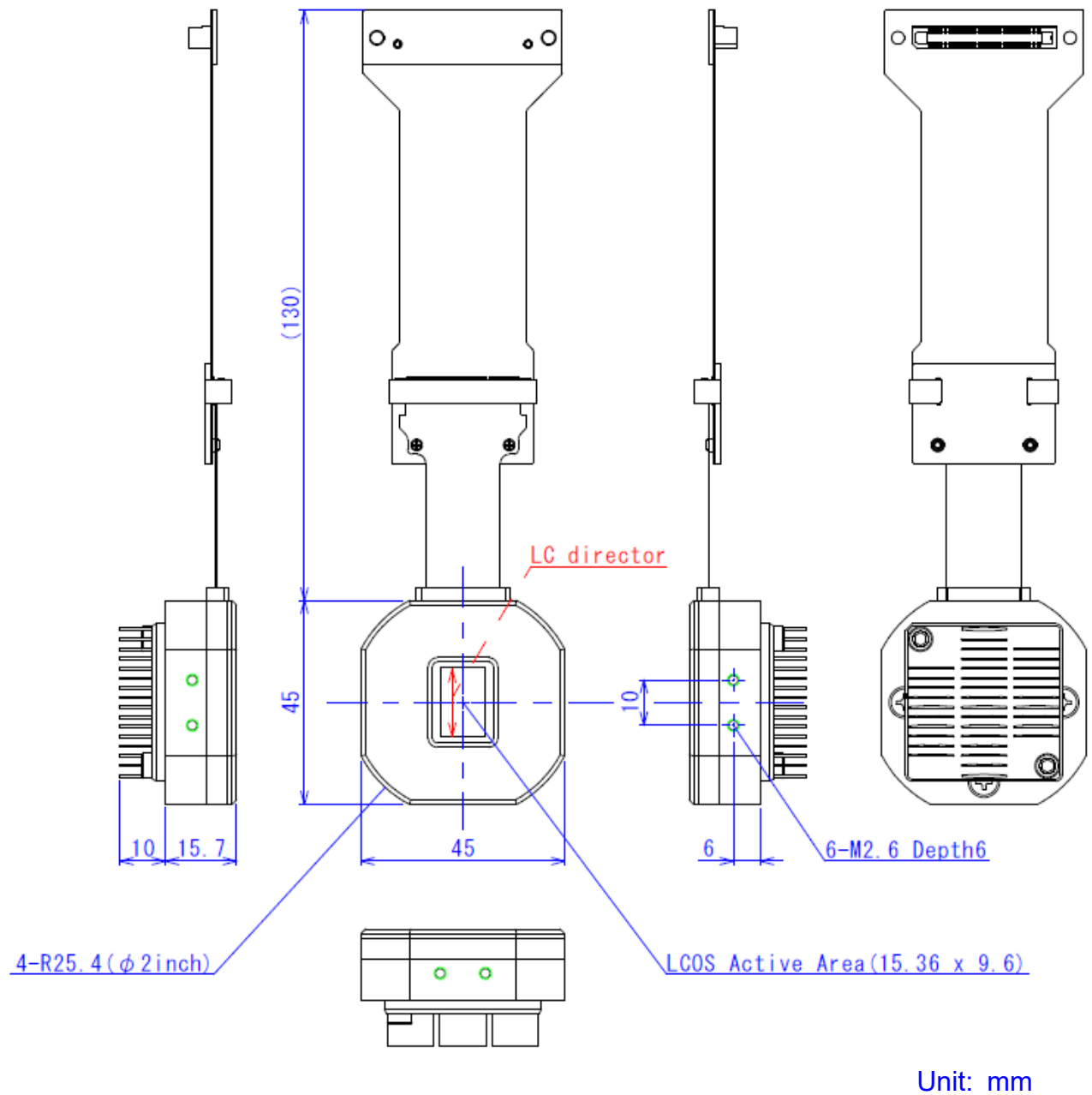
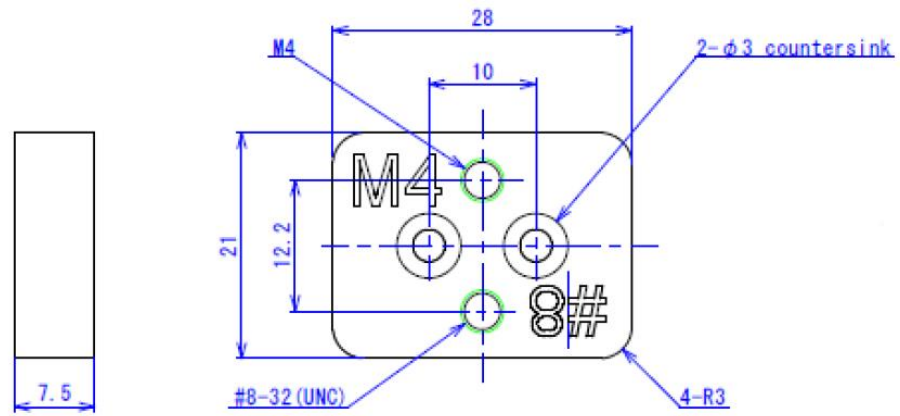
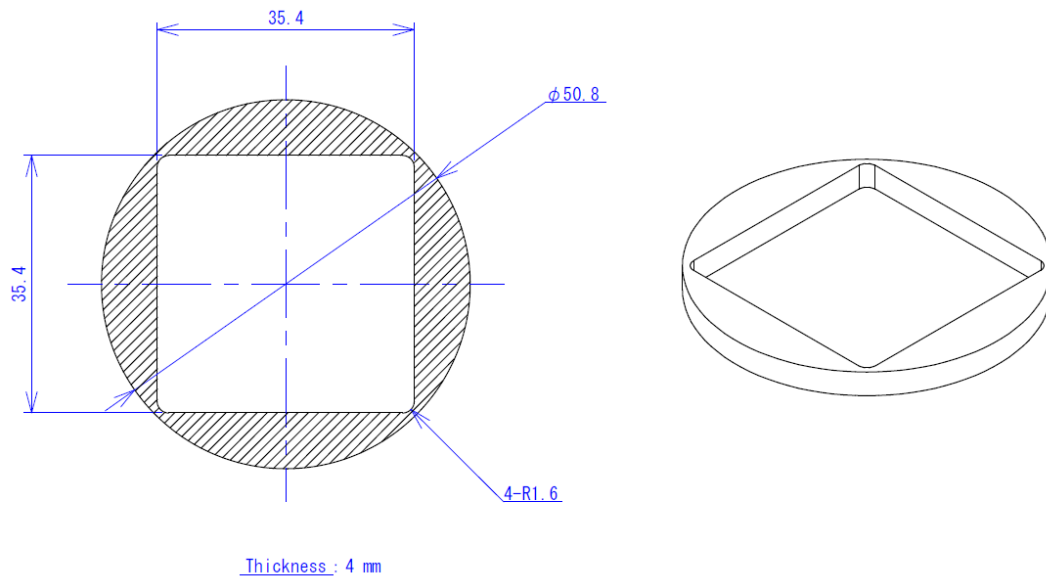


Figure 12: Dimensions (LCOS unit).



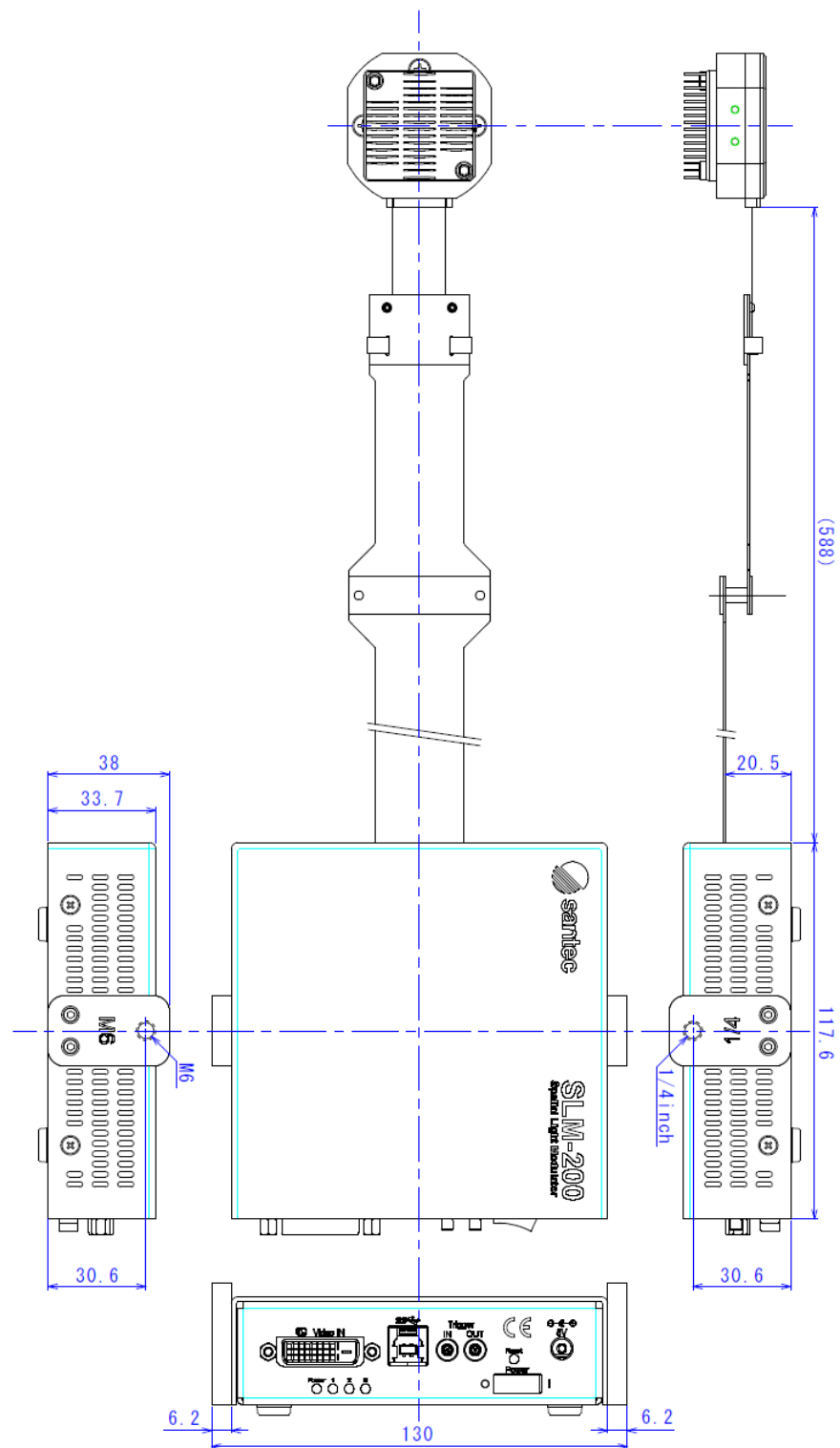
Unit: mm

Figure 13: Dimensions (Bracket for LCOS unit).



Unit: mm

Figure 14: Dimensions (Attachment for LCOS unit).



Unit: mm

Figure 15: Dimensions (Separate model - LCOS unit connection).

3.3 Operating principle

The SLM is an optical intensity or phase modulator using LCOS display technology that consists of a silicon-based reflective, active matrix CMOS driver which uses a nematic liquid crystal. LCOS has several million 2-dimensional pixels separated by a few tenths of a micron and can address pixel-by-pixel. Optical phase or intensity can be controlled spatially by the voltage level applied to each pixel.

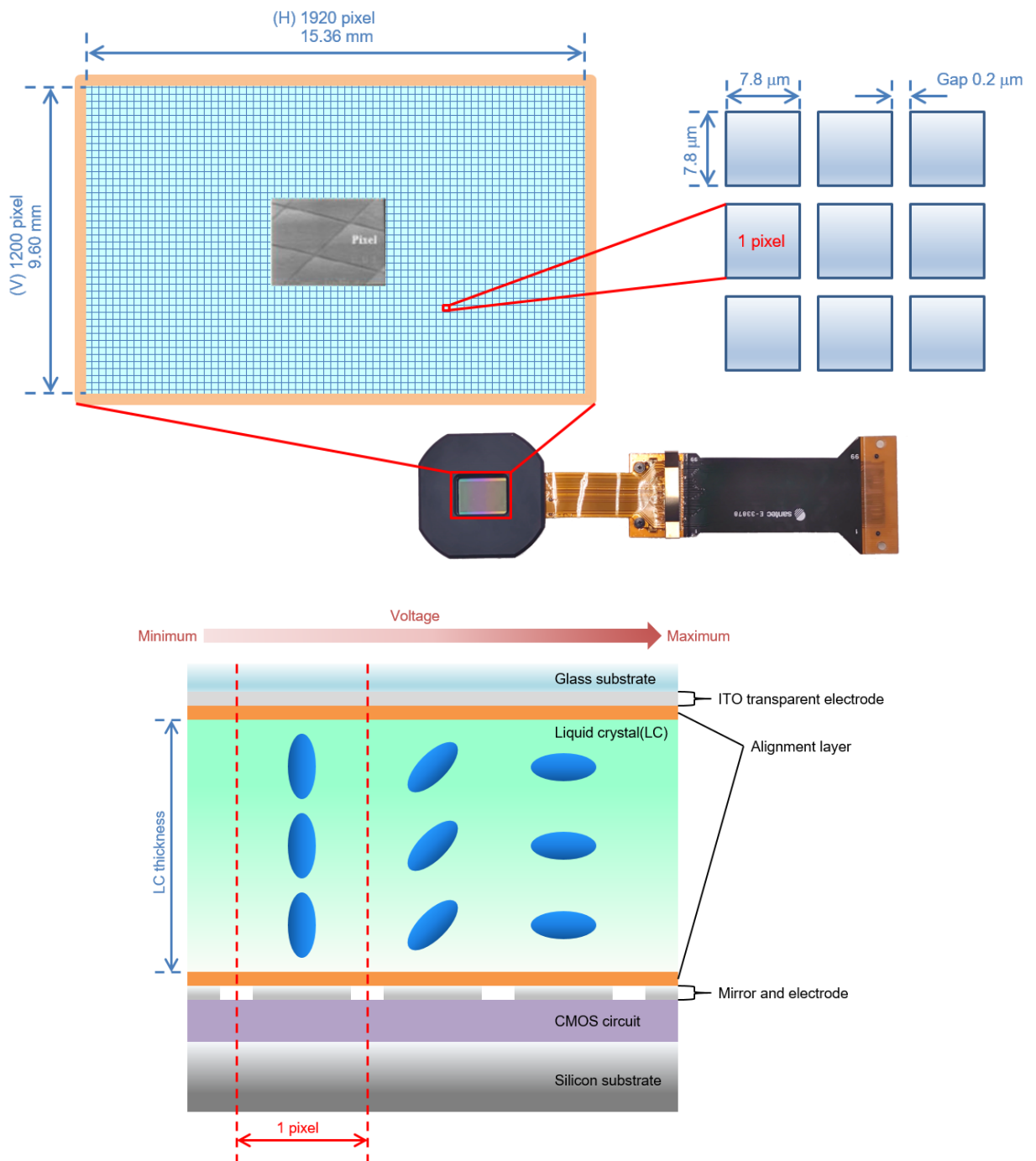


Figure 16: LCOS structure. *Vertical alignment (VA)

The SLM can be controlled via a PC using the DVI and USB interface. Easy control using the attached optical phase (intensity) editor and graphical user interface (GUI), wavelength independent precise and linear phase modulation with software built-in gamma control can be accomplished.

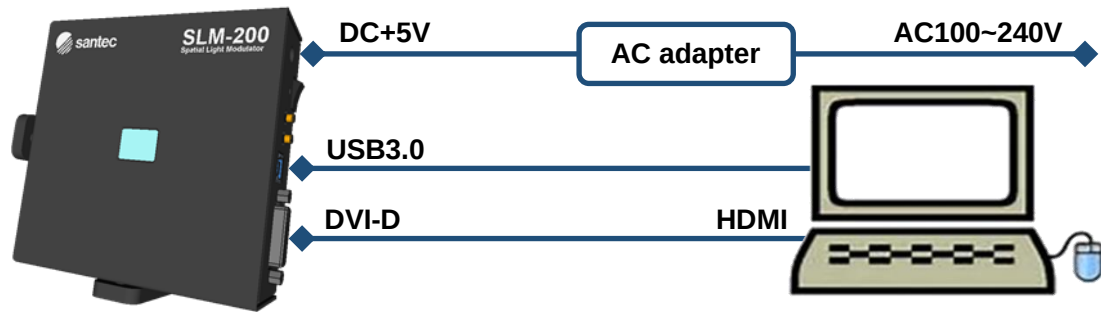


Figure 17: SLM connection.

3.4 Part names and functions

All-in-one model

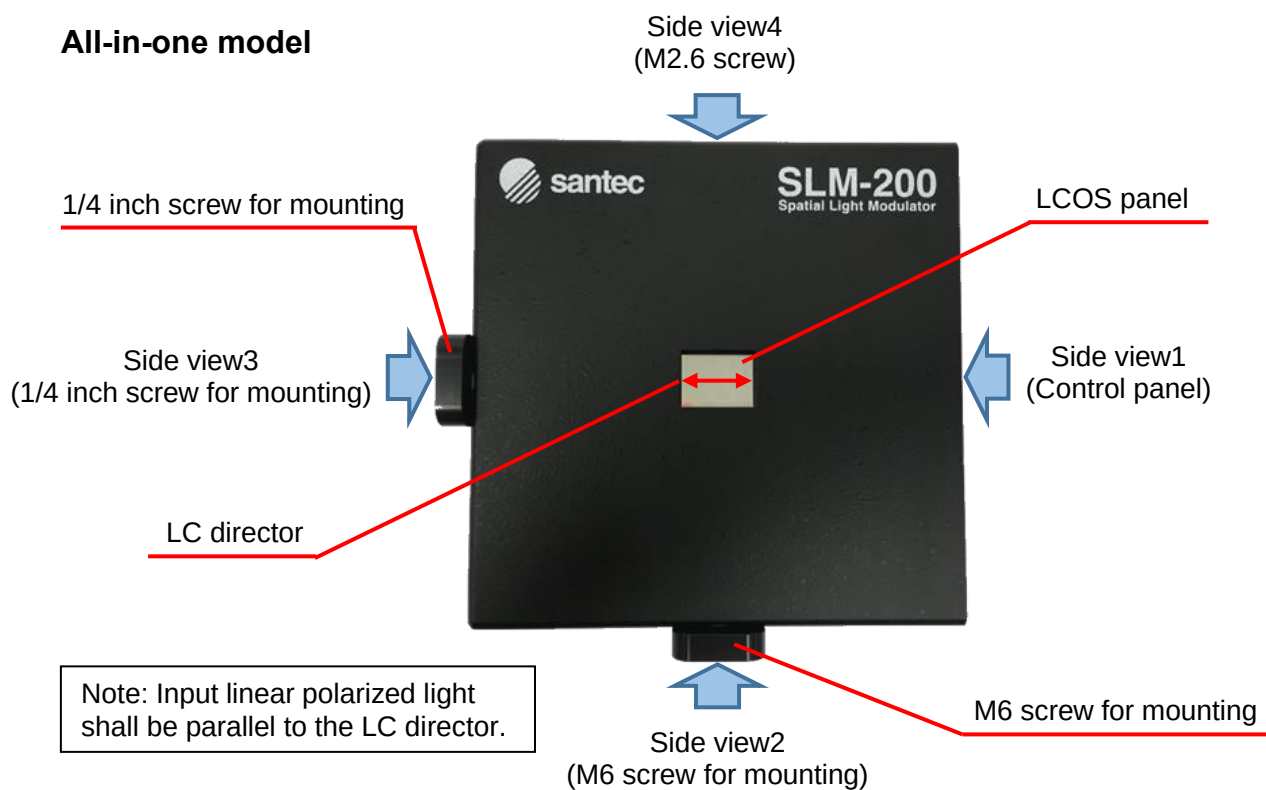


Figure 18: Front view.



Figure 19: Back view.

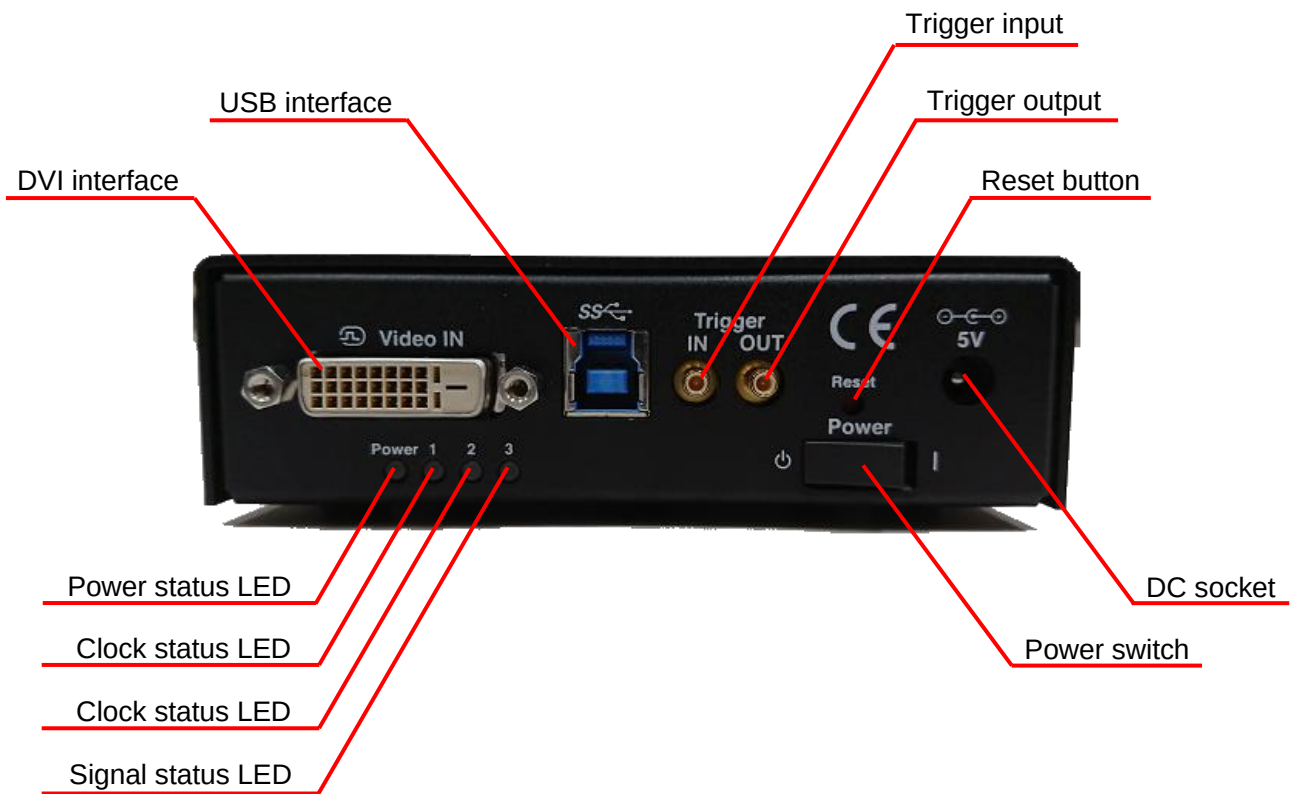


Figure 20: Side view 1 (Control panel).

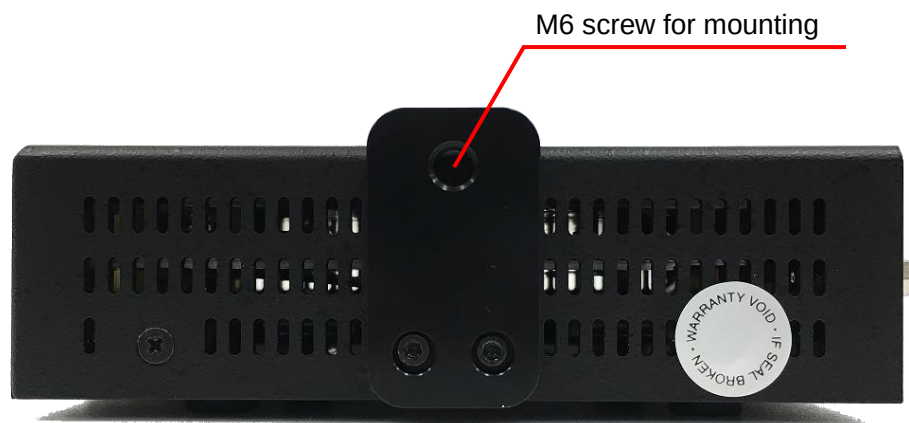


Figure 21: Side view 2 (M6 screw for mounting).

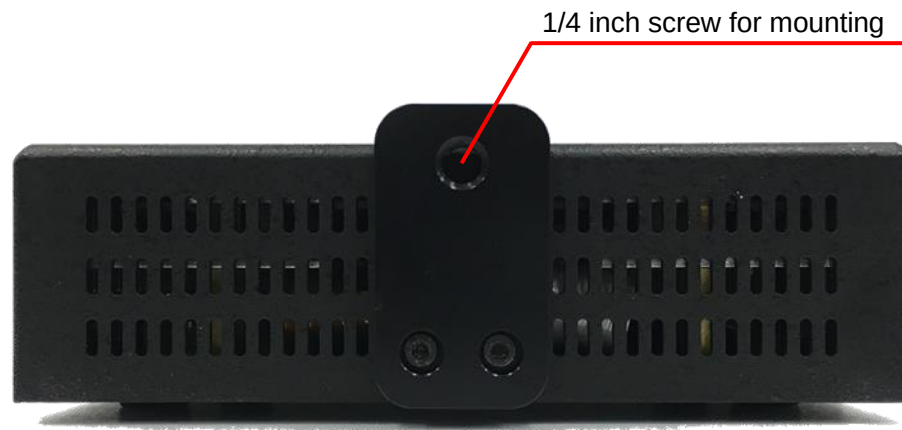


Figure 22: Side view 3 (1/4 inch screw for mounting).

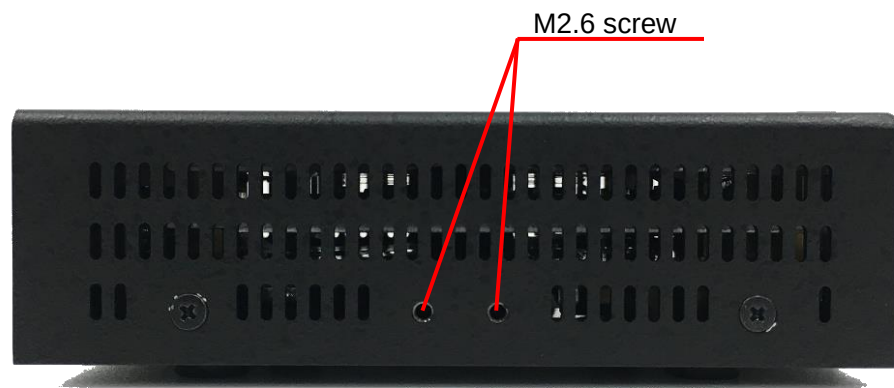


Figure 23: Side view 4 (M2.6 screw).

M6 and 1/4 inch mounting blocks can be fixed to user preferable sides.

Separate model

Figure 24: Front view.

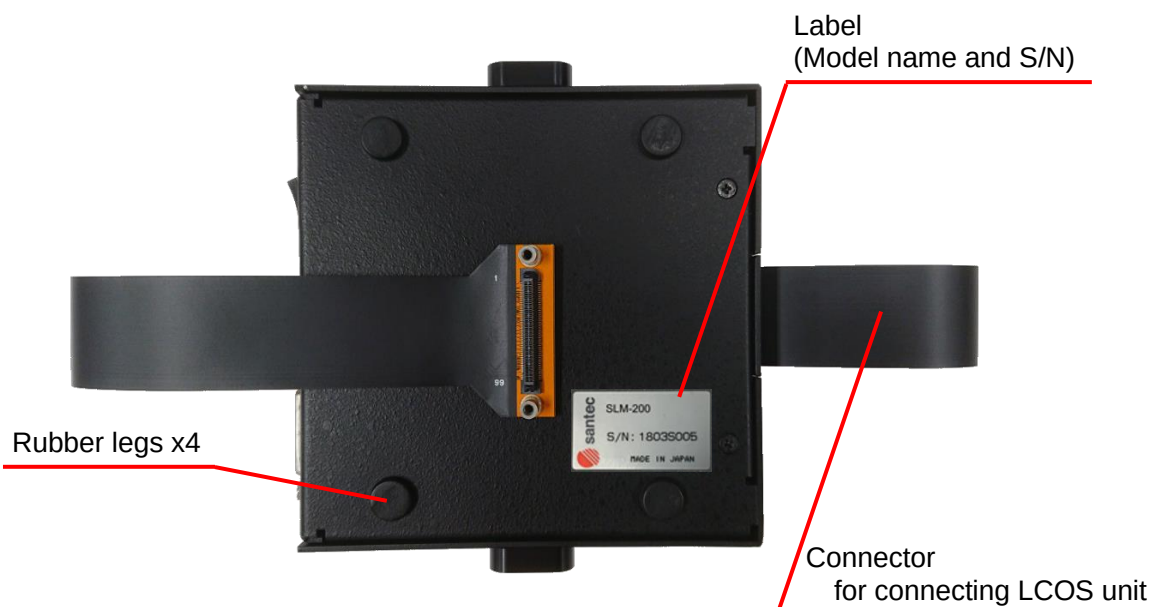


Figure 25: Back view.

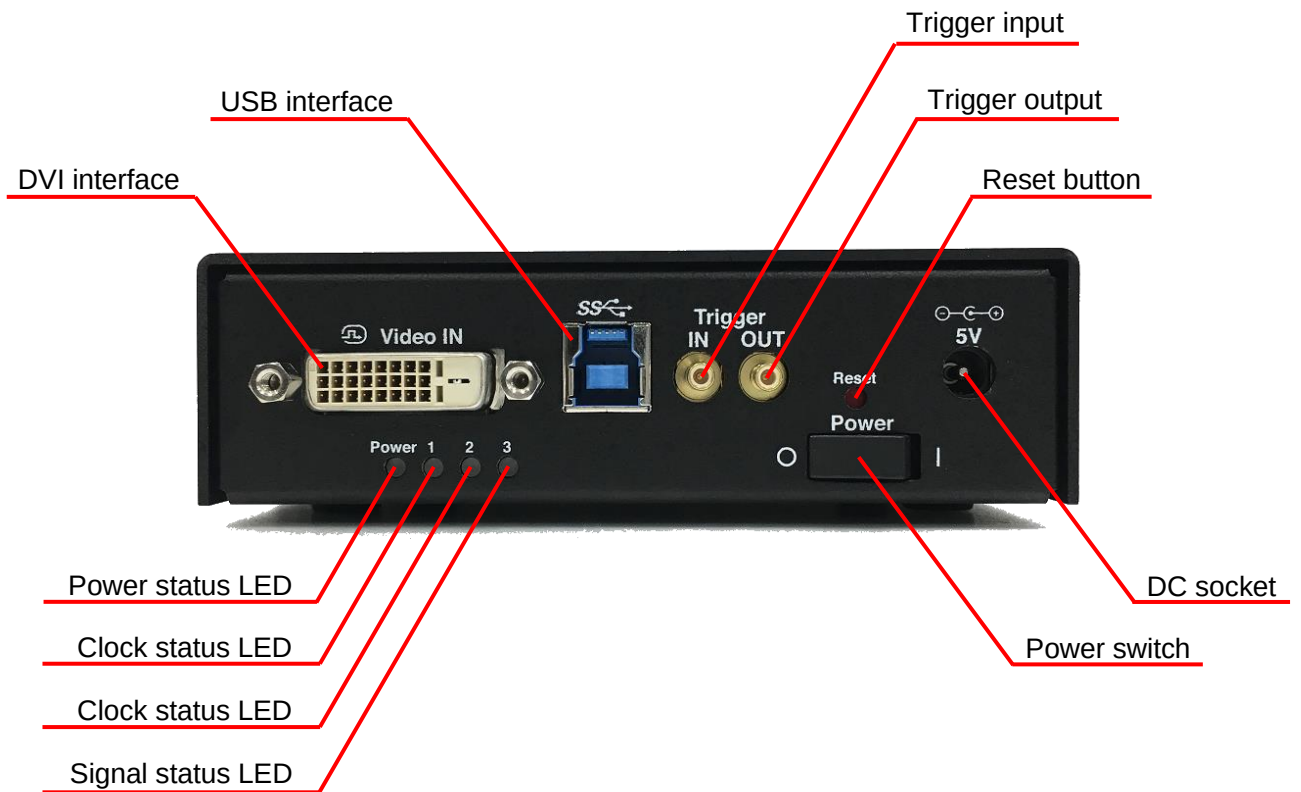


Figure 26: Side view 1 (Control panel).

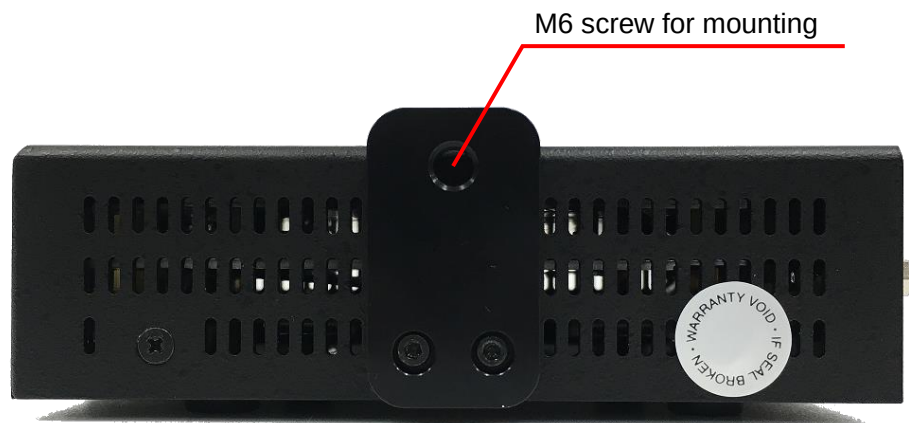


Figure 27: Side view 2 (M6 screw for mounting).

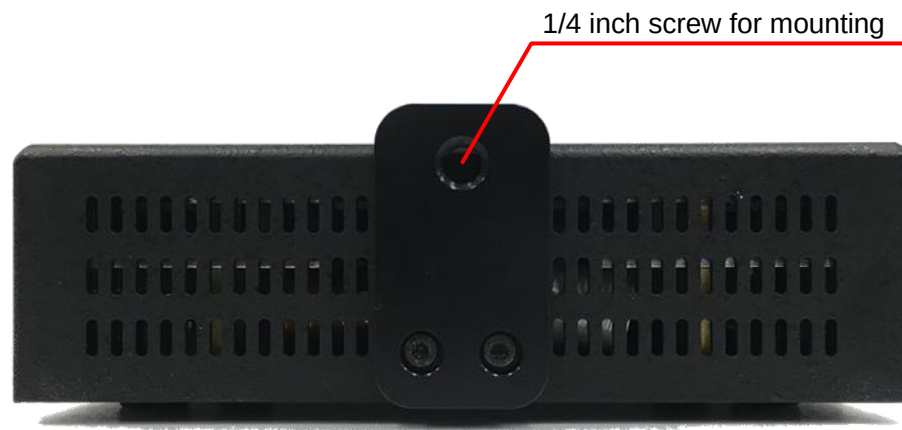
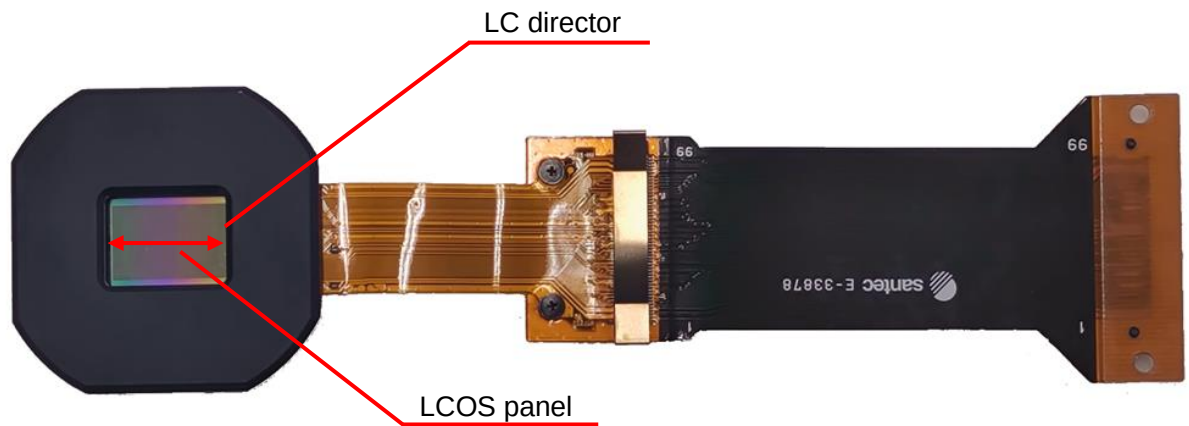


Figure 28: Side view 3 (1/4 inch screw for mounting).

M6 and 1/4 inch mounting blocks can be fixed to user preferable sides.

LCOS unit

Note: Input linear polarized light shall be parallel to the LC director for optical phase control.

Figure 29: Front view.

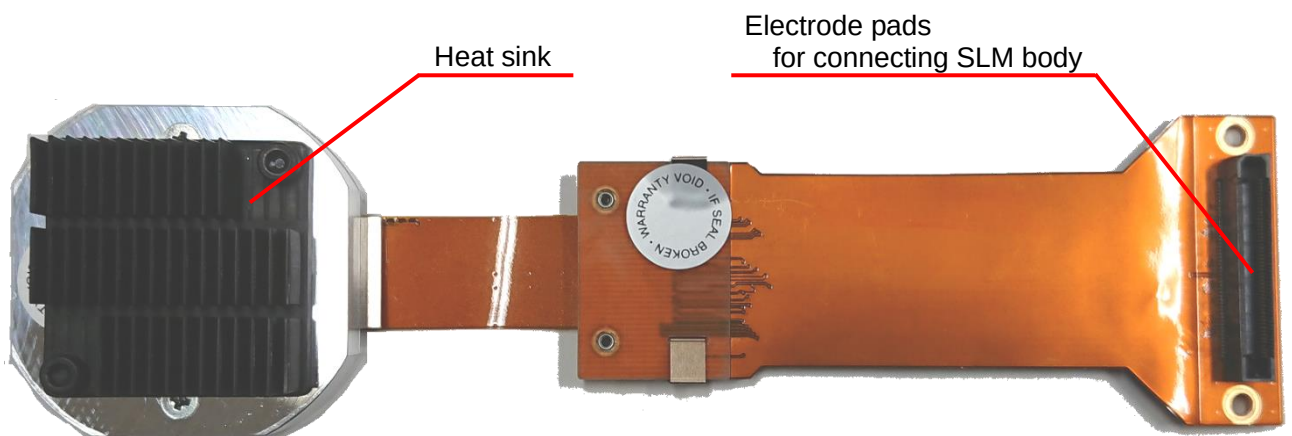


Figure 30: Back view.



Figure 31: Side view.

Polarizer

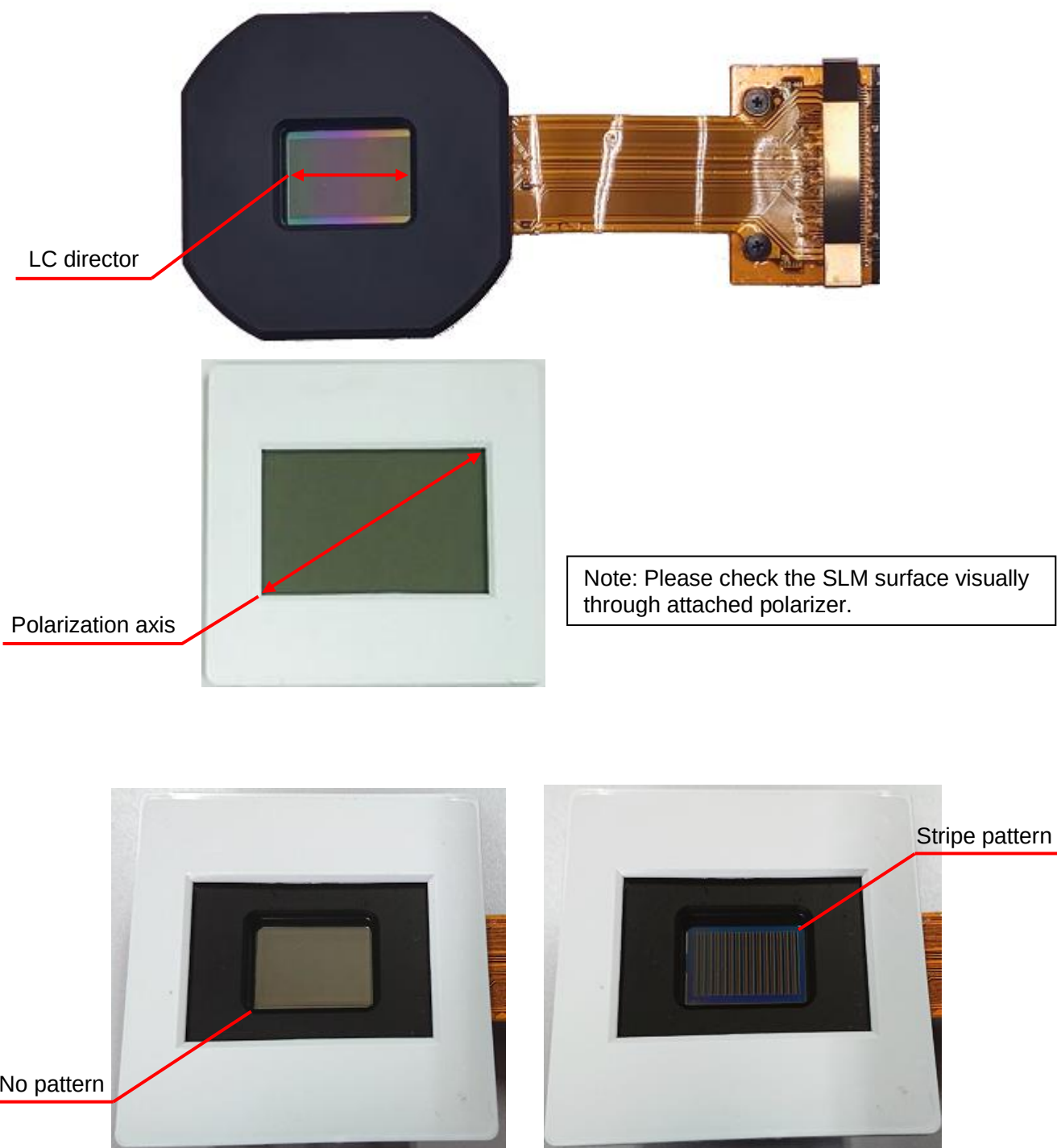


Figure 32: Polarizer setting.

3.5 LED status notifications

The SLM has four LED indicators, which inform the status of POWER, signal clocks and DVI video signals. Operation status for each LED is shown in Table 5.

Table 5: Operation status for LED indicators

Label	Color	Normal	Switching status
Power	Green	LED ON	LED OFF when the power supply voltage is less than 4.5V (5V-10%)
LED1	Green	LED ON	LED OFF when clock signal condition of LCOS display driver is abnormal
LED2	Green	LED ON	LED OFF when clock signal condition of Video data is abnormal
LED3	Red	LED OFF	LED ON when video signal to DVI interface is abnormal

3.6 Trigger signal specifications

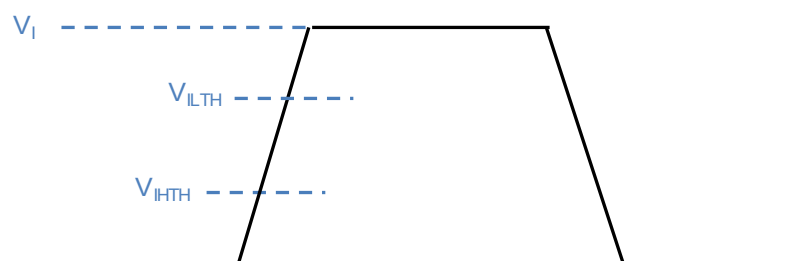
[Trigger input]

Absolute Maximum Ratings

	Symbol	MIN	MAX	UNIT
Input voltage	V_I	0	5	V

Electrical Characteristics

	Symbol	MIN	MAX	UNIT
High-level input voltage	$V_{I\text{H}\text{TH}}$	3.5	-	V
Low-level input voltage	$V_{I\text{L}\text{TH}}$	-	1.5	V



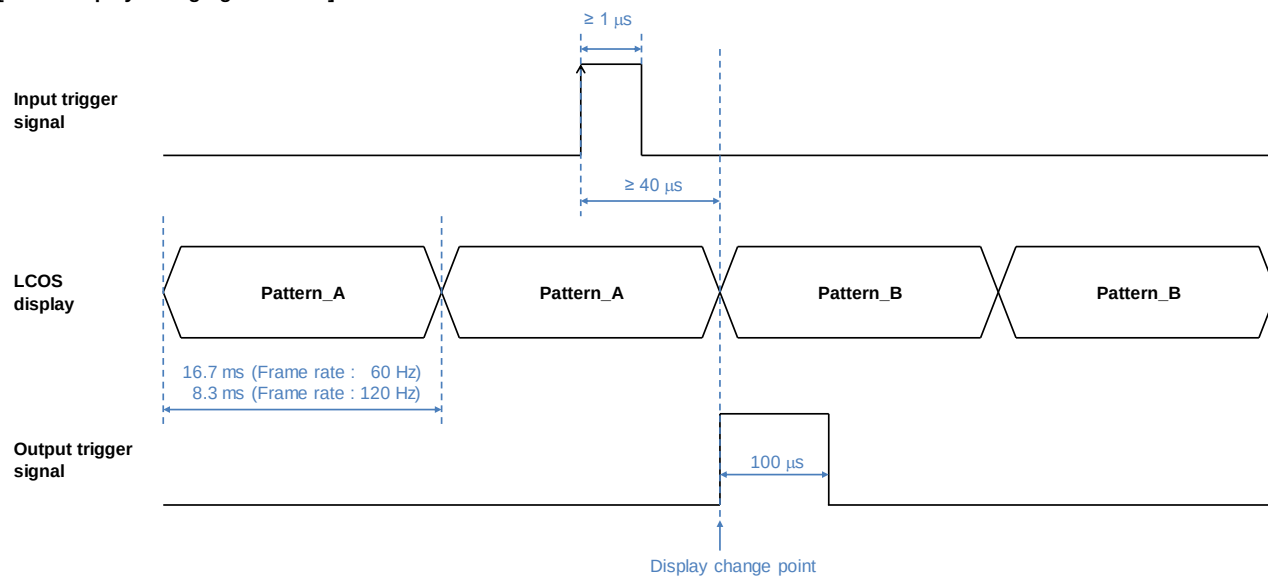
[Trigger output]

Electrical Characteristics

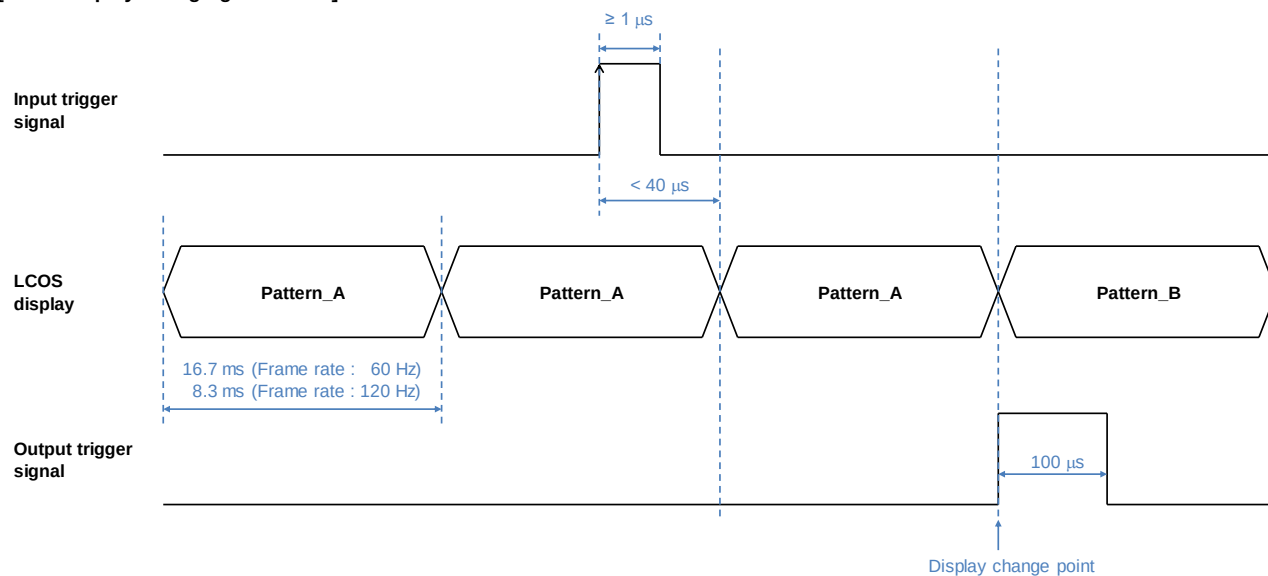
Parameter	Conditions	MIN	MAX	UNIT
V_{OH}	$I_{OH} = 34 \text{ mA}$	4.2	-	V
V_{OL}	$I_{OL} = 34 \text{ mA}$	-	0.6	V

LCOS display changing timing (Trigger input)

[LCOS display changing minimum]



[LCOS display changing maximum]



4. SLM handling instructions

4.1 Connections

Please connect attached components as shown in figure 33 and 34. Please make sure you connect the device to the computer before connecting it to the power supply.

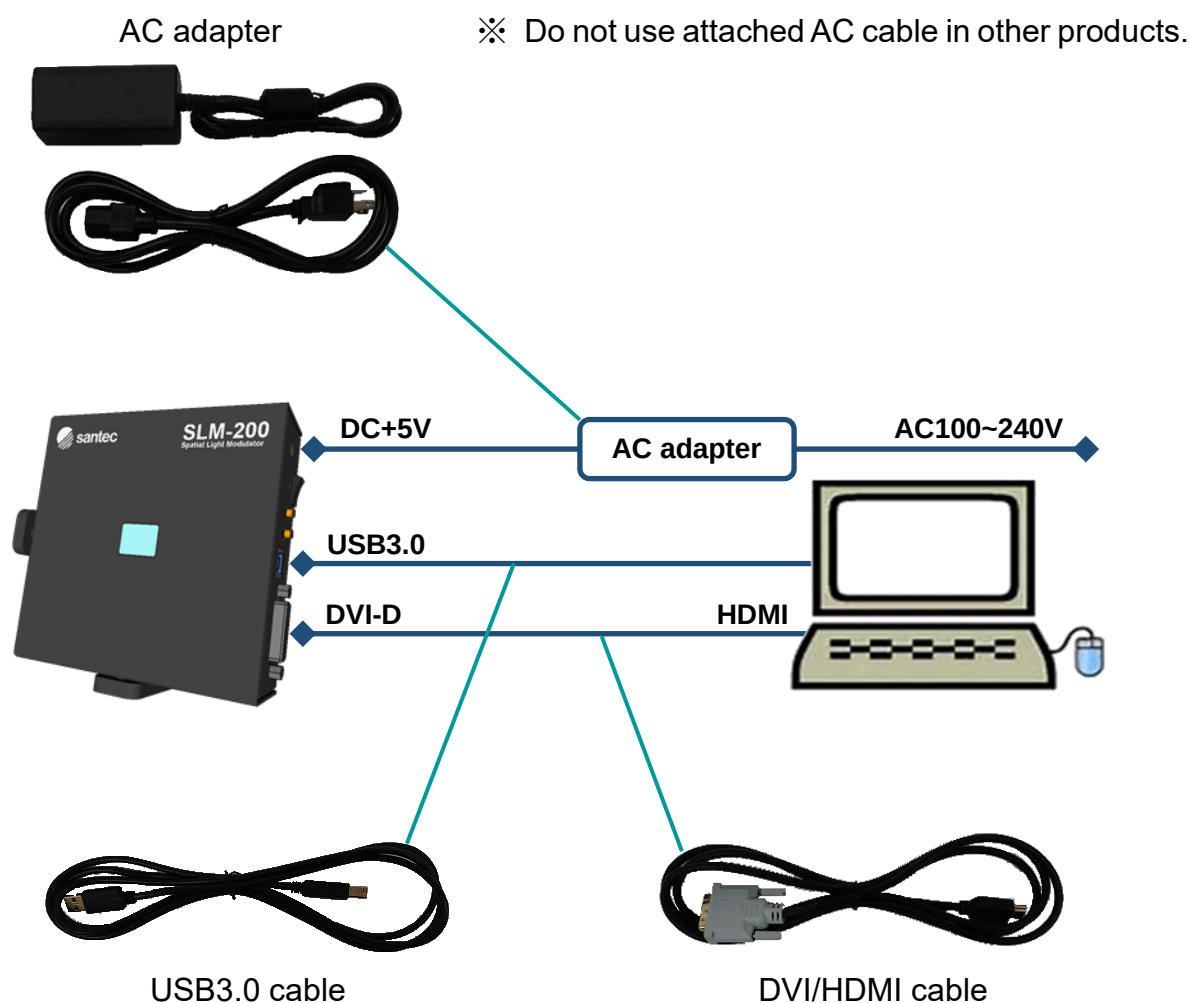
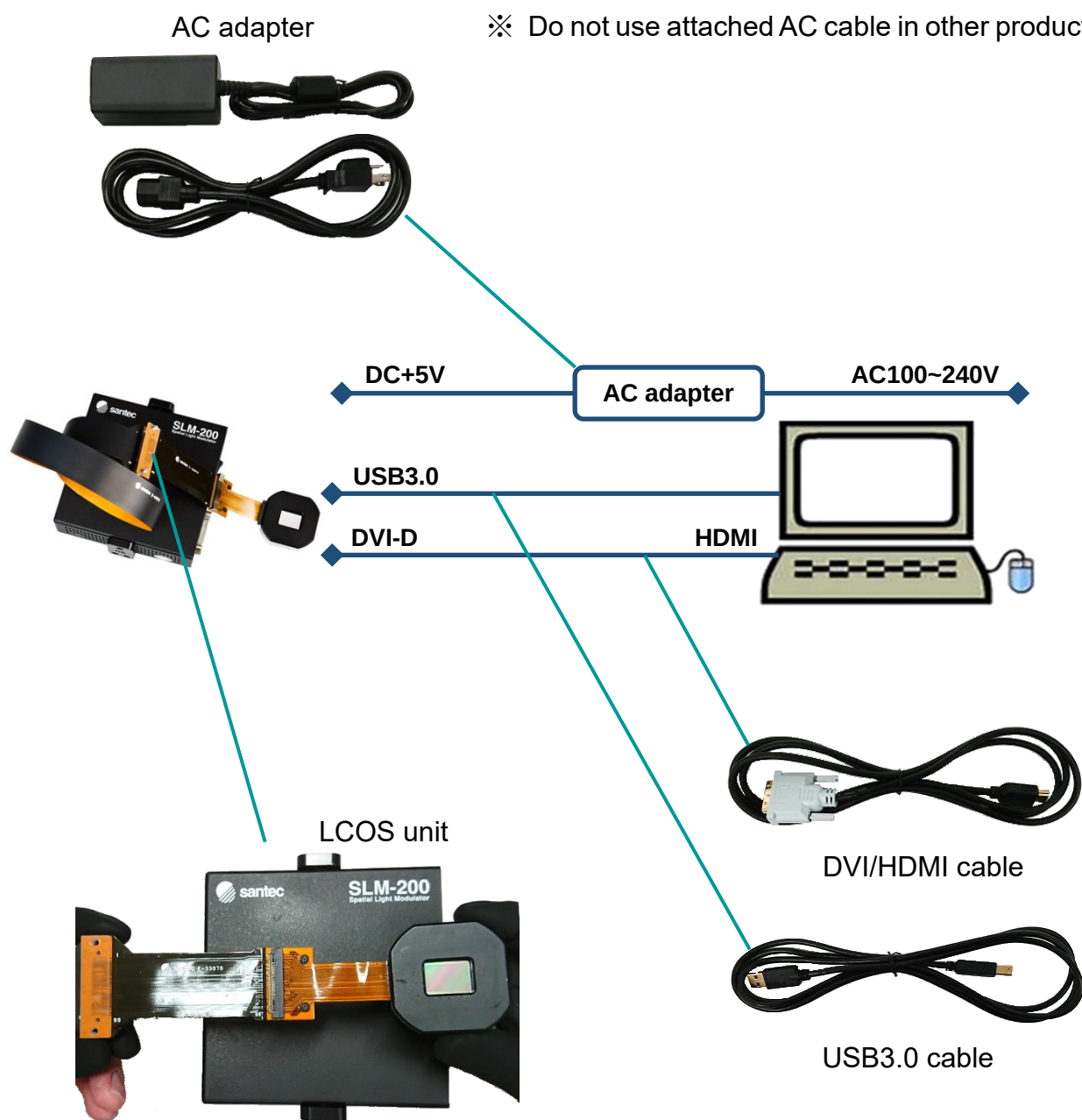


Figure 33: SLM power connections (All-in-one model).

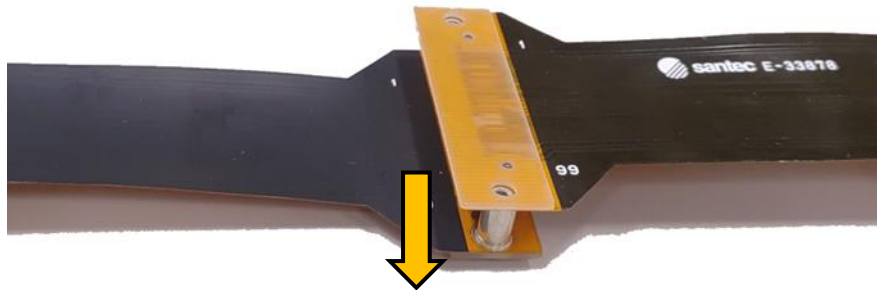


※ Refer to the next page how to connect LCOS unit.

Figure 34: SLM power connections (Separate model).



(Step A) Insert the FPC connector.



(Step B) Insert the FPC connector into the end.



(Step C) Secure with the accessory pre-assembled washers and screws.

Figure 35: Connection method of LCOS unit.

4.2 Power up sequence

4.2.1 HDMI/DVI communication

(1) Boot up the PC. And connect all components according to figure 33 and 34 above.
Again, please note: make all connections before connecting the AC adapter.

(2) Connect adapter plug to AC inlet and start voltage supply.

There is power switch on the SLM body. SLM is powered on by the power switch.

(3) The LCOS DVI interface is displayed on the PC screen.

You can check the pattern directly on the LCOS using a polarizer.

If the LCOS panel is not displayed on the PC screen, please check PC monitor setting status. The recommended setting is shown in figure 36.

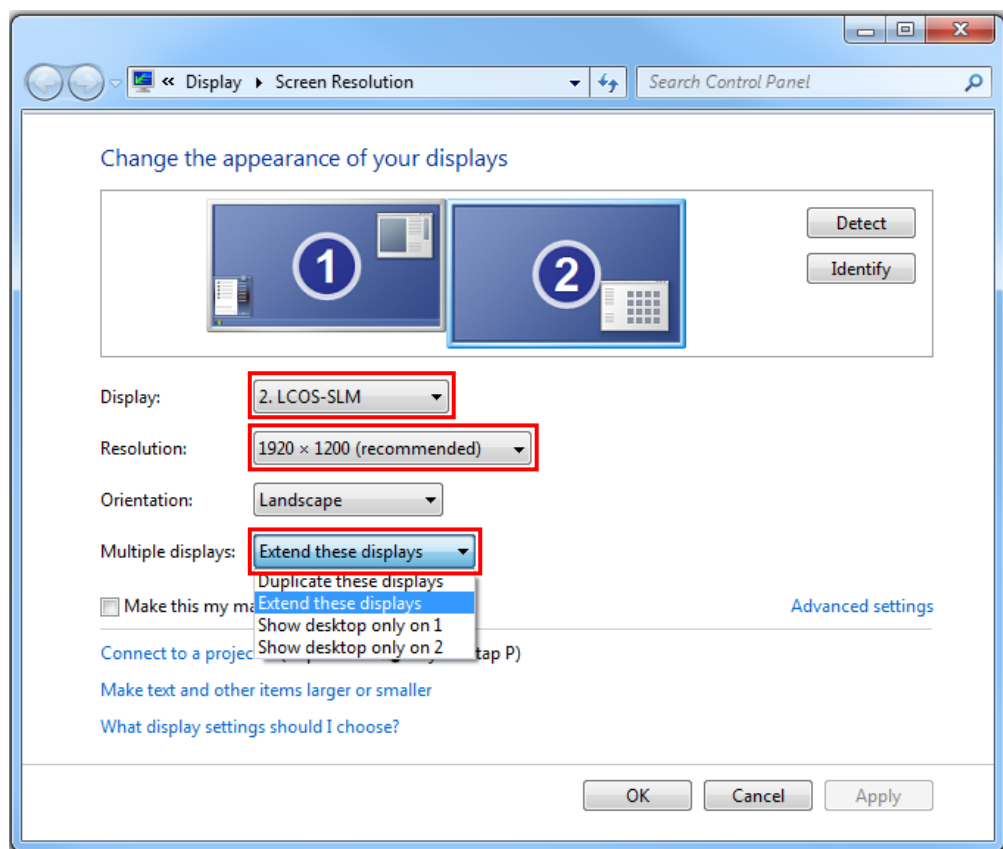


Figure 36: Recommended screen resolution setting on PC.

4.3 Mounting

4.3.1 All-in-one model

The SLM body has two screw holes (1/4 inch and M6) for mounting to experimental stages. For example, the SLM can be mounted to a slide stage using a commercially available screw bar as shown in figure 37.



Figure 37: Example of SLM mounting (All-in-one model).

4.3.2 Separate model

The LCOS unit has two screw holes (M4 and #8) for mounting to experimental stages. For example, the LCOS unit can be mounted to a slide stage using a commercially available screw bar and ϕ 2 inch holder as shown in figure 38.



Figure 38: Example of LCOS unit mounting (Separate model).

4.4 FPC bending radius

Do not make the bending radius of FPC less than 5 mm. There is a possibility of damage to the internal electrical wires.

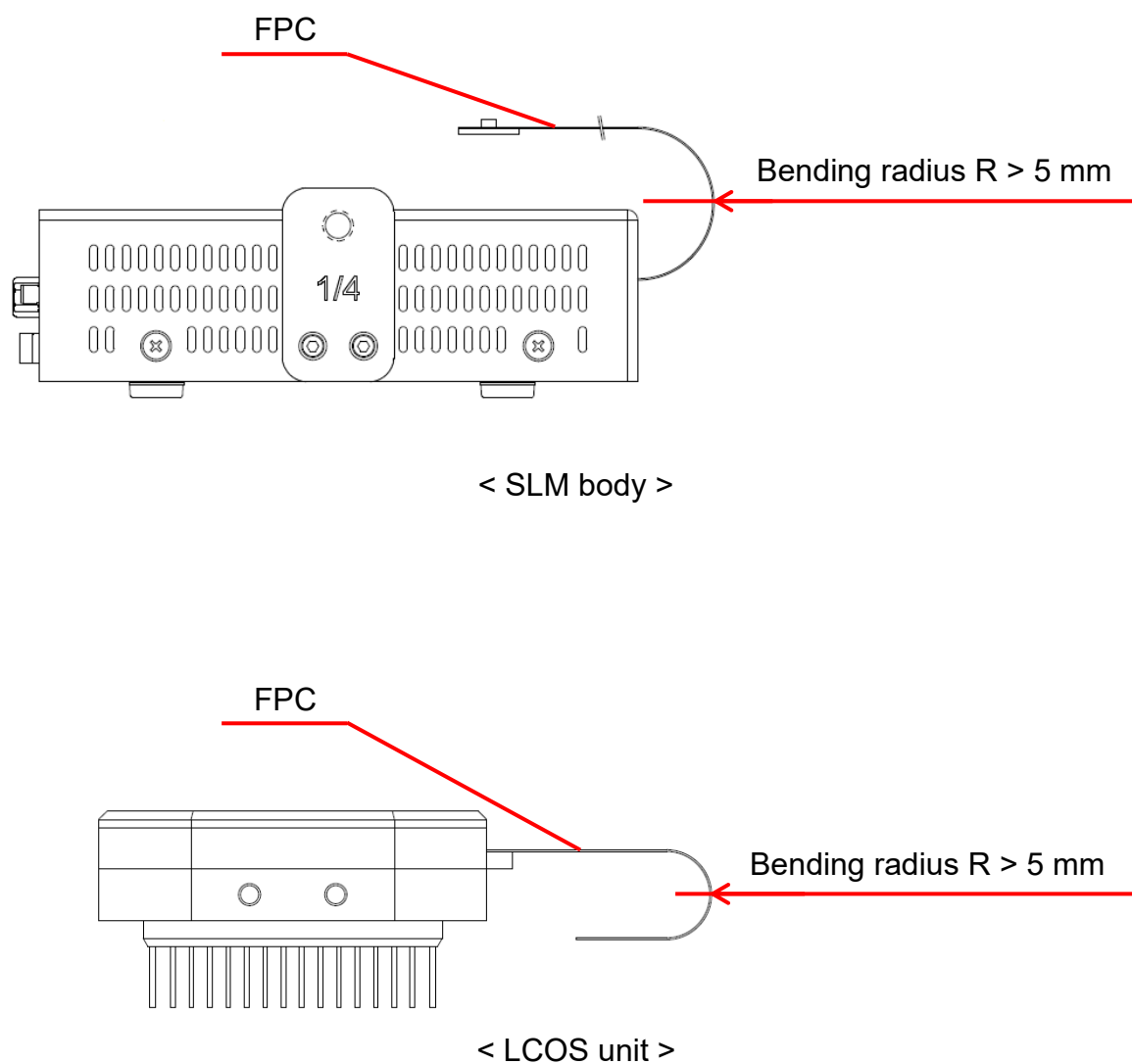


Figure 39: FPC bending radius (Separate model).

4.5 Cleaning

If the LCOS panel is dirty, please clean by hand using lint-free optical wipes, air blower or a liquid solvent such as isopropyl alcohol. Do not use acetone.



Figure 40: Cleaning the LCOS display window by lint-free cotton swab



Figure 41: Cleaning the LCOS display window by air blower.

5. Software package

5.1 Installation

Please install LCOS display software in accordance with the procedure of 5.1.1.

The recommended PC specifications for this software are as follows:

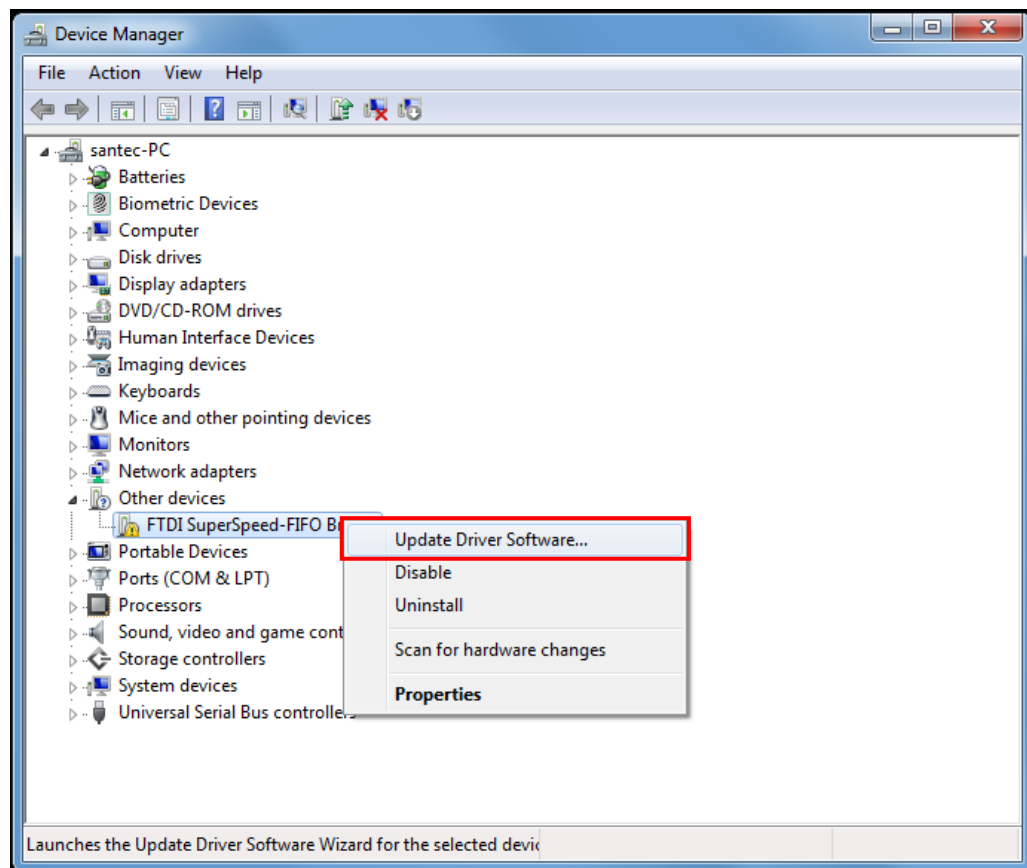
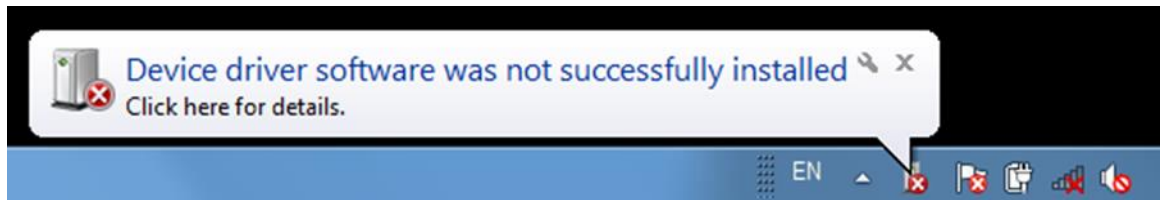
- OS : Windows10, 64bit
- Display resolutions : 1024x768(XGA) or higher
- Video output : HDMI
- Interface : USB2.0/3.0
- SLM version : SLM-200, SLM-210, SLM-250, SLM-300
(SLM-100 and 150 is not supported)

Before launching the software, please make sure to be connected correctly in reference to “4.1 Connections”, page 36-38.

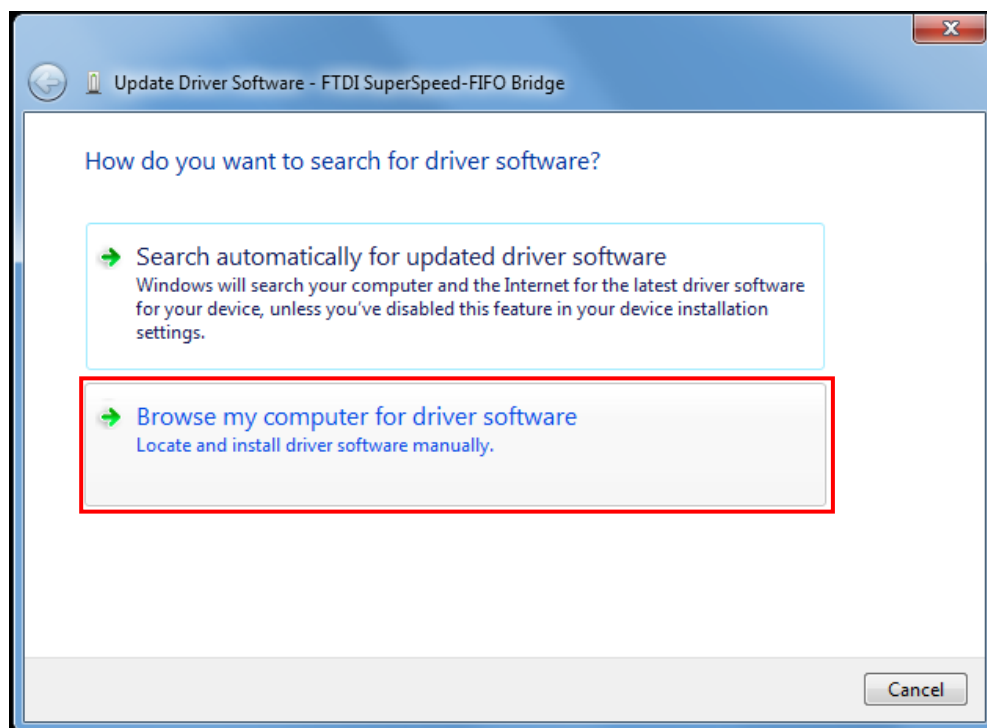
5.1.1 USB driver installation procedure

(1) SLM Power ON after connecting control PC and SLM via USB interface

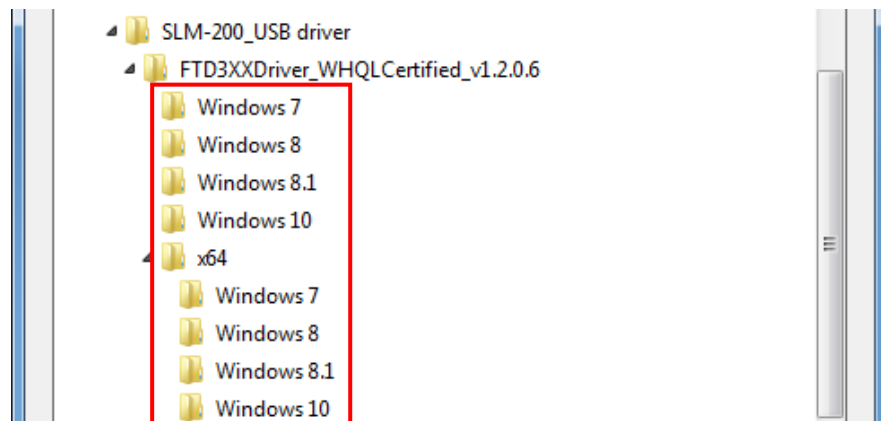
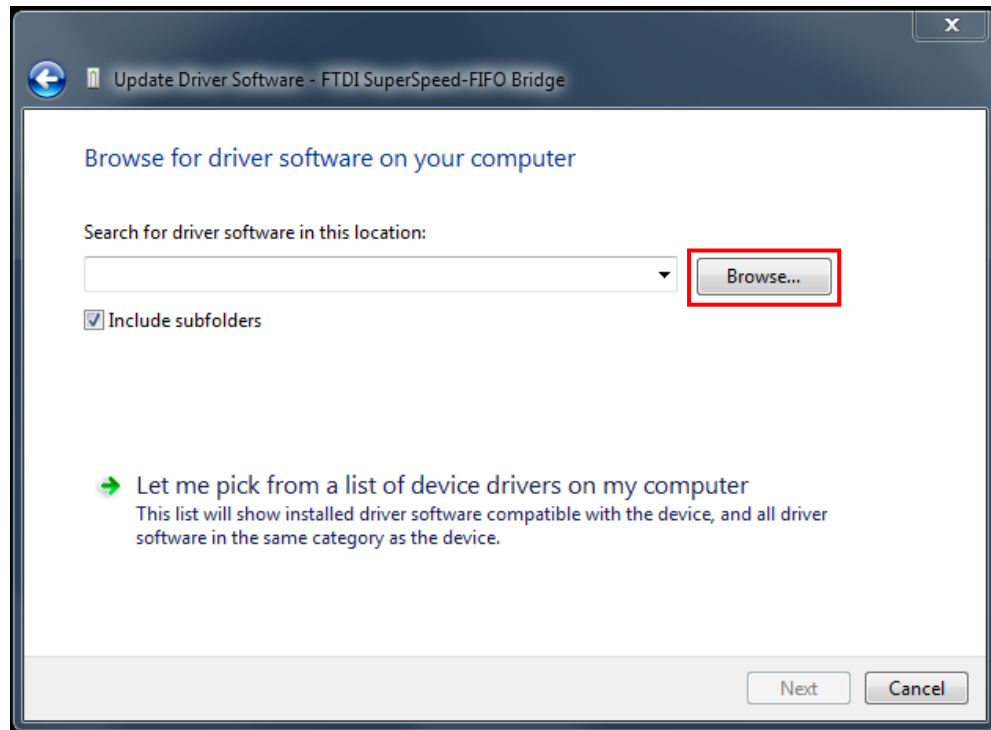
Please install the device driver in attached USB flash drive when you see following error message.



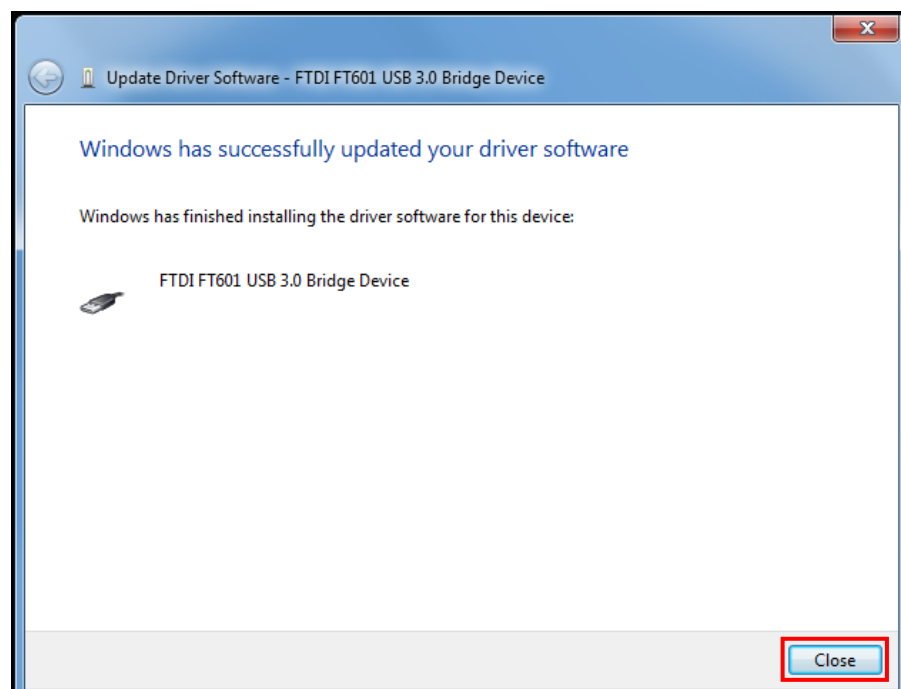
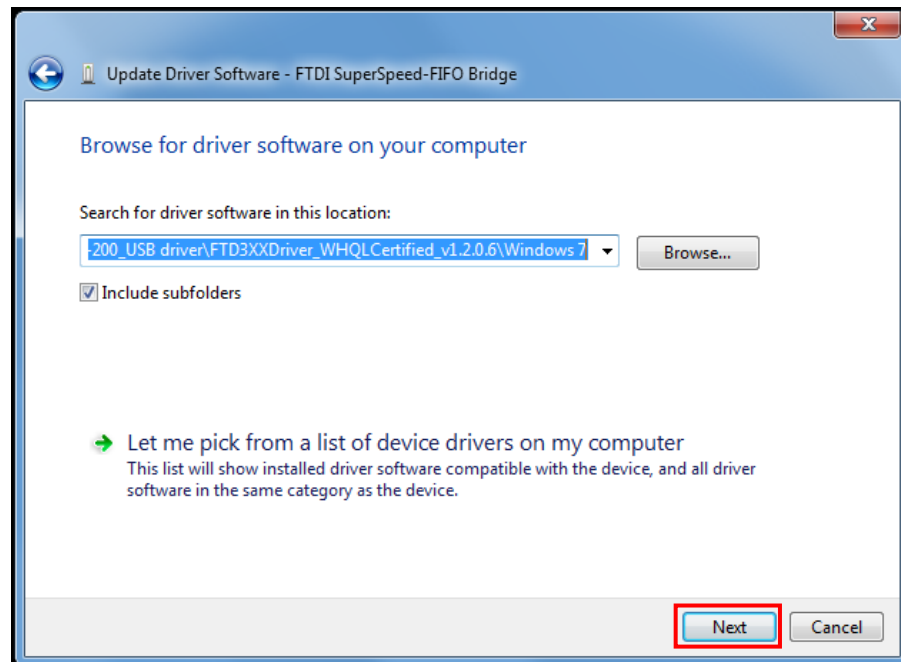
(2) Please select “Browse my computer for driver software”.



(3) Select adequate OS folder on attached USB flash drive contents.



(4) Installation of USB driver.



5.1.2 LCOS display software installation procedure

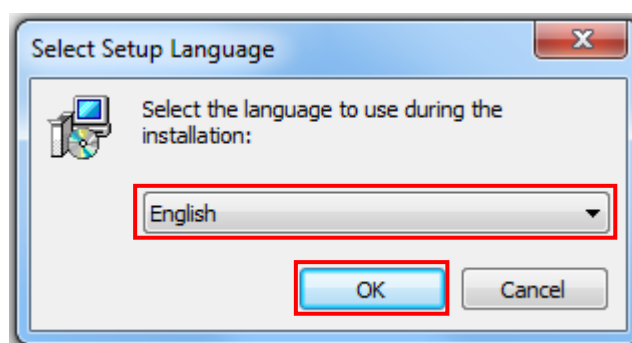
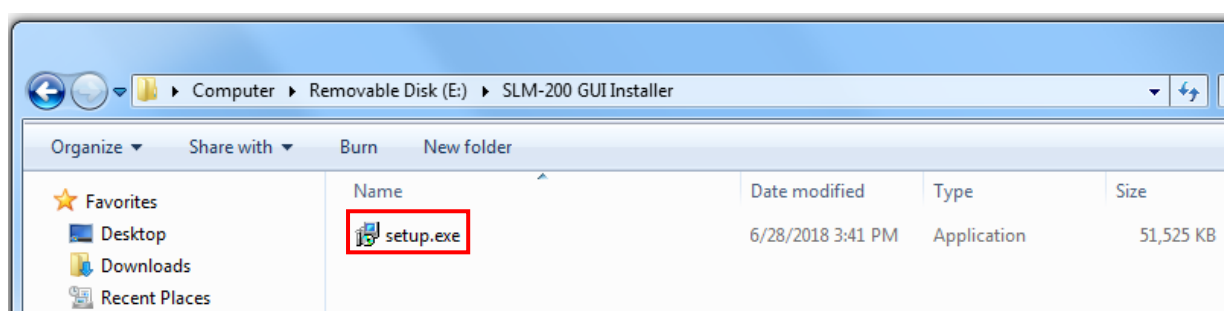
(1) Install the control software by starting the setup.exe.

Please select adequate language for installation.

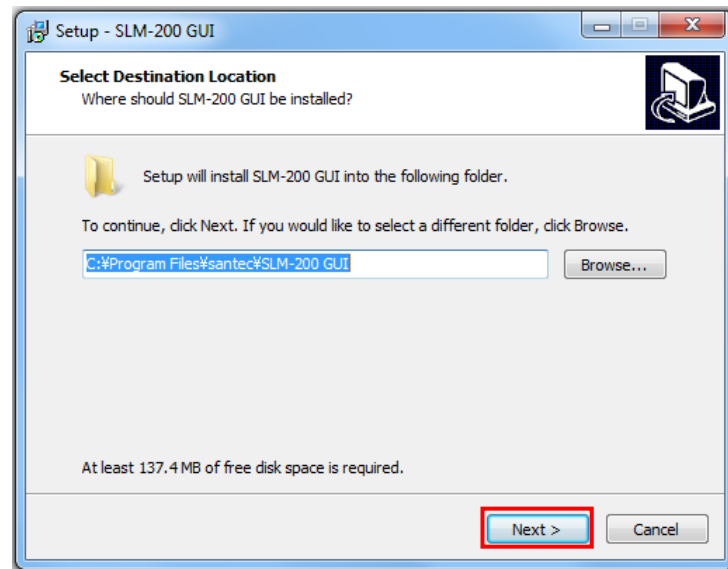
(Attached USB flash drive, E:¥SLM-200 GUI Installer¥setup.exe)

*Download the latest version installer from the following URL.

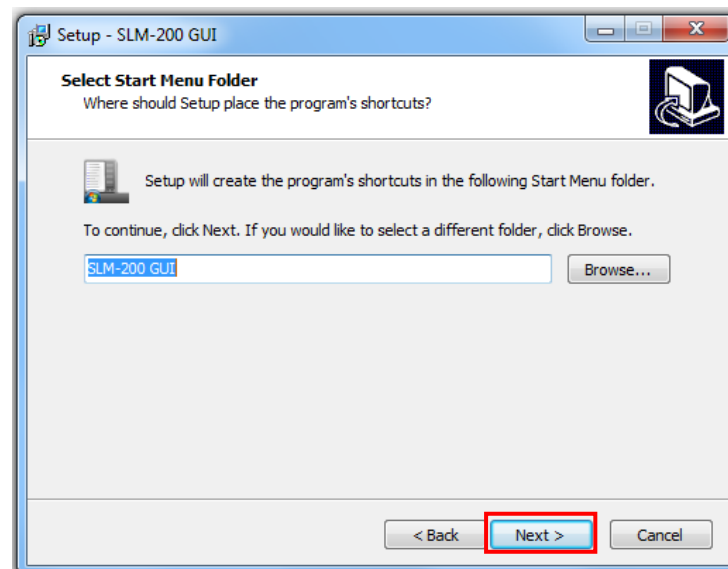
<URL> <https://www.santec.com/en/support>



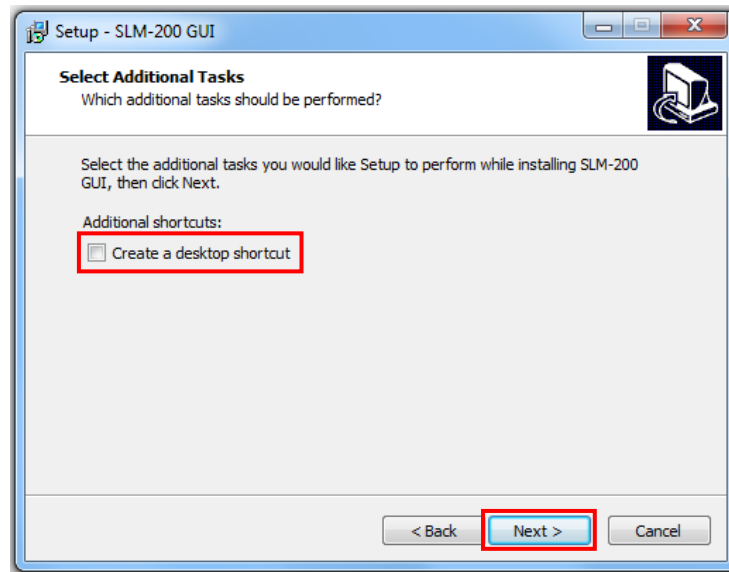
- (2) Specify the folder to install software.
(Default: C:\Program Files\santec\SLM-200 GUI)
Click “next”.



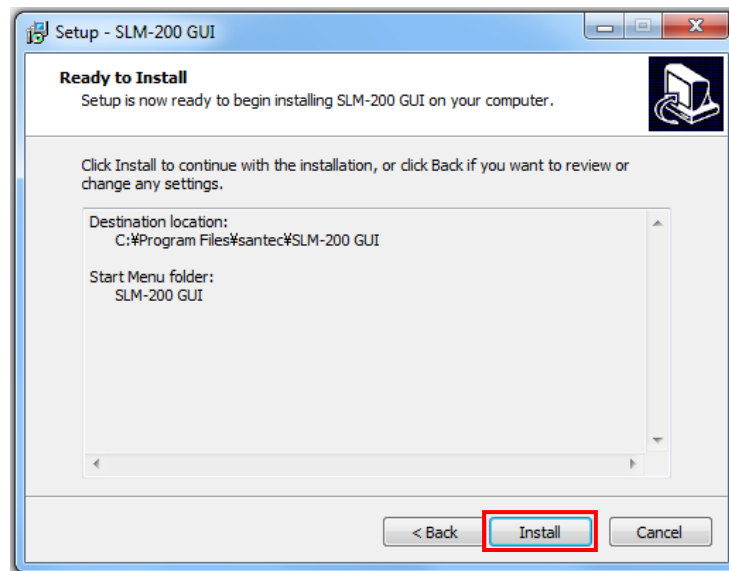
- (3) Specify and confirm a folder for the program shortcuts.



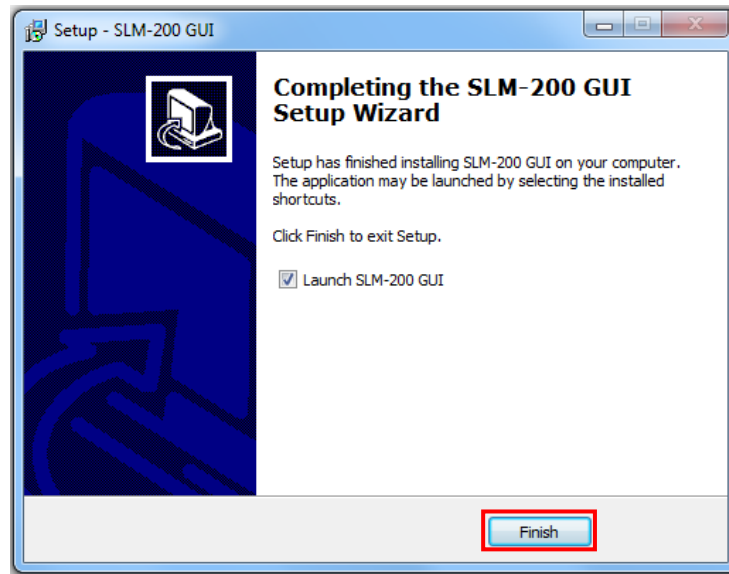
(4) If you create the file shortcut on desktop, click the checkbox and click “next”.



(5) Click “install” and start installation.

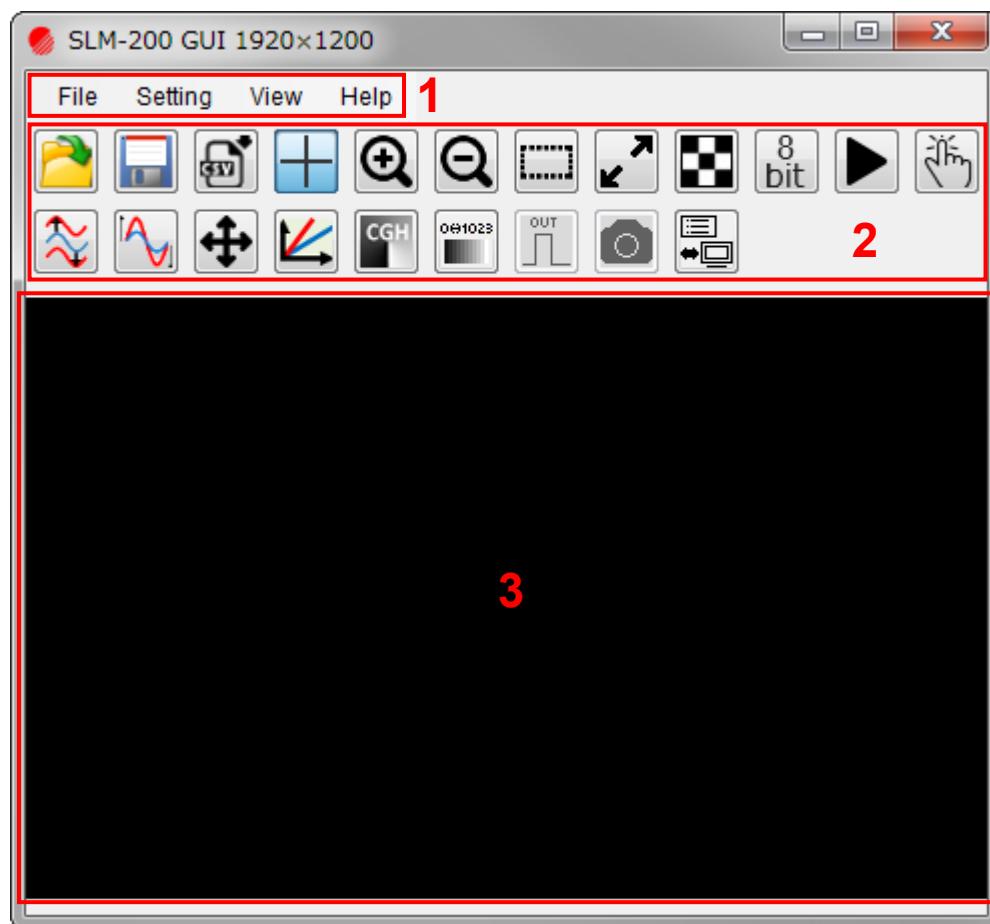


(6) If you launch the GUI after installation, click the checkbox and click “Finish”.



5.2 Software operation

5.2.1 GUI window



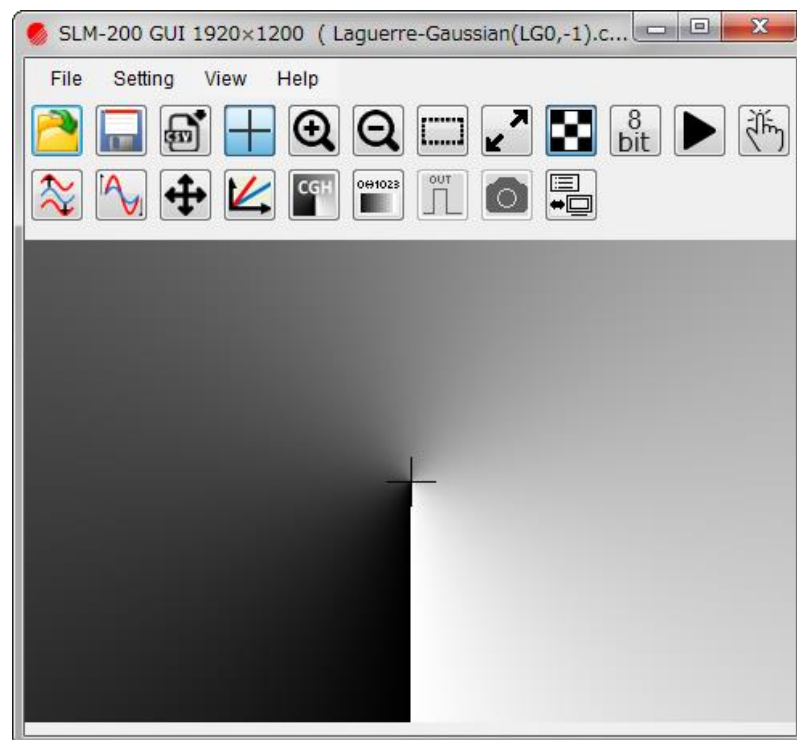
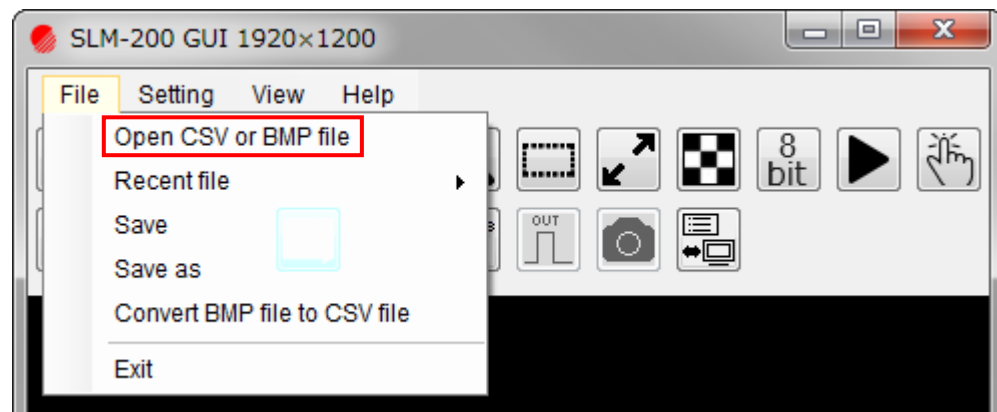
1. Menu tab
2. Shortcut menu
3. Preview of LCOS displayed phase pattern

5.2.2 Menu tab

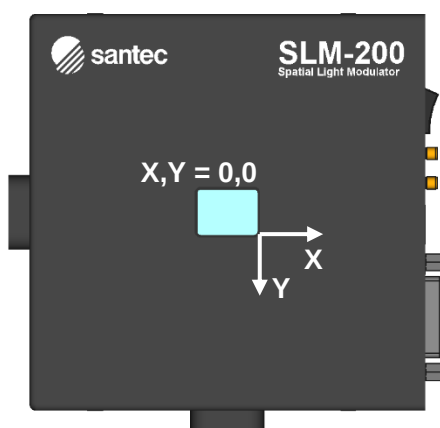
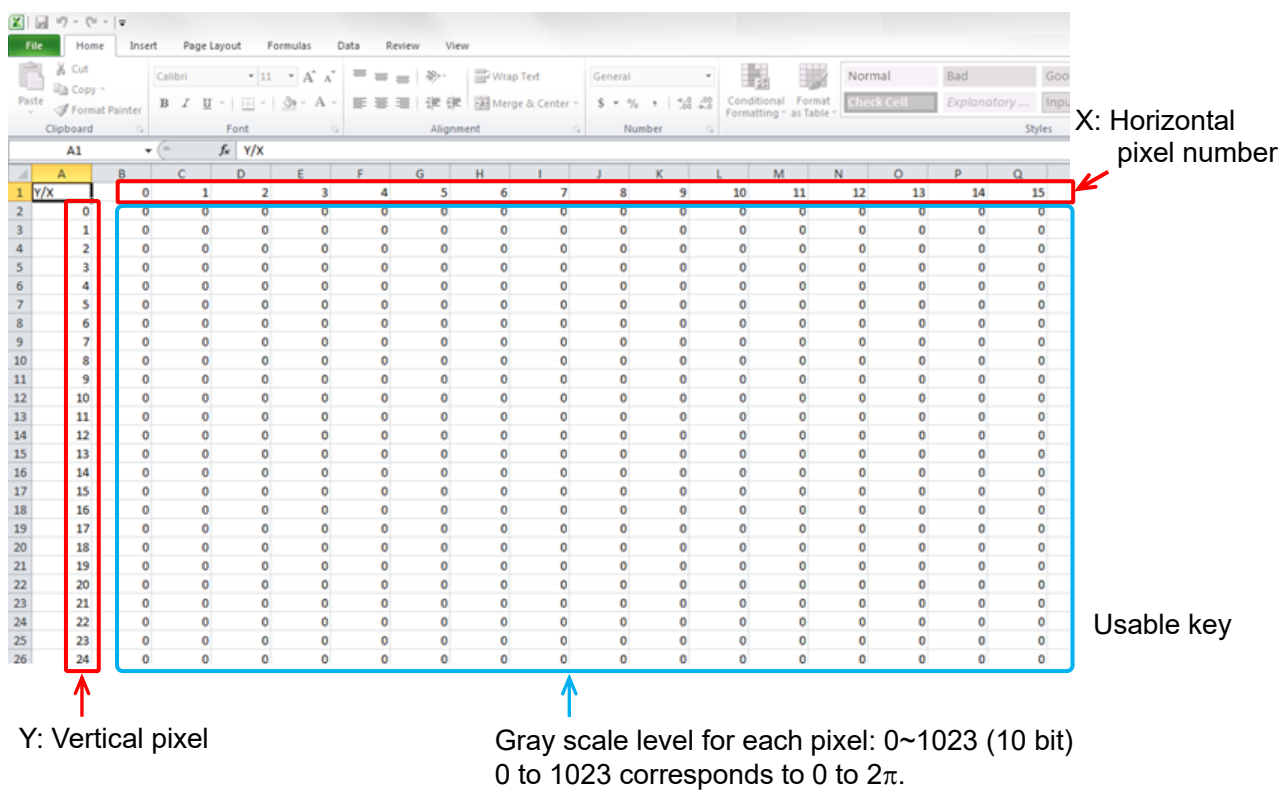
5.2.2.1 Open CSV file



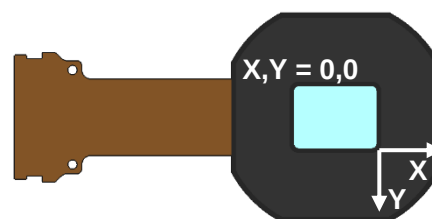
The CSV file with data format described in figure 42 can be opened. The data can be edited using spreadsheet like Microsoft excel.



The CSV file with data format made in accordance with figure 42 can be opened.
The data can be edited using spreadsheet like Microsoft excel.



(a) All-in-one model

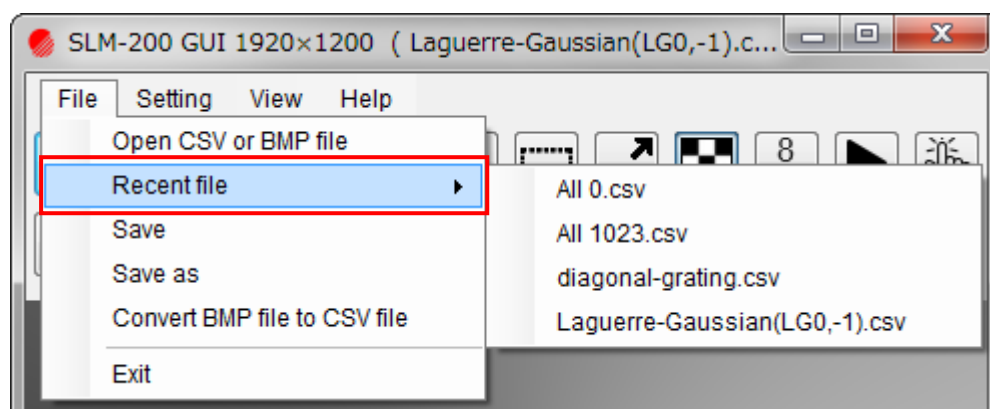


(b) Separate model (LCOS unit)

Figure 42: Data format of pattern files.

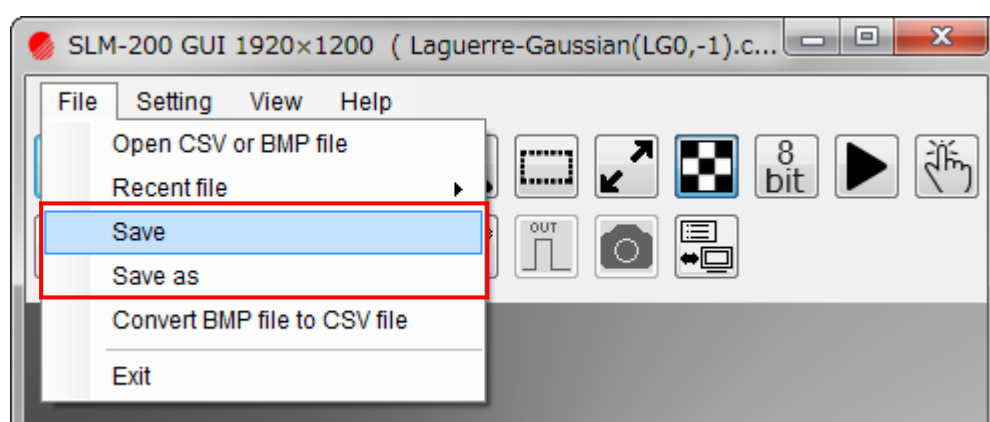
5.2.2.2 Recent file

The GUI memorizes the 4 past files. They can be displayed on the LCOS panel.



4.2.2.3 Save / Save as

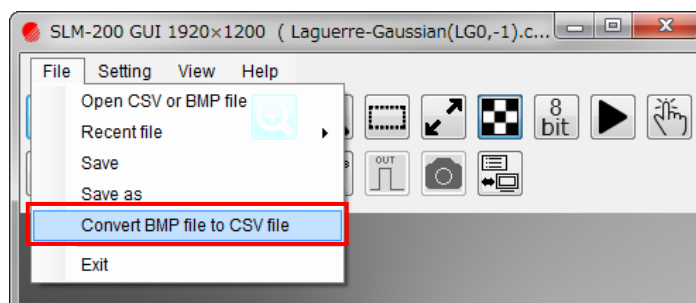
Current preview on GUI can be saved or saved as a new file.



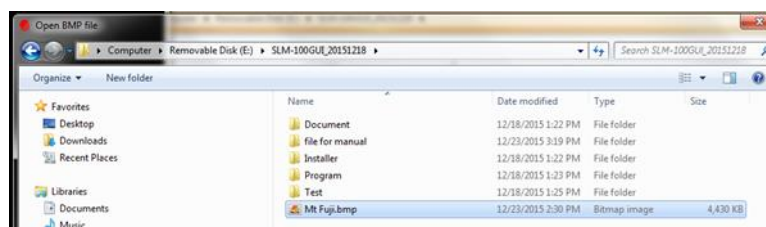
5.2.2.4 Convert BMP file to CSV file



Specified bit map (BMP) file can be converted to CSV format file.
BMP files for conversion should be in 24 bit format.



Select the BMP file for converting.



If the BMP file is converted in rescaling of screen resolution, magnification and gray scale of blank area can be specified.

※Maximum screen resolution of converted BMP file and CSV file is 5760 x 4200 respectively, which is quadruple of LCOS panel screen resolution 1920x1200.

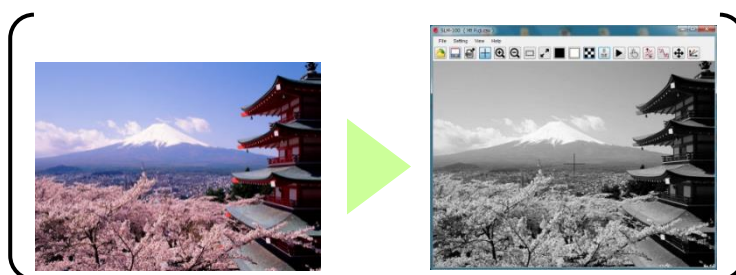
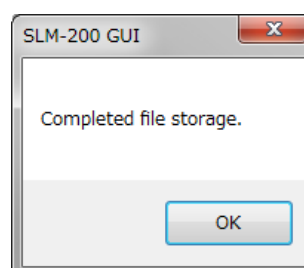
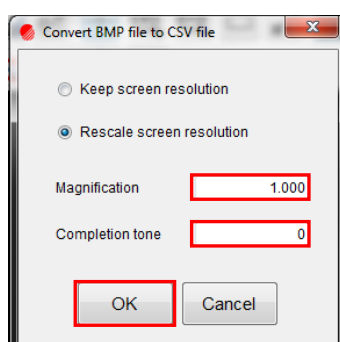


Table 6: 10 bit encoding format in RGB color (Frame rate: 60Hz)

bit	7	6	5	4	3	2	1	0
Red	3 bit							
Green	3 bit							
Blue	4 bit							

bit	9	8	7	6	5	4	3	2	1	0
Display data	R7	R6	R5	G7	G6	G5	B7	B6	B5	B4

RGB color table (Frame rate: 60Hz)

10 bit phase steps (0-1023)	Red	Green	Blue
0	0	0	0
1	0	0	16
2	0	0	32
3	0	0	48
4	0	0	64
⋮			
126	0	224	224
127	0	224	240
128	32	0	0
129	32	0	0
130	32	0	0

510	96	224	224
511	96	224	240
512	128	0	0
513	128	0	16
514	128	0	32
⋮			
1019	224	224	176
1020	224	224	192
1021	224	224	208
1022	224	224	224
1023	224	224	240

Table 7: 10 bit encoding format in RGB color (Frame rate: 120Hz)

	First frame				Next frame			
bit	7	6	5	4	3	2	1	0
Red	3 bit				3 bit			
Green	3 bit				3 bit			
Blue	4 bit				4 bit			

bit	9	8	7	6	5	4	3	2	1	0
Display data <1>	R7	R6	R5	G7	G6	G5	B7	B6	B5	B4

bit	9	8	7	6	5	4	3	2	1	0
Display data <2>	R7	R6	R5	G7	G6	G5	B7	B6	B5	B4

SLM display

Display data <1>

Display data <2>

First frame

Next frame

RGB color table <1> (Frame rate: 120Hz)

[First frame]

10 bit phase steps (0-1023)	Red	Green	Blue
0	0	0	0
1	0	0	16
2	0	0	32
3	0	0	48
4	0	0	64
⋮			
126	0	224	224
127	0	224	240
128	32	0	0
129	32	0	0
130	32	0	0
⋮			
510	96	224	224
511	96	224	240
512	128	0	0
513	128	0	16
514	128	0	32
⋮			
1019	224	224	176
1020	224	224	192
1021	224	224	208
1022	224	224	224
1023	224	224	240

*Next frame 10bit phase steps: 0

[Next frame]

10 bit phase steps (0-1023)	Red	Green	Blue
0	0	0	0
1	0	0	0
2	0	0	0
3	0	0	0
4	0	0	0
⋮			
126	0	0	0
127	0	0	0
128	0	0	0
129	0	0	0
130	0	0	0
⋮			
510	0	0	0
511	0	0	0
512	0	0	0
513	0	0	0
514	0	0	0
⋮			
1019	0	0	0
1020	0	0	0
1021	0	0	0
1022	0	0	0
1023	0	0	0

RGB color table <2> (Frame rate: 120Hz)

[First frame]

10 bit phase steps (0-1023)	Red	Green	Blue
0	0	0	0
1	0	0	0
2	0	0	0
3	0	0	0
4	0	0	0
⋮			
126	0	0	0
127	0	0	0
128	0	0	0
129	0	0	0
130	0	0	0
⋮			
510	0	0	0
511	0	0	0
512	0	0	0
513	0	0	0
514	0	0	0
⋮			
1019	0	0	0
1020	0	0	0
1021	0	0	0
1022	0	0	0
1023	0	0	0

*First frame 10bit phase steps: 0

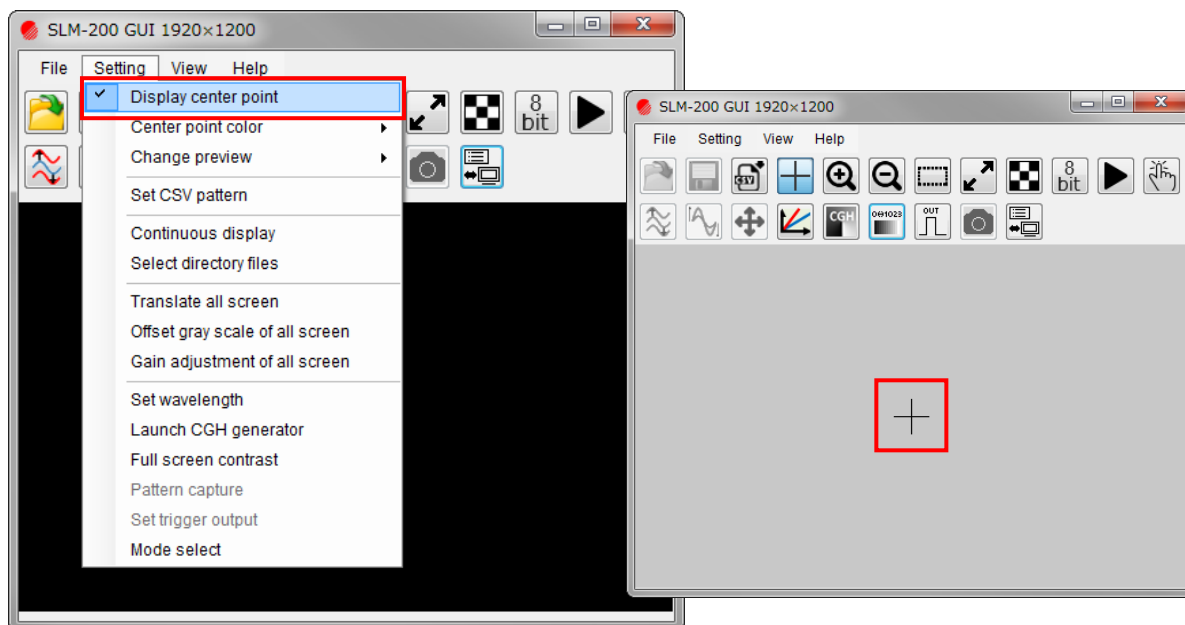
[Next frame]

10 bit phase steps (0-1023)	Red	Green	Blue
0	0	0	0
1	0	0	1
2	0	0	2
3	0	0	3
4	0	0	4
⋮			
126	0	14	14
127	0	14	15
128	1	0	0
129	1	0	1
130	1	0	2
⋮			
510	6	14	14
511	6	14	15
512	8	0	0
513	8	0	1
514	8	0	2
⋮			
1019	14	14	11
1020	14	14	12
1021	14	14	13
1022	14	14	14
1023	14	14	15

5.2.2.5 Display center point

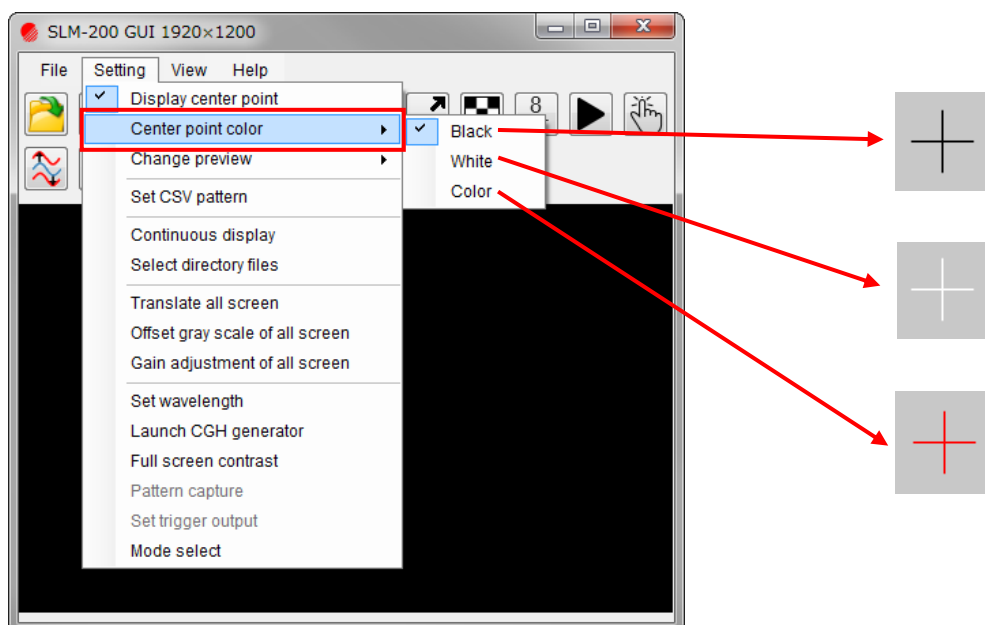


The cross mark indicating display center can be selected whether it is displayed or not.



5.2.2.6 Center point color

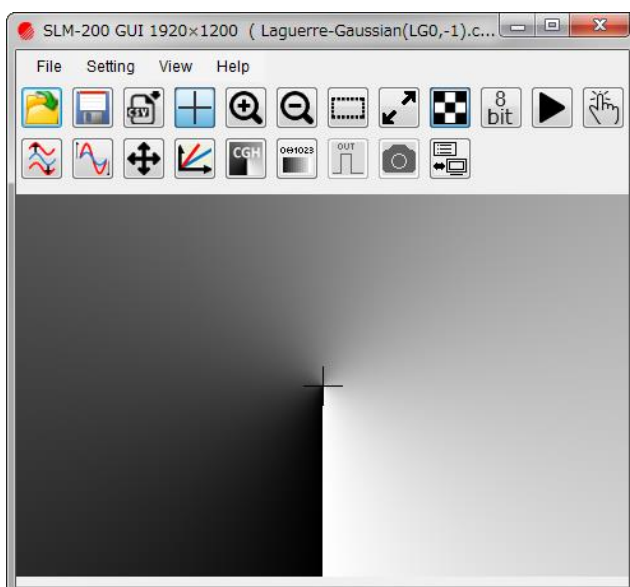
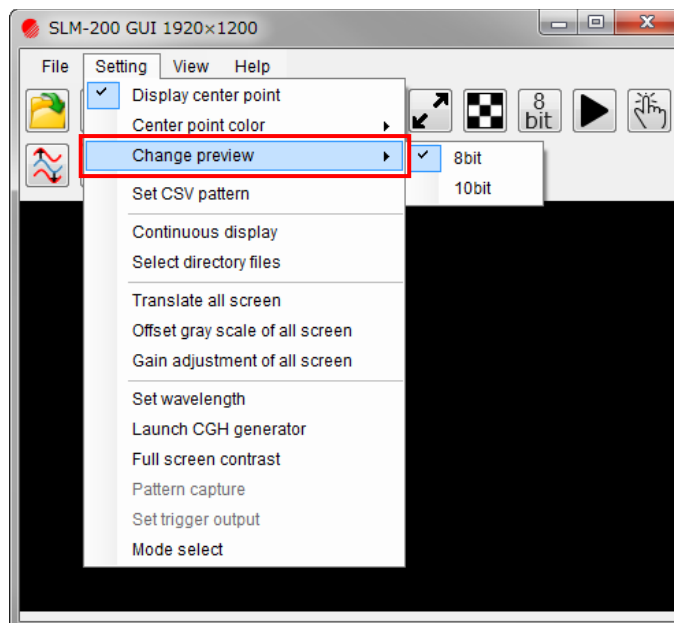
The color of cross mark can be changed.



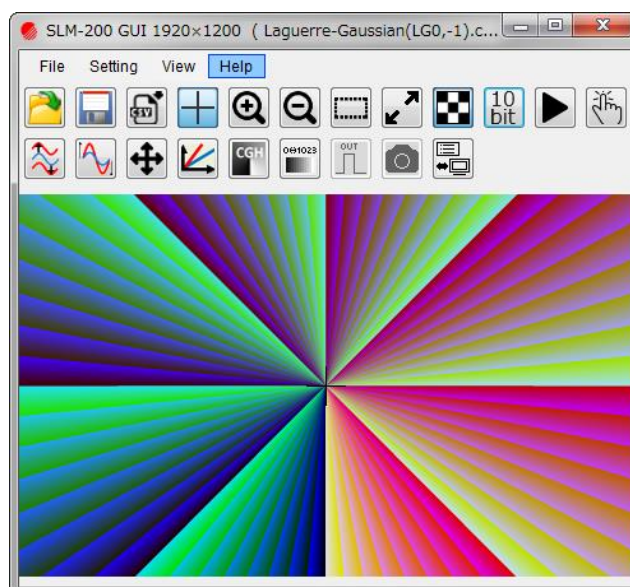
5.2.2.7 Change preview



The gray level for preview of 8 bit or 10 bit can be selected.



8 bit preview



10 bit preview

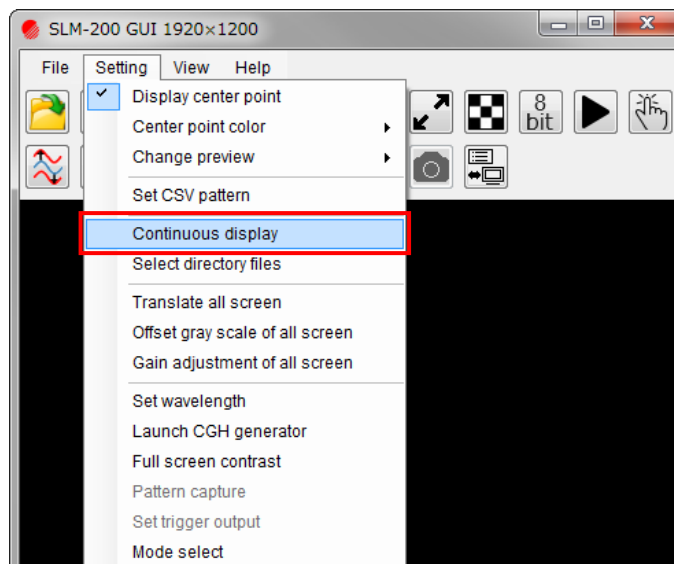
5.2.2.8 Set CSV pattern

The preview of specified display data is displayed.

5.2.2.9 Continuous display

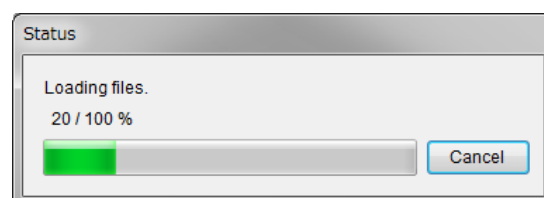
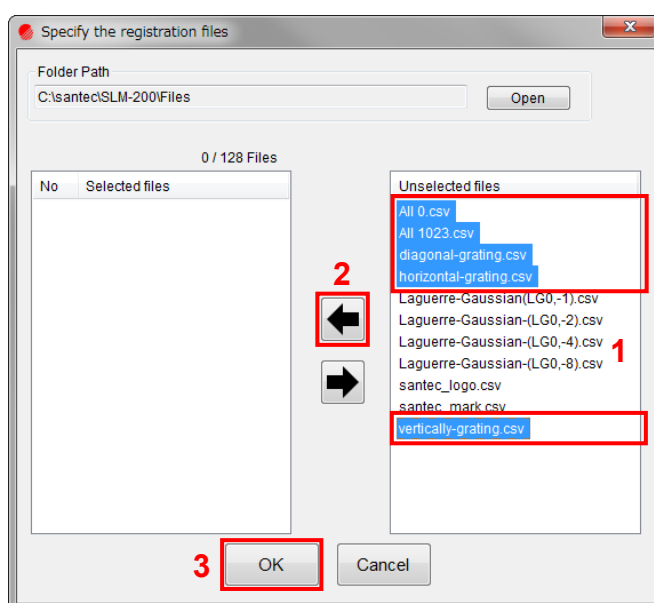


The specified set of multiple CSV files can be displayed continuously.

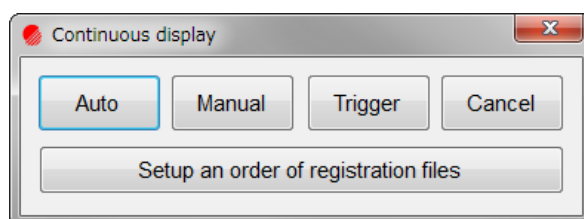


Continuously displayed image files are selected and moved to “Selected files” box as shown in the figure below. Maximum number of selected files at a time is 128. CSV files stored in reference folder described in following below are displayed on “Unselected folder” box.

※Reference folder : C:¥santec¥SLM-200¥Files



After selecting the image files, the following dialog box for selecting play-modes is displayed.



5.2.2.9.1 “Auto” mode

“Auto” mode displays image files continuously with specified regular interval and loop count.

※Available range for interval and loop count

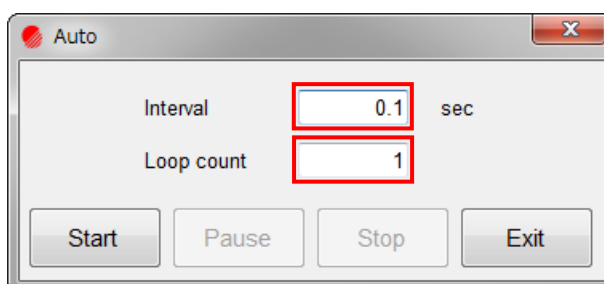
DVI mode

$0 \text{ sec} \leq \text{Interval} \leq 10, 0 \leq \text{Loop count} \leq 500 (0 = \infty)$

Memory mode, Pattern capture mode

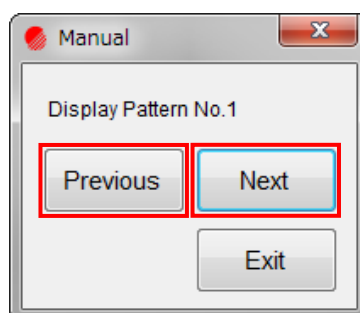
<Frame rate 60Hz> : $0 \text{ sec} \leq \text{Interval} \leq 2$

<Frame rate 120Hz> : $0 \text{ sec} \leq \text{Interval} \leq 1$



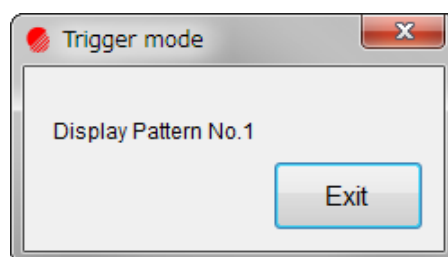
5.2.2.9.2 “Manual” mode

“Manual” mode displays the image files in arbitral timing by clicking the “previous” or “Next” button.



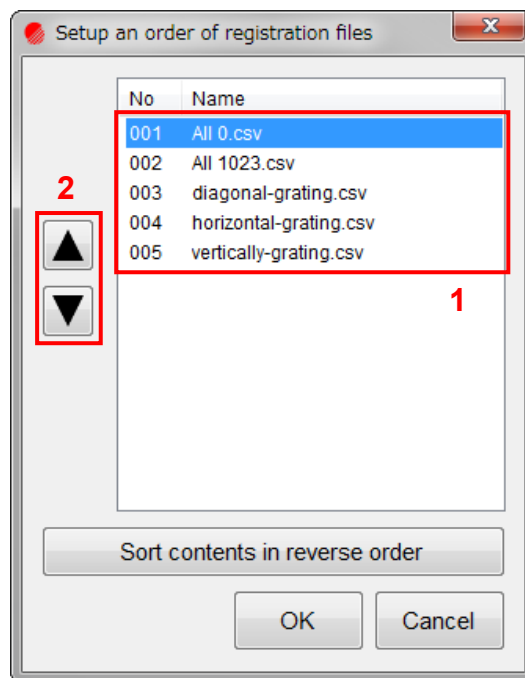
5.2.2.9.3 “Trigger” mode

“Trigger” mode displays the image files in input trigger timing and continue until clicking “Exit” button.



5.2.2.9.4 Change the order of registered files

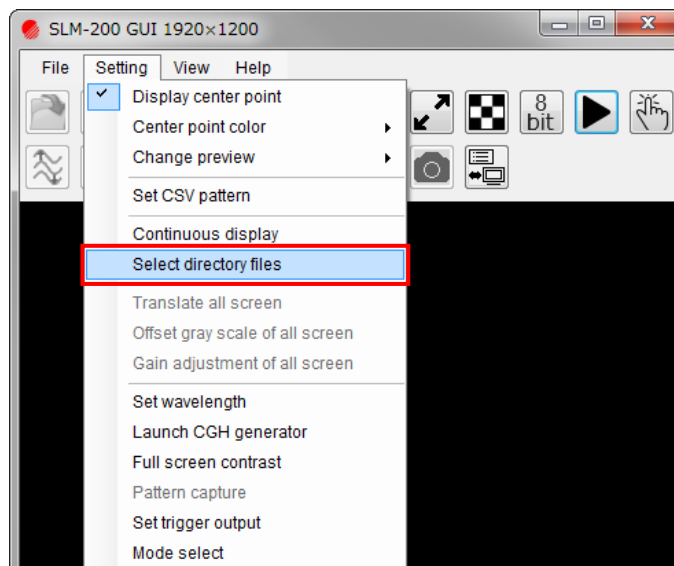
The order of displayed image files can be changed from the dialog box.



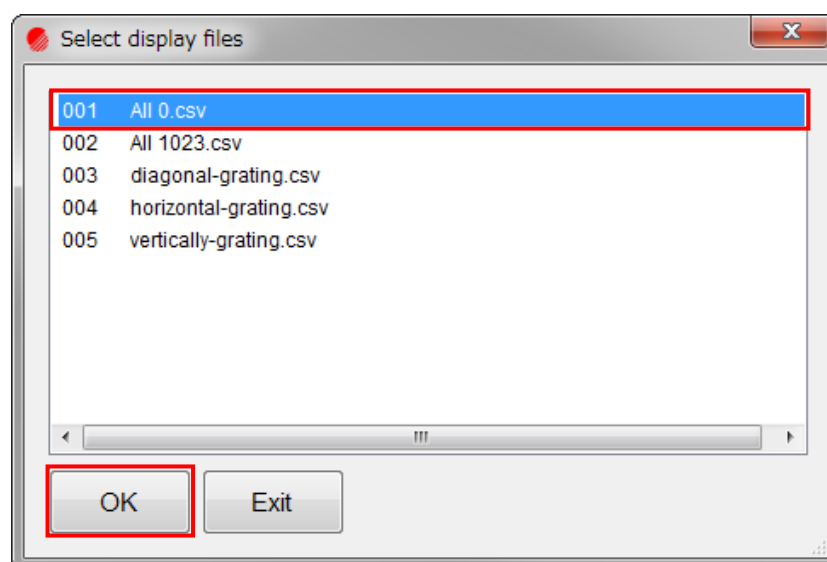
5.2.2.10 Selection directory files



In accordance with the same procedure as 4.2.2.9, displayed image files are selected.



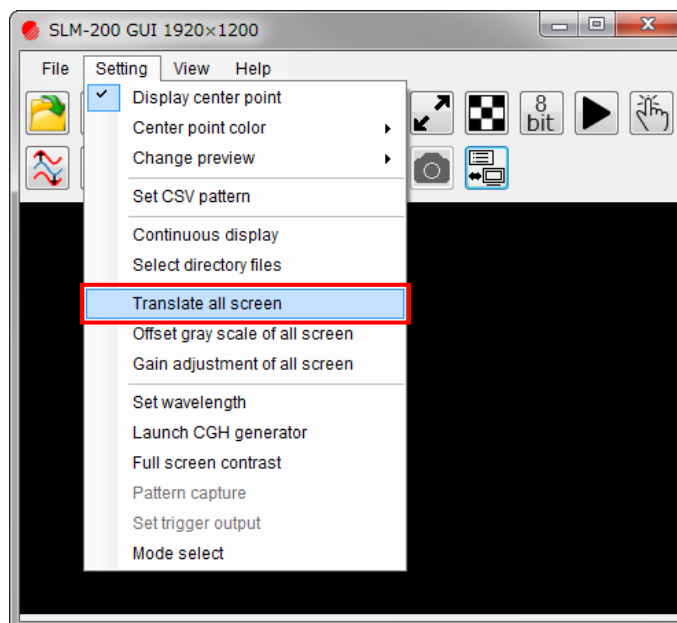
After registering the image files, the following dialog box is displayed. The image file selected by double-click or pushing the displayed “OK” button.



5.2.2.11 Translate all screen

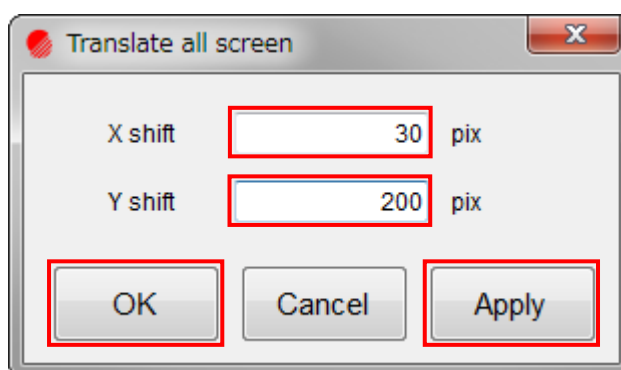


The position of the image displayed on the LCOS panel can be changed.



By specifying the pixel number for translation on the X and Y axes, the position of the displayed image can be shifted.

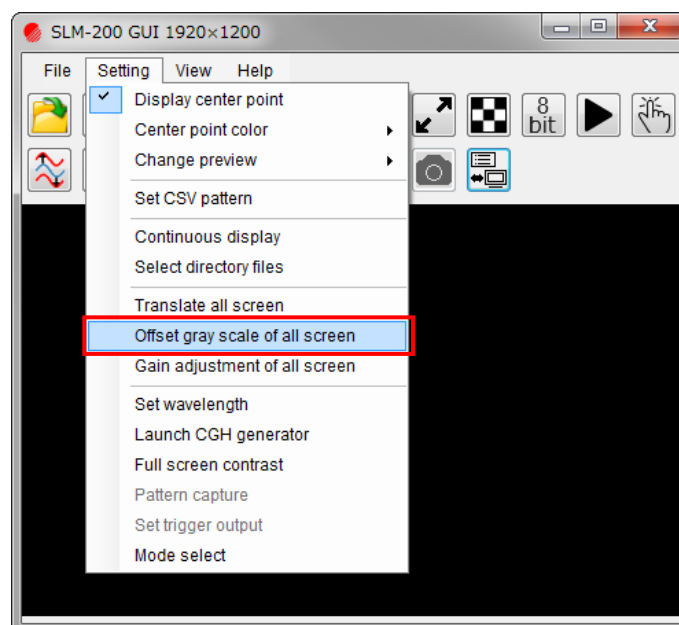
※Available translation range : $-1920 \leq X \text{ shift} \leq 1920$, $-1200 \leq Y \text{ shift} \leq 1200$



5.2.2.12 Offset gray scale of all screen

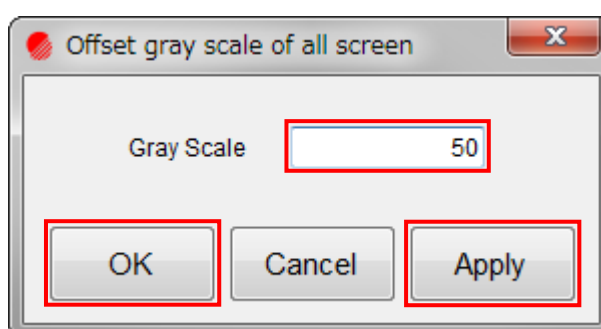


The gray scale of all pixels for the displayed image can be offset at a specified level.



By specifying the offset level of the gray scale on the dialog box below, the offset image is displayed.

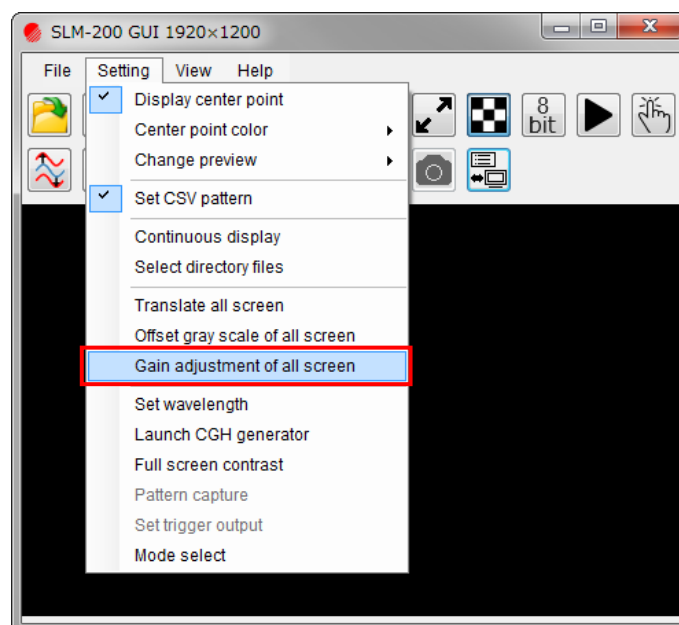
※Available offset range : $-1023 \leq \text{Gray Scale} \leq 1023$



5.2.2.13 Gain adjustment of all screen

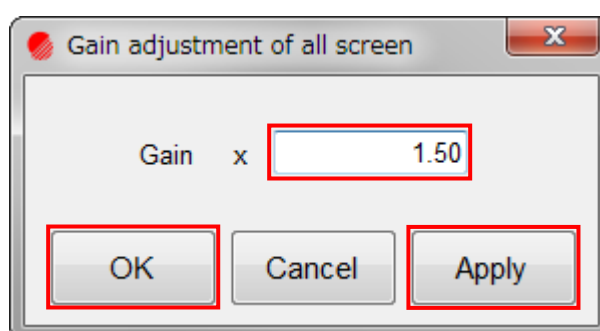


The gray scale of all pixels for the displayed image can be rescaled with a specified value.



By specifying the gain value for rescaling on dialog box below, the displayed image can be rescaled.

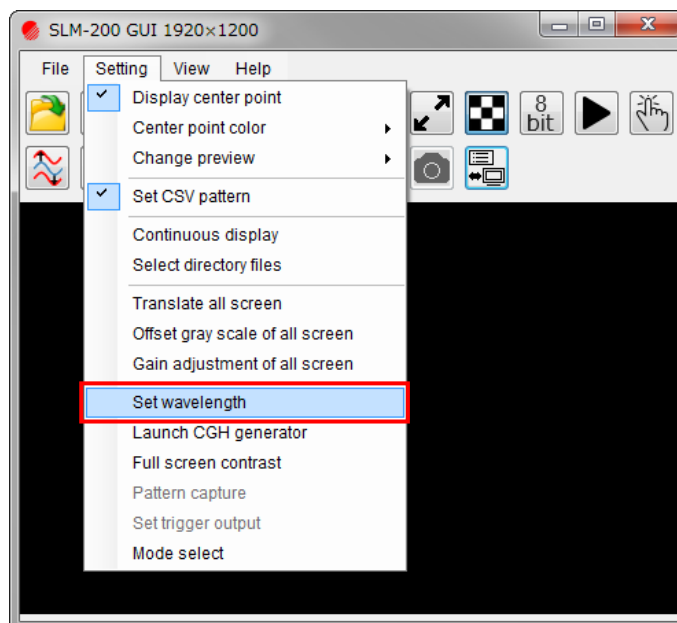
※Available gain range : $0.01 \leq \text{Gain} \leq 1023$



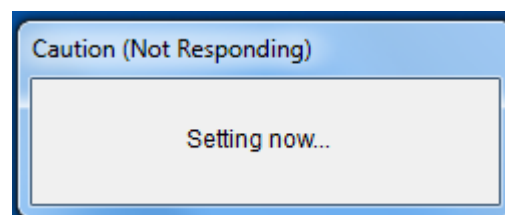
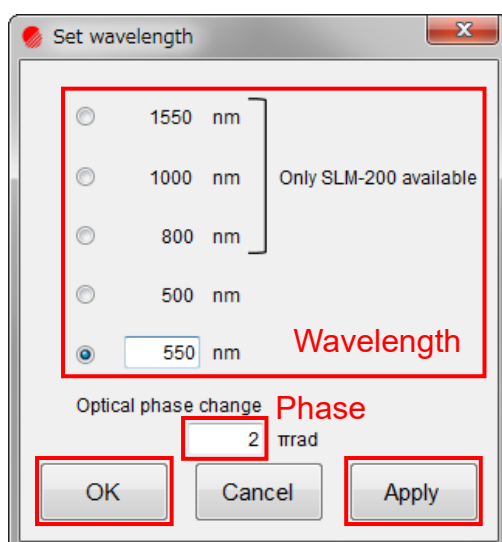
5.2.2.14 Set wavelength



The calibration data of phase level can be adjusted with a user specified wavelength.



The arbitral wavelength within 450nm to 1600nm can be set at a 1 nm interval. Maximum optical phase at specified wavelength can be adjusted to the inputted value in "Optical phase change". The setting change takes about 30 to 40 second.

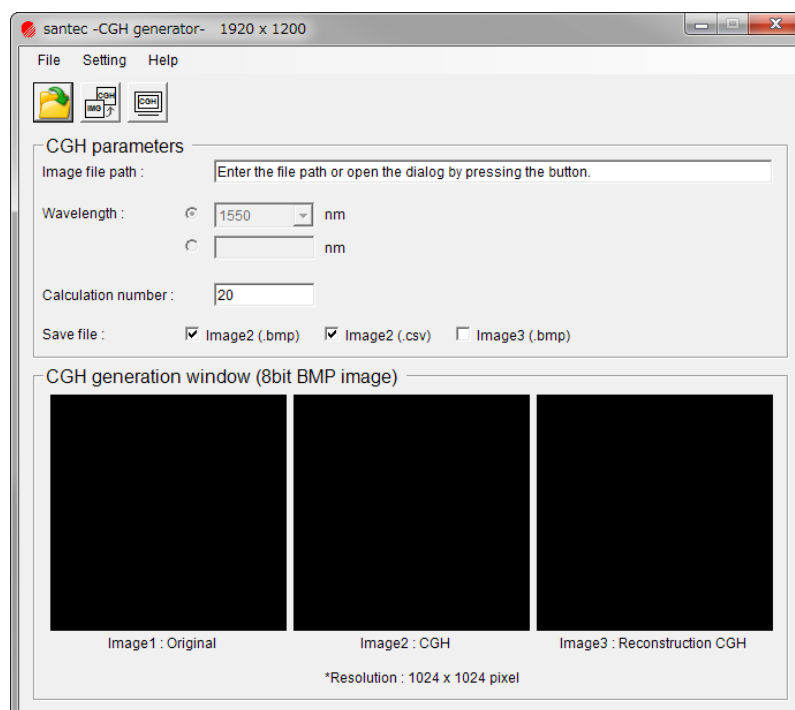
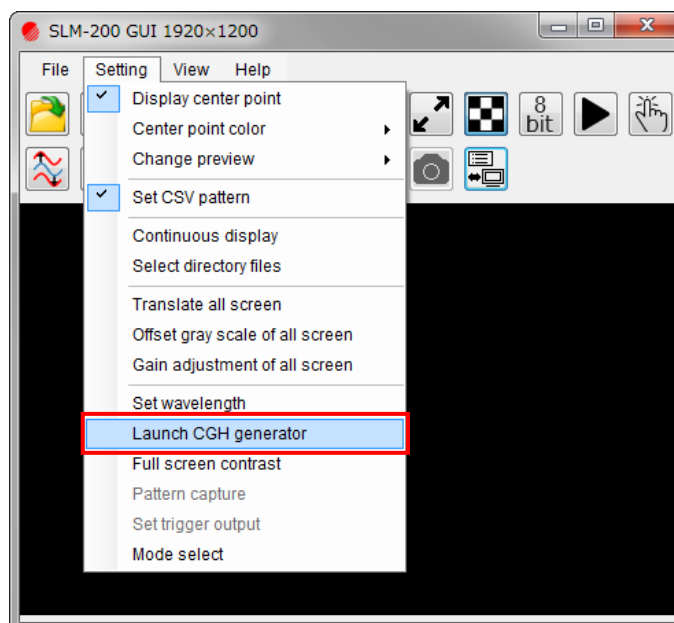


Message under calibration setting

5.2.2.15 Launch CGH generator



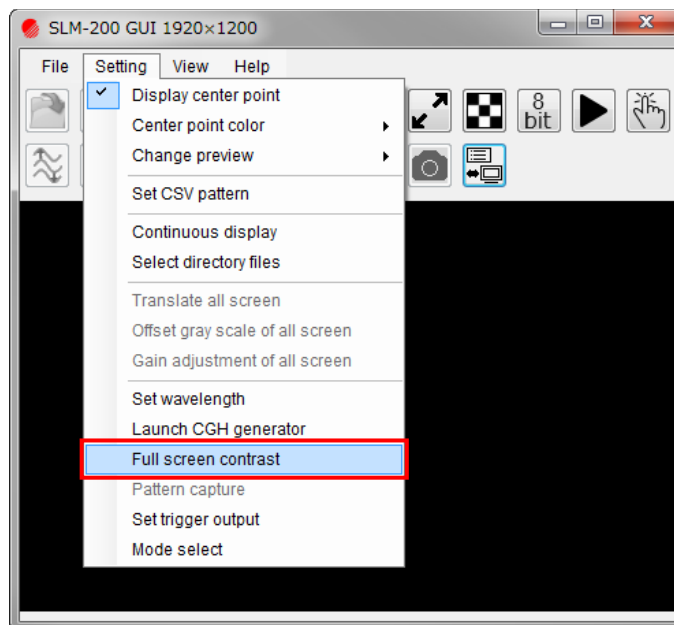
The CGH (Computer Generated Hologram) phase pattern can be generated from specified image file.



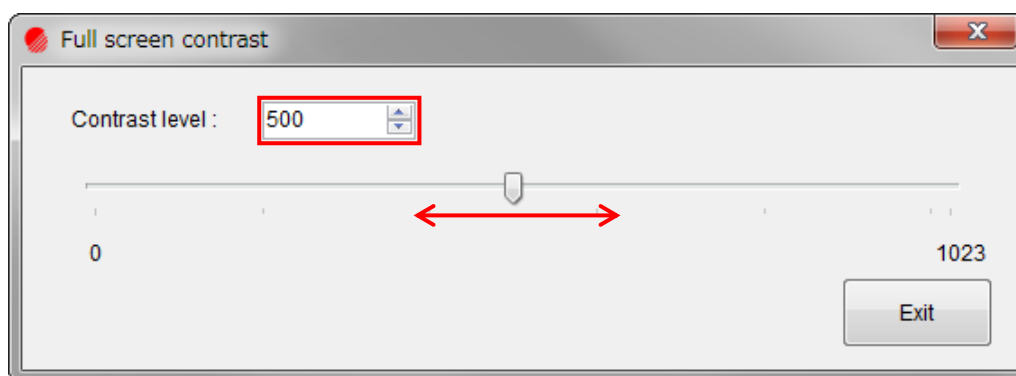
5.2.2.16 Full screen contrast



The entire LCOS display can be controlled to specified 10bit contrast level from 0 to 1023.



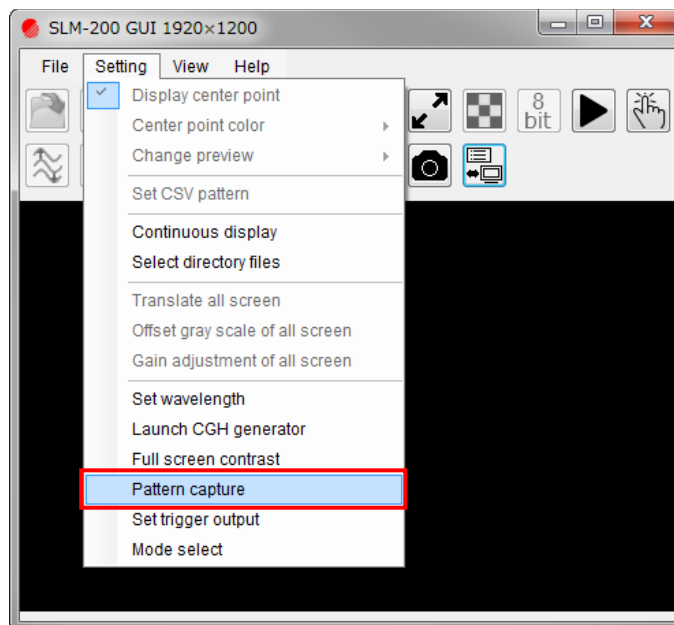
The contrast level can change slider control or entering specified value.



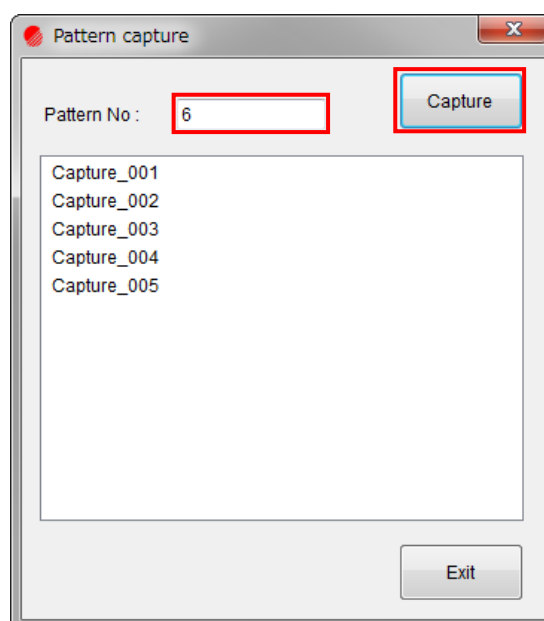
5.2.2.17 Pattern capture



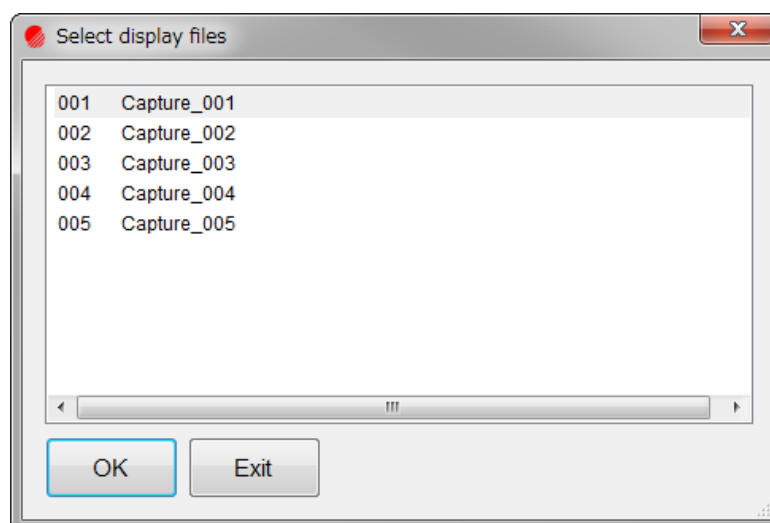
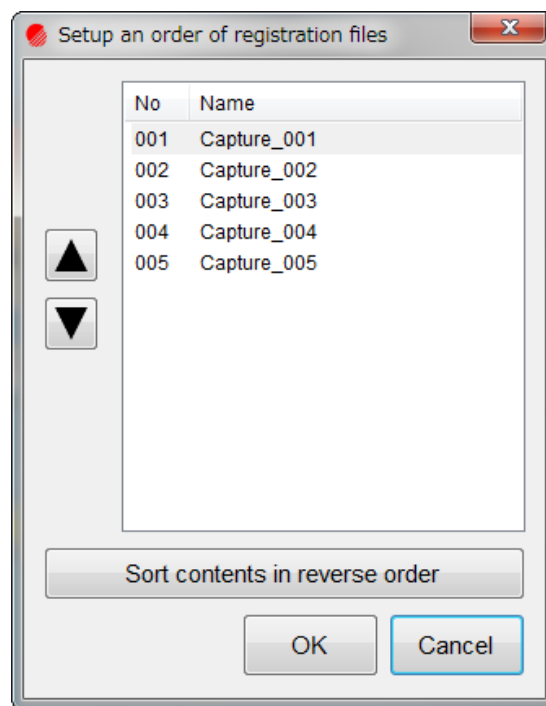
Displayed pattern on LCOS can be captured and stored in memory.



The displayed pattern on LCOS in the timing of clicking “Capture” button is captured. Captured pattern is numbered according to specified number in “Pattern No” and can be stored maximum 128 patterns.



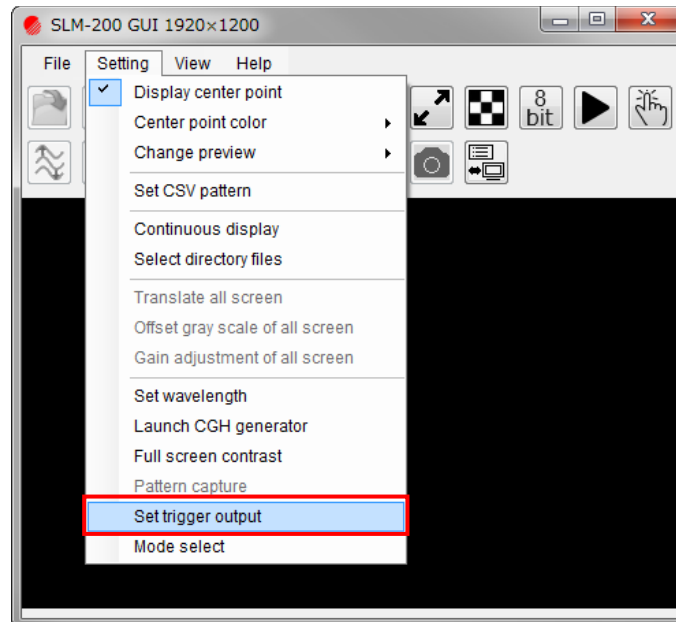
Captured patterns can be displayed in “Set continuous mode” or selecting display files directly.



5.2.2.18 Set trigger output

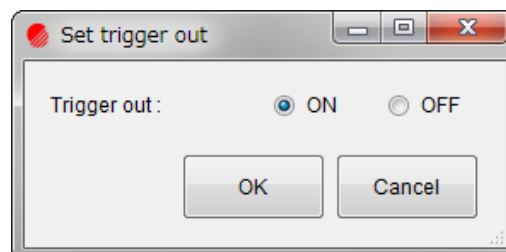


The output trigger can be set ON or OFF.



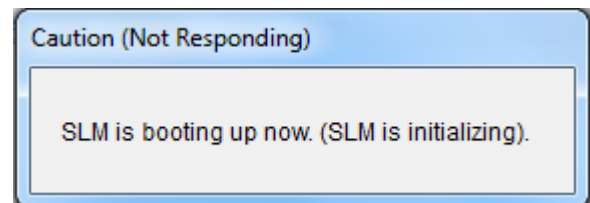
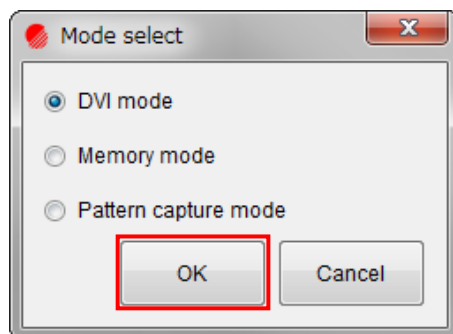
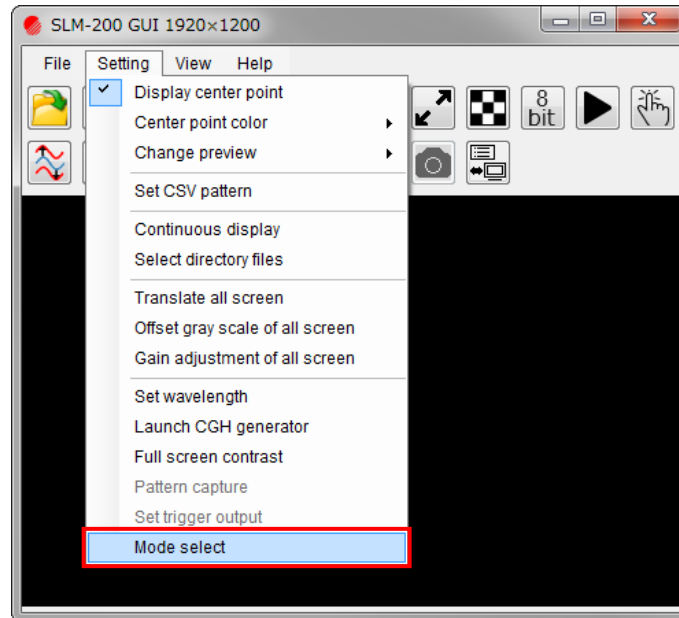
“ON” : Trigger signal output in timing of display change

“OFF” : No trigger output



5.2.2.19 Mode select

The SLM display mode can be selected.



Message under changing display mode

<DVI mode>

Displayed input patterns via DVI-D DVI.

<Memory mode>

Displayed memory patterns stored in SLM via USB (Max.: 128 patterns).

<Pattern capture mode>

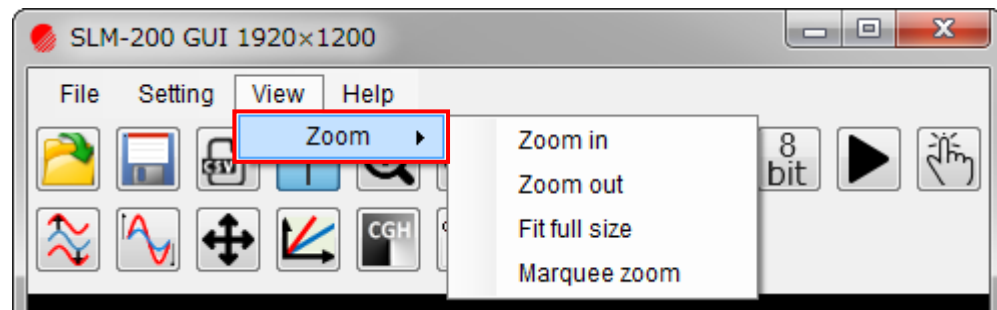
Captured patterns displayed on LCOS via DVI-D and stored in SLM internal memory (Max.: 128 patterns).

Available functions for each mode.

	DVI mode	Memory mode	Pattern capture mode
Open CSV or BMP file	✓		
Convert BMP file to CSV file	✓	✓	
Display center point	✓	✓	
Center point color	✓	✓	
Change preview	✓	✓	
Set CSV pattern	✓	✓	
Set continuous mode	✓	✓	✓
Selection directory files	✓	✓	✓
Translate all screen	✓		
Offset gray scale of all screen	✓		
Gain adjustment of all screen	✓		
Set wavelength	✓	✓	✓
Launch CGH generator	✓	✓	✓
Full screen contrast	✓	✓	✓
Pattern capture			✓
Set trigger output		✓	✓
Mode select	✓	✓	✓
Zoom	✓	✓	

5.2.2.20 Zoom in

Preview display is expanded at the center of cross mark.



5.2.2.21 Zoom out

Preview display is reduced at the center of cross mark.

5.2.2.22 Fit full size

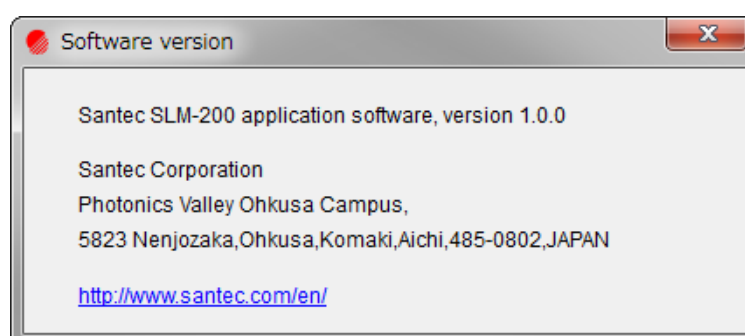
Preview is displayed full-screen.

5.2.2.23 Marquee zoom

Selected area is expanded on preview display.

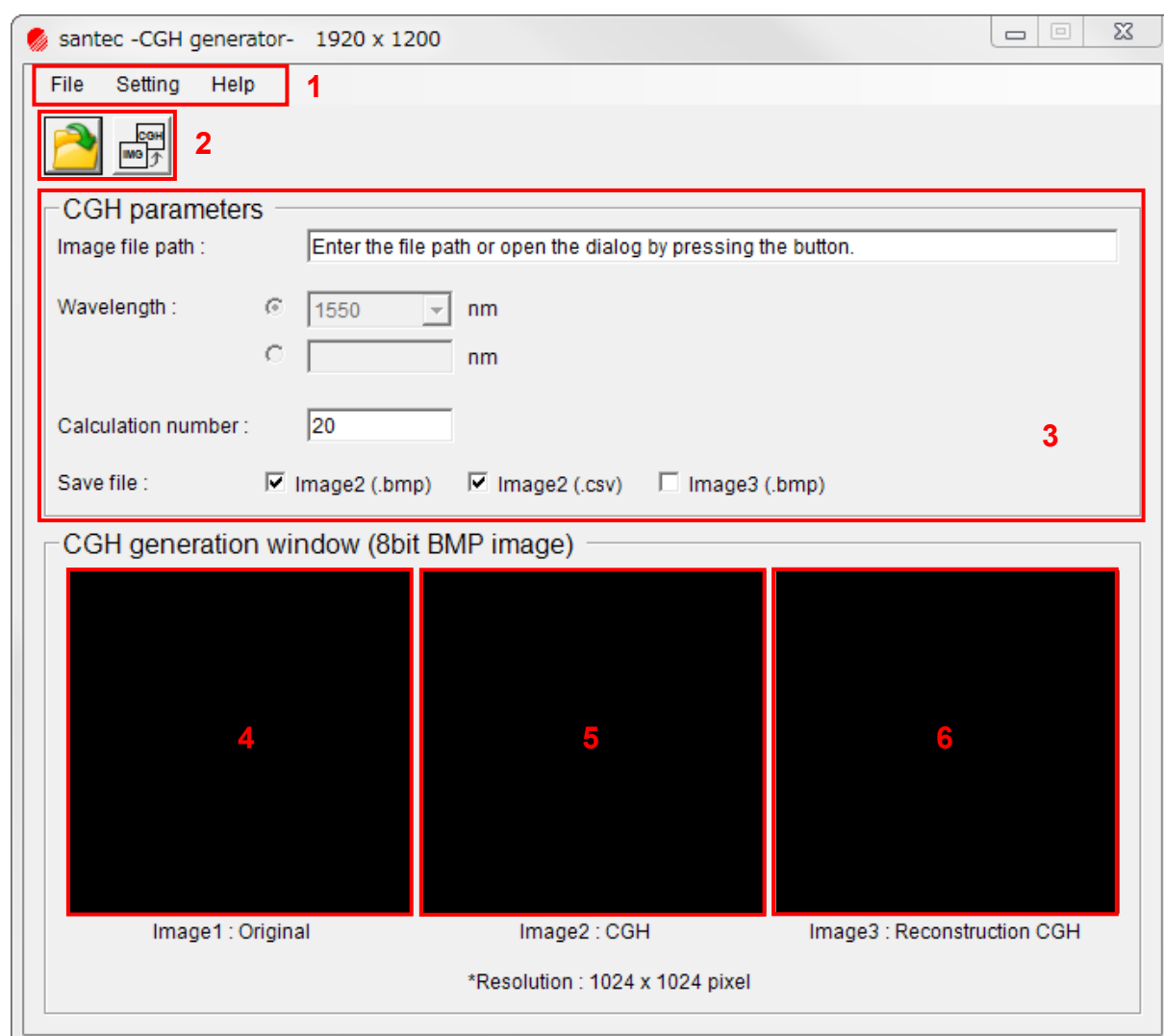
5.2.2.24 Software version

Software version is displayed.



5.3 CGH generator

5.3.1 GUI window

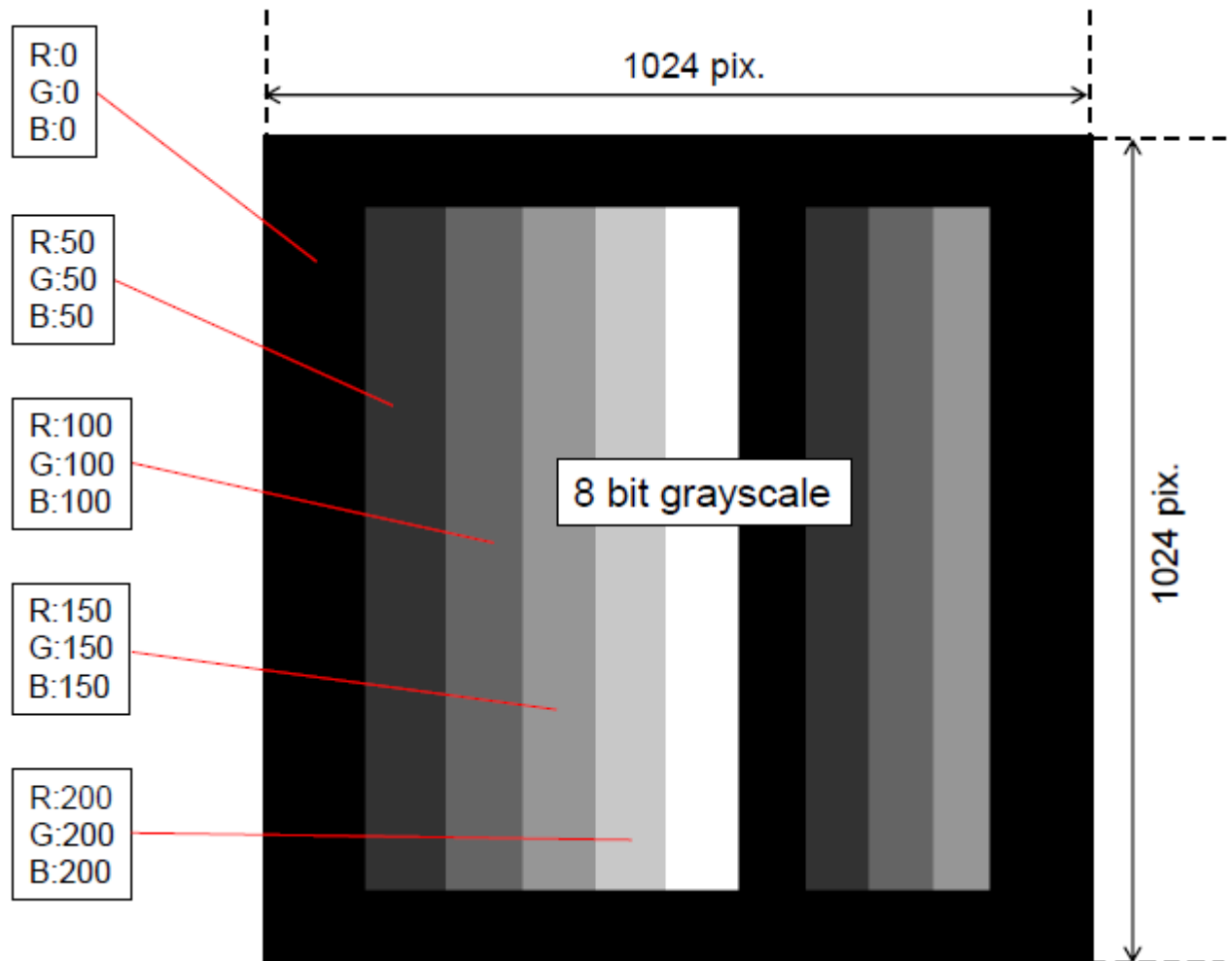


1. Menu tab
2. Shortcut menu
3. CGH parameters
4. Original image
5. CGH image
6. Computed reconstruction image

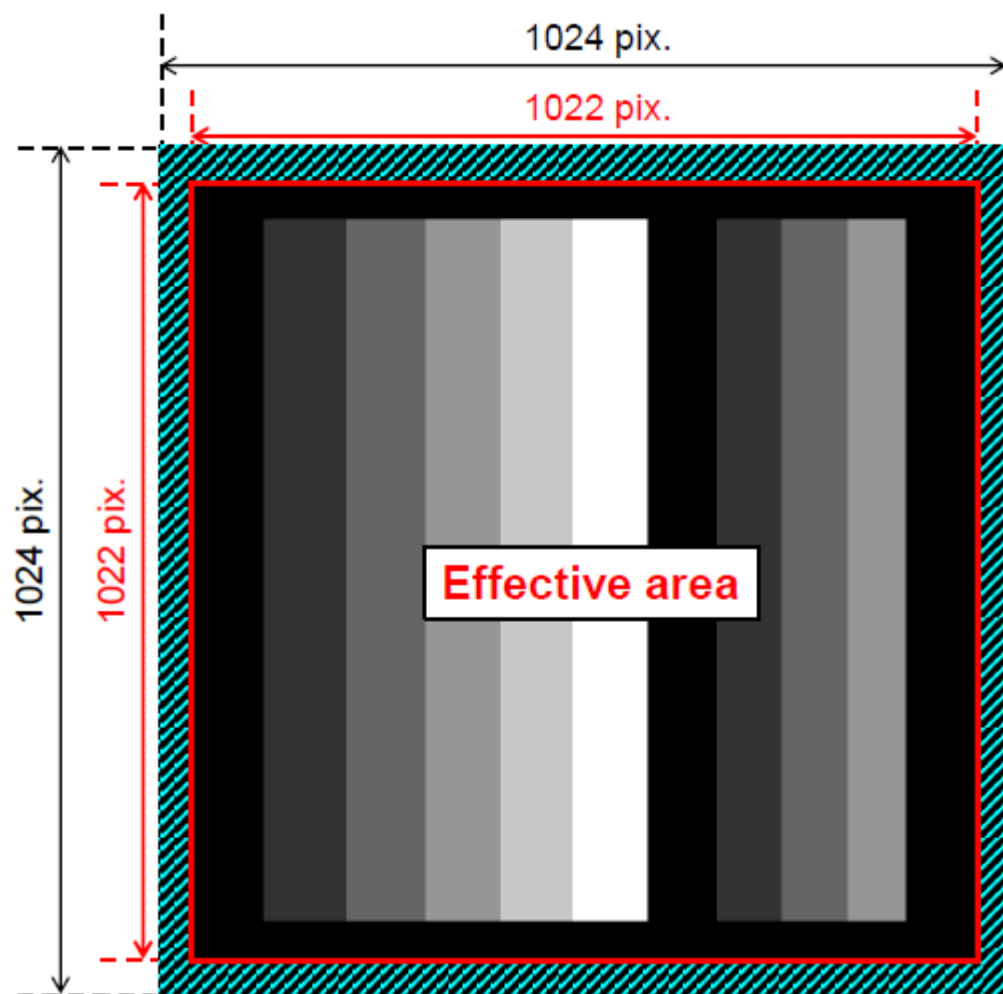
5.3.2 BMP file format for CGH generation

Prepare BMP file for CGH generation

- Effective image size is 1024×1024 pixel
- 8 bit grayscale

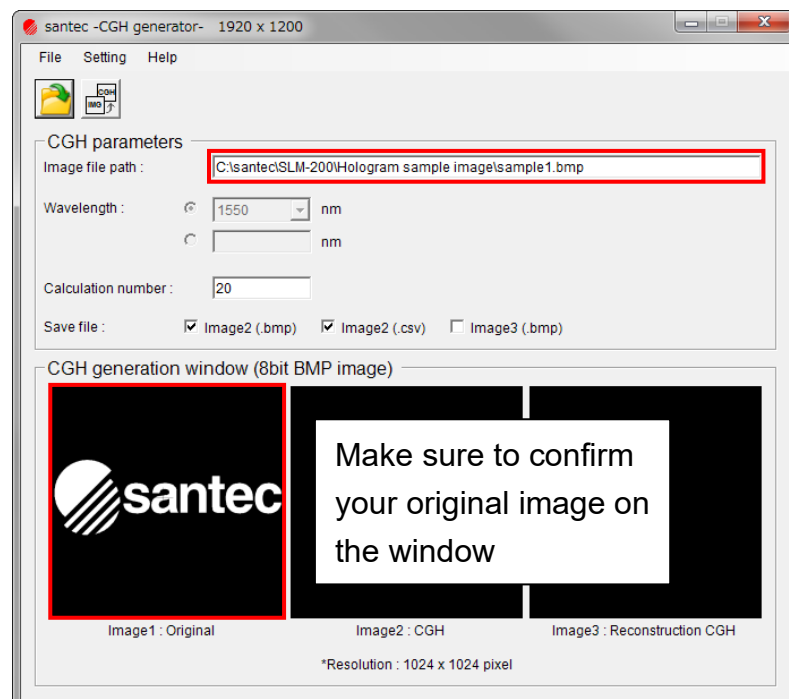
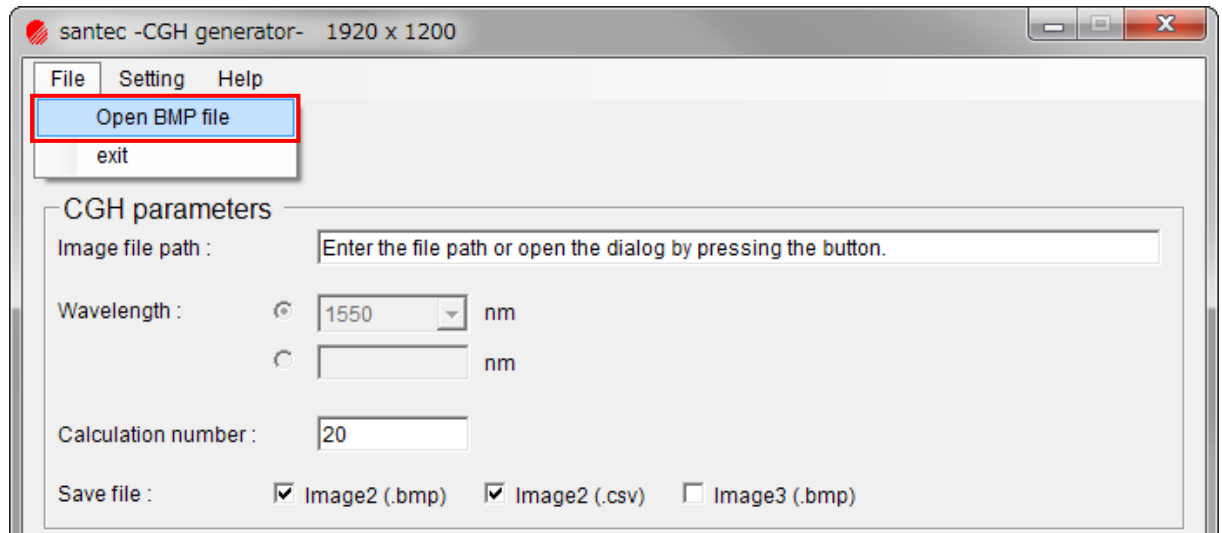


5.3.3 Effective area of BMP image

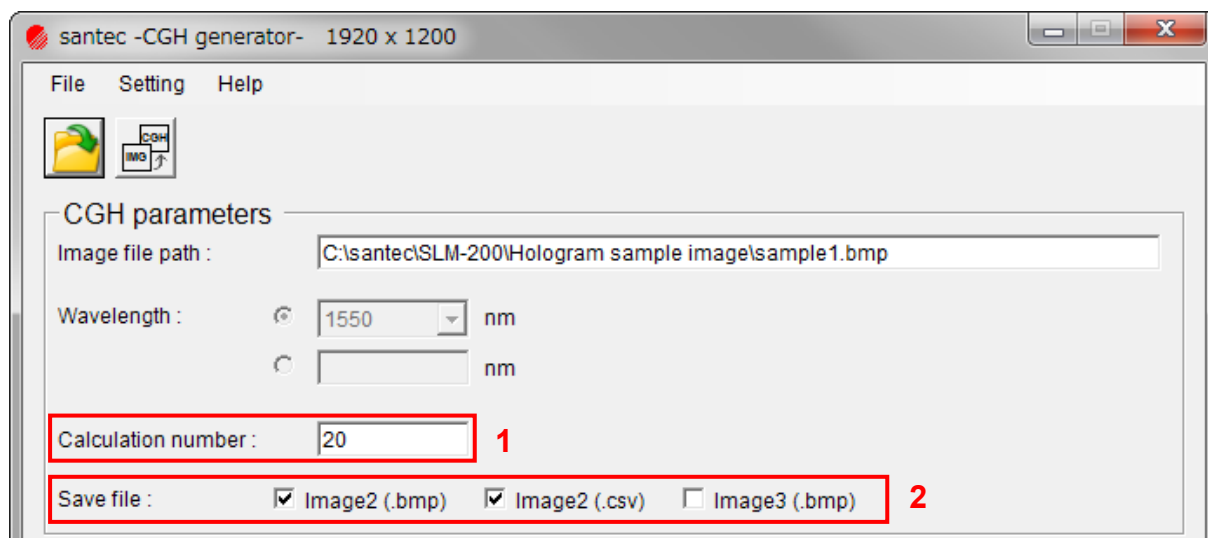


5.3.4 CGH generation

(1) Open BMP file for target image

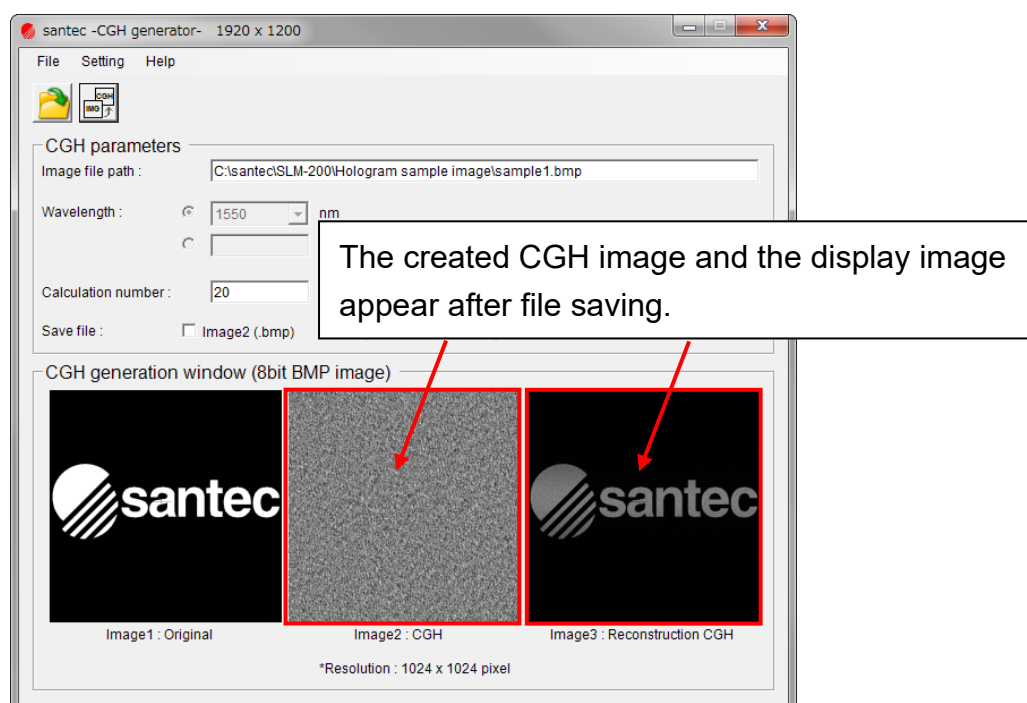
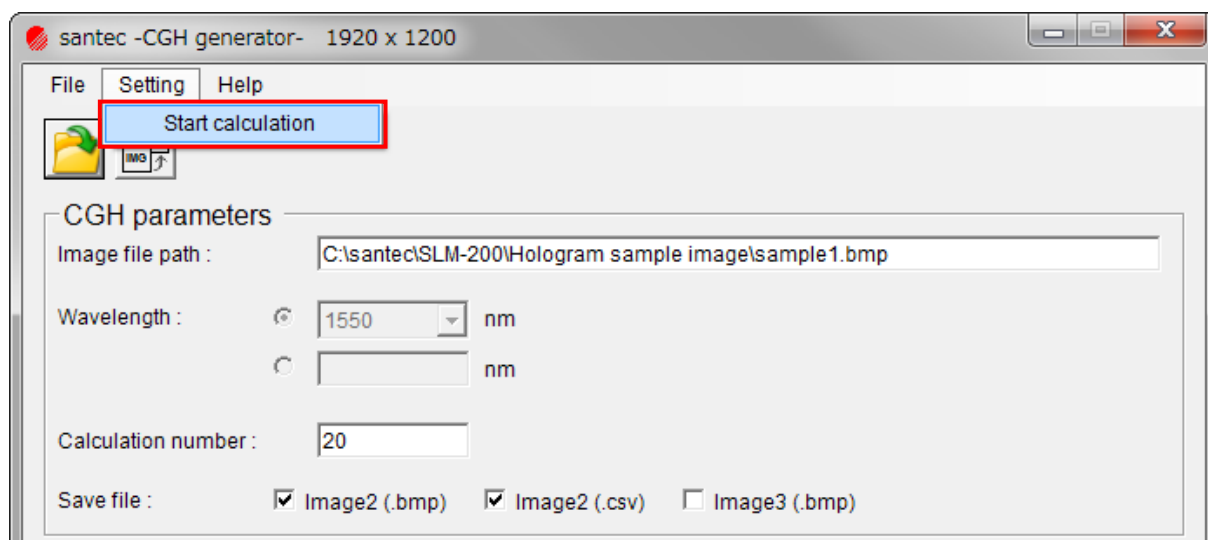


(2) Set CGH parameters



1. Calculation loop number for CGH generation, integers 1 to 1000 are available.
Calculation time strongly depends on the loop number.
The setting 20 is recommended for an adequate accuracy.
2. Select files type for saving
 - Image2 (.bmp) : Saved calculated CGH pattern as BMP file type
 - Image2 (.csv) : Saved calculated CGH pattern as CSV file type
 - Image3 (.bmp) : Saved display image of calculated CGH pattern

(3) Start calculation



5.4 Wavefront Correction

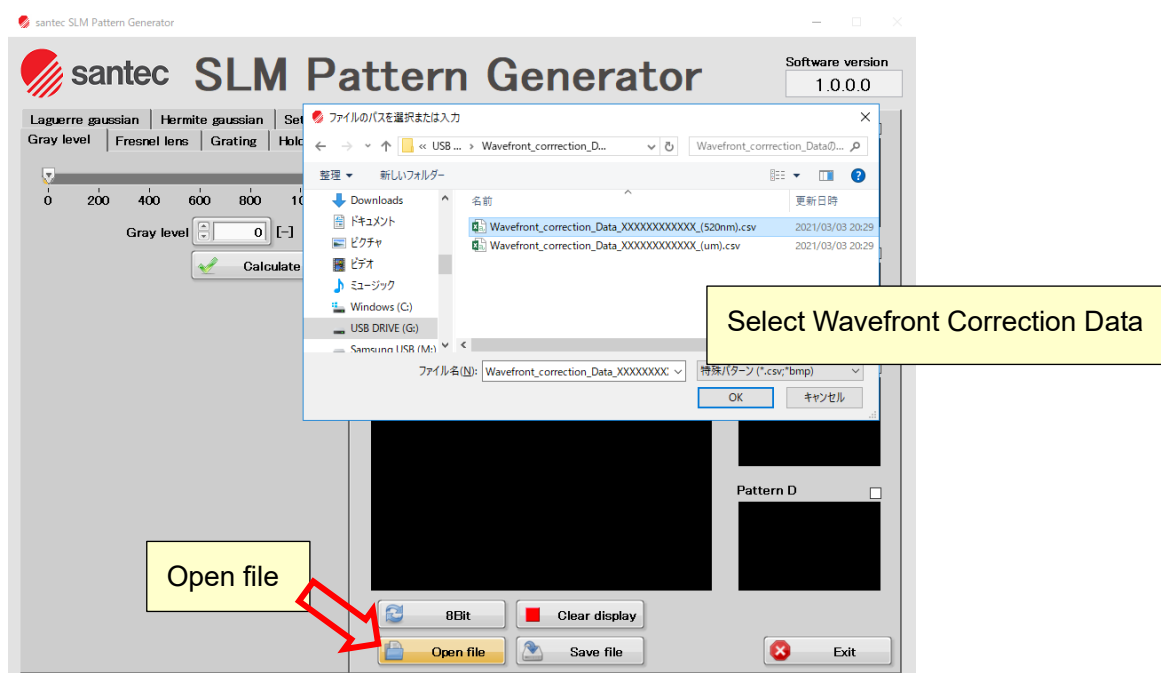
5.4.1 Wavefront Correction Data Location

See attached USB flash drive.

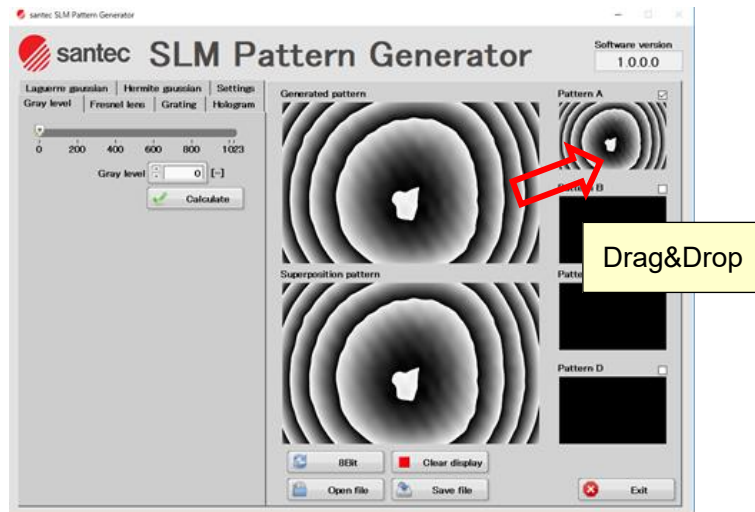


5.4.2 How to use a Wavefront Correction Data (Using SLM Pattern Generator)

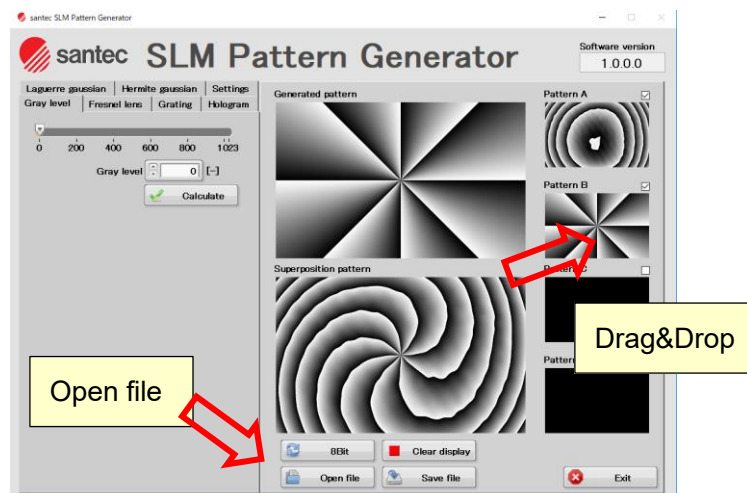
1. Run the SLM Pattern Generator.
(See. "santec SLM Pattern Generator Software Operation Manual v*. *.pdf")
2. Open the CSV file of wavefront correction with PatternGenerator.



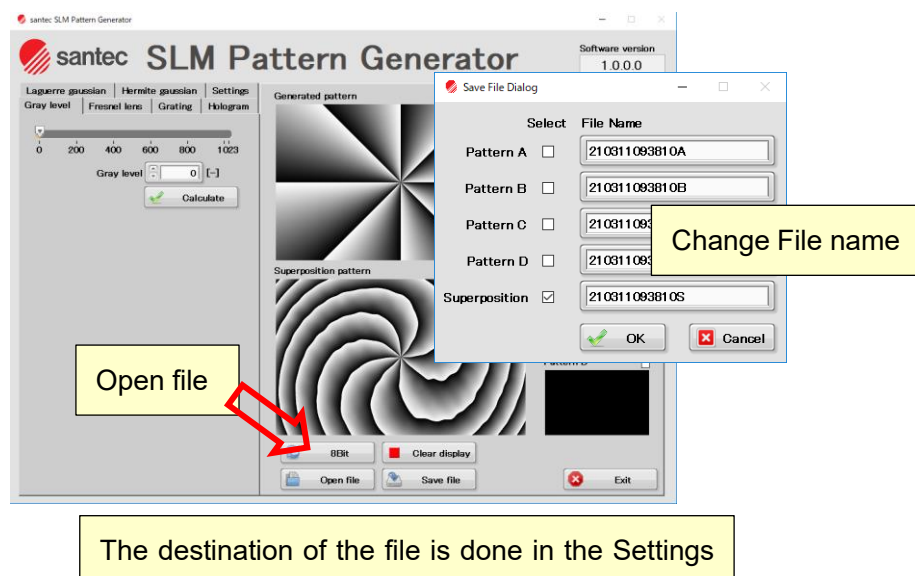
3. Drag and drop from Generated pattern to PatternA.



4. Open the file you want to correct and drag and drop it onto patternB.

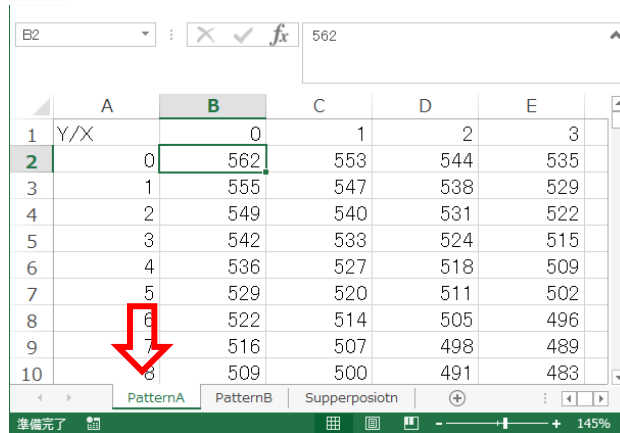


5. Save Superposition pattern.

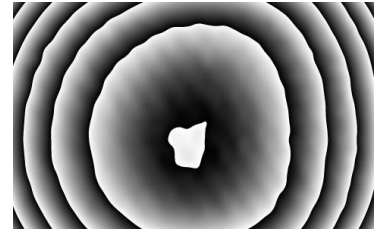


5.4.3 How to use a Wavefront Correction Data (Using Excel)

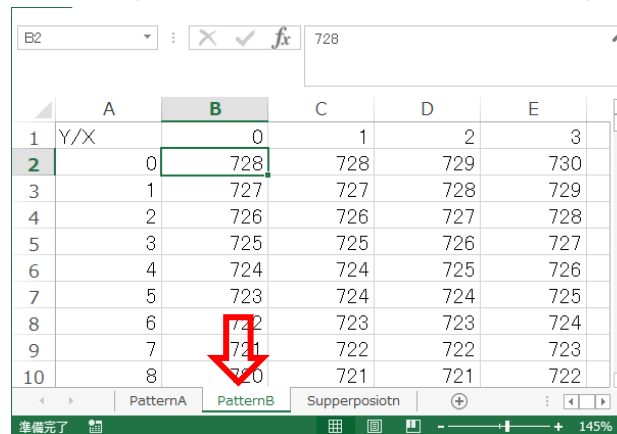
1. Open the CSV file of wavefront correction csv in Excel and copy it to the PatternA sheet.



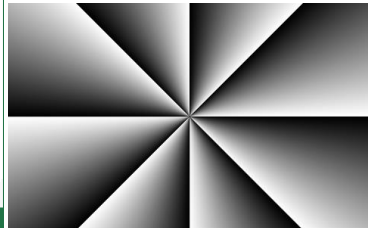
	A	B	C	D	E
1	Y/X	0	1	2	3
2		562	553	544	535
3	1	555	547	538	529
4	2	549	540	531	522
5	3	542	533	524	515
6	4	536	527	518	509
7	5	529	520	511	502
8	6	522	514	505	496
9	7	516	507	498	489
10	8	509	500	491	483



2. Open the file you want to correct in Excel and copy it to the PatternB sheet.



	A	B	C	D	E
1	Y/X	0	1	2	3
2		728	728	729	730
3	1	727	727	728	729
4	2	726	726	727	728
5	3	725	725	726	727
6	4	724	724	725	726
7	5	723	724	724	725
8	6	722	723	723	724
9	7	721	722	722	723
10	8	720	721	721	722

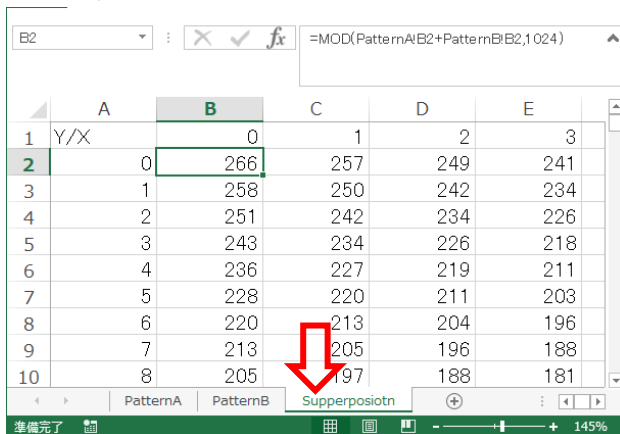


3. Use the sheet to calculate as follows.

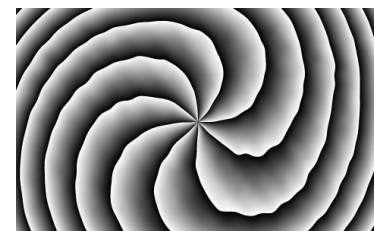
In "cell B2" on new sheet, enter the following formula.

$\text{=MOD}(\text{PatternA!B2}+\text{PatternB!B2},1024)$

Then, copy "cell B2" to "B2:BUW1201".



	A	B	C	D	E
1	Y/X	0	1	2	3
2		266	257	249	241
3	1	258	250	242	234
4	2	251	242	234	226
5	3	243	234	226	218
6	4	236	227	219	211
7	5	228	220	211	203
8	6	220	213	204	196
9	7	213	205	196	188
10	8	205	197	188	181



6. Contact



In the event of any trouble with this product, turn the unit off in accordance with the procedures to shut off the power described in this operation manual, disconnect the power source cord, record the product name and serial number described on the name plate of the product, and then contact our dealer at your place or directly contact us at santeC Photonics Laboratories. Our telephone number and facsimile number are shown below. However, we are not responsible for any trouble arising from your own repair or modification on this product.

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